

Project Data Page:

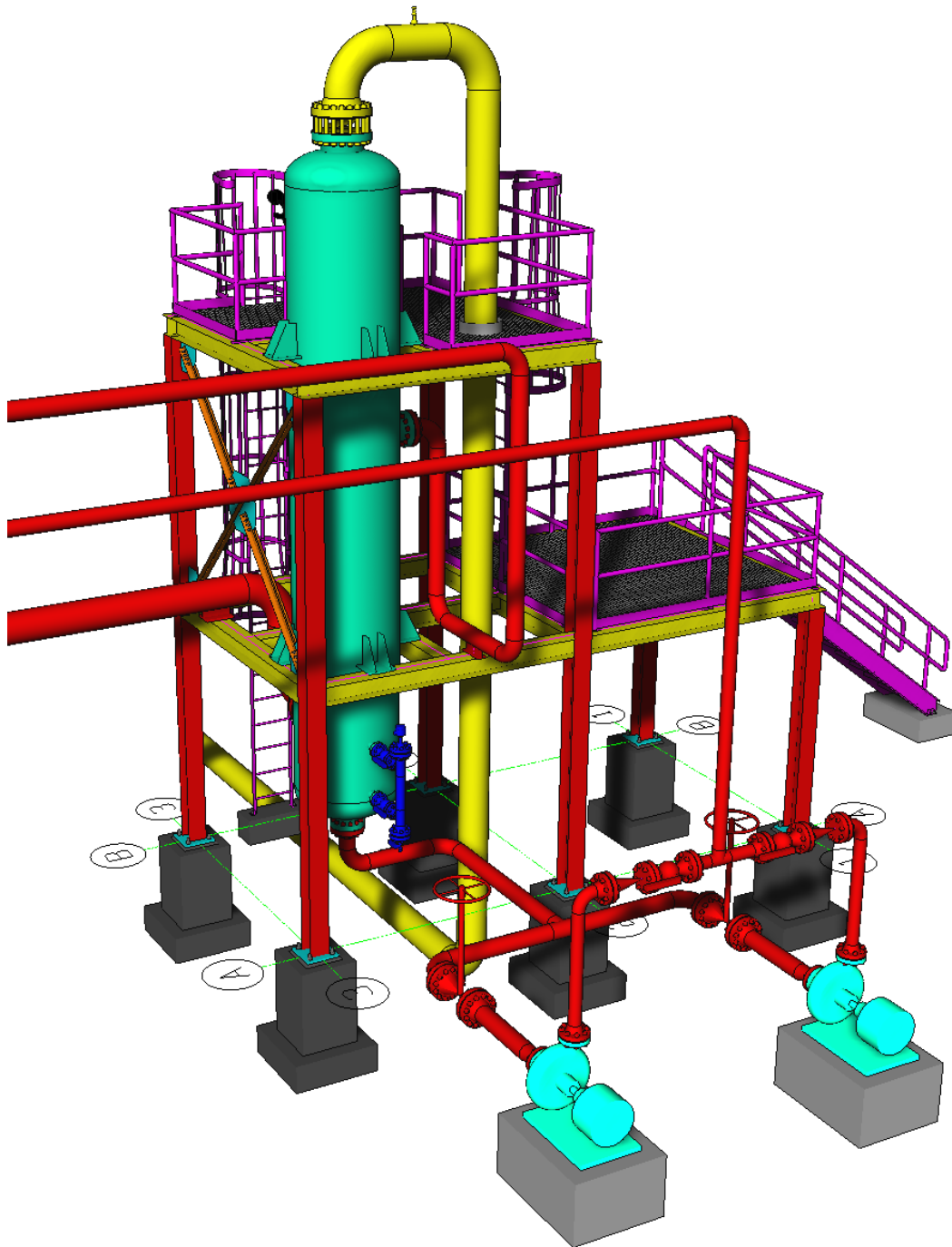




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\C-1901R00pe.pvdb

Date of Analysis : Sep 9, 2026 2:08pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 5506.000 x 5506.000 x 3760.000 mm OD
 Vessel Design Length, Tangent to Tangent 13971.17 mm

Distance of Bottom Tangent above Grade 17000.00 mm
 Distance of Base above Grade 300.00 mm
 Specified Datum Line Distance 16700.00 mm

Internal Design Temperature 160 °C
 Internal Design Pressure 0.348 MPa

External Design Temperature 0 °C
 External Design Pressure 0.103 MPa

Maximum Allowable Working Pressure 0 MPa
 External Max. Allowable Working Pressure 0 MPa
 Shop Test Pressure 0.588 MPa
 Field Test Pressure 0.493 MPa

Required Minimum Design Metal Temperature -1.0 °C
 Warmest Computed Minimum Design Metal Temperature -26.0 °C

Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Cone	SA-516 70	...	...	K02700	
Flange	SA-250 T1b	...	...	K11422	
Flange	SA-266 2	...	...	K03506	+IT
Skirt	E250A	...	t > 40	SA/IS 20	
Skirt	SA-516 60	...	...	K02100	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-106 B	...	...	K03006	
Rings	SA-516 70	...	...	K02700	
Basering	E250A	...	t < 20	SA/IS 20	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-1	...	...	...	1.0000	N/A	N/S
Skirt-2	...	...	...	1.0000	N/A	N/S
Skirt-3	...	...	...	1.0000	N/A	N/S
Skirt-4	...	...	...	1.0000	N/A	N/S
Skirt-5	...	...	...	1.0000	N/A	N/S
Skirt-6	...	...	...	1.0000	N/A	N/S
Skirt-7	...	...	...	1.0000	N/A	No
BTM DISH	0.402	0.10	1.000	1.0000	No	No



Shell-1	0.388	0.10	1.000	1.0000	N/A	No
Shell-2	0.358	0.10	1.000	1.0000	N/A	No
Shell-3	0.348	0.10	1.000	1.0000	N/A	No
Shell-4	0.348	0.10	1.000	1.0000	N/A	No
Cone 1	0.348	0.10	1.000	1.0000	N/A	No
Cone 2	0.348	0.10	1.000	1.0000	N/A	No
Port shell	0.348	0.10	1.000	1.0000	N/A	No
N4 Flange	0.348	0.10	0.000	1.0000	N/A	No
N2A Flange	0.358	0.10	1.000	1.0000	N/A	No
N2B Flange	0.358	0.10	0.000	1.0000	N/A	No

N/S in the Creep Range column specifies that the Creep Range Temperature was not specified in the ASME Code Section II Part D for this material.

Liquid Level: 5446.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 1.000

Stiffener Ring Specifications:

Elevation mm	Selected Type	Ring Description
250.00	Bar 285.0 x 16.0	Ring R1 Fr90
2900.00	Bar 285.0 x 16.0	Ring R2 Fr90
5900.00	Bar 337.0 x 14.0	Ring R1 Fr100
8550.00	Bar 287.0 x 14.0	Ring R1 Fr110
10755.00	Bar 300.0 x 20.0	R1
13318.00	Bar 300.0 x 14.0	R2

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-15700.0	1000.0	30.0	30.0	...	...	0.70	0.70
Skirt	-13000.0	2700.0	20.0	20.0	...	...	0.70	0.70
Skirt	-10000.0	3000.0	18.0	18.0	...	...	0.70	0.70
Skirt	-7000.0	3000.0	18.0	18.0	...	...	0.70	0.70
Skirt	-4000.0	3000.0	18.0	18.0	...	...	0.70	0.70
Skirt	-1000.0	3000.0	16.0	16.0	...	...	0.70	0.70
Skirt	-0.0	1000.0	16.0	16.0	...	...	0.70	0.70
Ellipse	50.0	50.0	20.0	16.0	9.0	...	1.00	1.00
Cylinder	3050.0	3000.0	16.0	16.0	8.7	...	1.00	1.00
Cylinder	6050.0	3000.0	16.0	16.0	8.2	...	1.00	1.00
Cylinder	9050.0	3000.0	16.0	16.0	8.0	...	1.00	1.00
Cylinder	10800.0	1750.0	14.0	14.0	8.0	...	1.00	1.00
Conical	12098.0	1298.0	16.0	14.0	8.4	...	1.00	1.00
Conical	13298.0	1200.0	16.0	14.0	7.1	...	1.00	1.00
Cylinder	13698.0	400.0	12.0	12.0	5.7	...	1.00	1.00
Body Flg	13971.2	270.0	260.0	240.0	234.1	...	1.00	1.00
Body Flg	3050.0	112.0	60.0	60.0	55.0	...	1.00	1.00
Body Flg	3050.0	112.0	60.0	60.0	55.0	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	...	No Calc	...	No Calc
50	60	...	No Calc	...	No Calc
60	70	...	No Calc	...	No Calc
70	80	...	No Calc	...	No Calc
80	90	16	15.3386	0.10342	0.11319
90	Ring	16	8.40974	0.10342	0.36509
Ring	Ring	16	14.0806	0.10342	0.14746
Ring	100	16	14.8176	0.10342	0.12936
100	Ring	16	14.8176	0.10342	0.12936
Ring	110	16	14.0806	0.10342	0.14746



110	Ring	16	14.0806	0.10342	0.14746
Ring	130	16	13.1002	0.10342	0.17907
130	Ring	14	13.0952	0.10342	0.12378
Ring	150	14	13.8877	0.10342	0.10563
150	160	14	13.8854	0.10342	0.10568
160	170	14	12.5988	0.10342	0.14018
170	Ring	12	11.2531	0.10342	0.12362
Ring	180	12	5.54304	0.10342	0.42945
180	190	240	234.137	0.10342	No Calc
190	200	60	55.0164	0.10342	No Calc
200	210	60	55.0164	0.10342	No Calc

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	16700	No Calc	No Calc	No Calc
20	30	16700	No Calc	No Calc	No Calc
30	40	16700	No Calc	No Calc	No Calc
40	50	16700	No Calc	No Calc	No Calc
50	60	16700	No Calc	No Calc	No Calc
60	70	16700	No Calc	No Calc	No Calc
70	80	16700	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc
90	Ring	708.835	3709.59	No Calc	No Calc
Ring	Ring	2650	3708.65	20258088	83331816
Ring	100	3000	3709.85	34071744	83331816
100	Ring	3000	3709.85	No Calc	No Calc
Ring	110	2650	3708.65	34053920	118103224
110	Ring	2650	3708.65	No Calc	No Calc
Ring	130	2205	3709.53	29236124	77327344
130	Ring	2205	2614.02	No Calc	No Calc
Ring	150	2563	2613.89	No Calc	No Calc
150	160	2563	2614.95	No Calc	No Calc
160	170	2563	3423.03	No Calc	No Calc
170	Ring	2563	3039.48	No Calc	No Calc
Ring	180	379.999	3039.48	No Calc	No Calc
180	190	No Calc	No Calc	No Calc	No Calc
190	200	No Calc	No Calc	No Calc	No Calc
200	210	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1 20"	WRC-107/537	0.583	n/a	Passed
N9 22"	WRC-297	0.961	Shell	Passed
N2A 104"	WRC-297	0.961	Shell	Passed
N2B 104"	WRC-297	0.961	Shell	Passed
N3 20"	WRC-297	0.961	Shell	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	500	150746	550005	2.437E+09	11.99E+09
20	30	2350	147074	549617	2.288E+09	11.44E+09
30	40	5200	132592	547981	1910744960	9.956E+09
40	50	8200	121599	545002	1529304320	8.316E+09
50	60	11200	107631	539269	1185319040	6.689E+09
60	70	14200	92911.8	531406	884382400	5.082E+09
70	80	16200	79376.7	522444	625844928	3.501E+09

80	90	16725	71529.1	506601	550361472	2.986E+09
90	100	18250	71343.3	465288	546788224	2.962E+09
100	110	21250	60197.6	343992	349396800	1747318272
110	130	24250	44483.2	244327	192311920	864481152
130	150	26625	28564.9	130952	82695392	301333792
150	160	28149	22067.9	65406.6	38373768	129450200
160	170	29398	12937.6	43619.2	15646001	58663748
170	180	30198	9558.33	37614.7	2142946	9903598
180	190	30536.2	681.359	7027.99	92019.9	949154

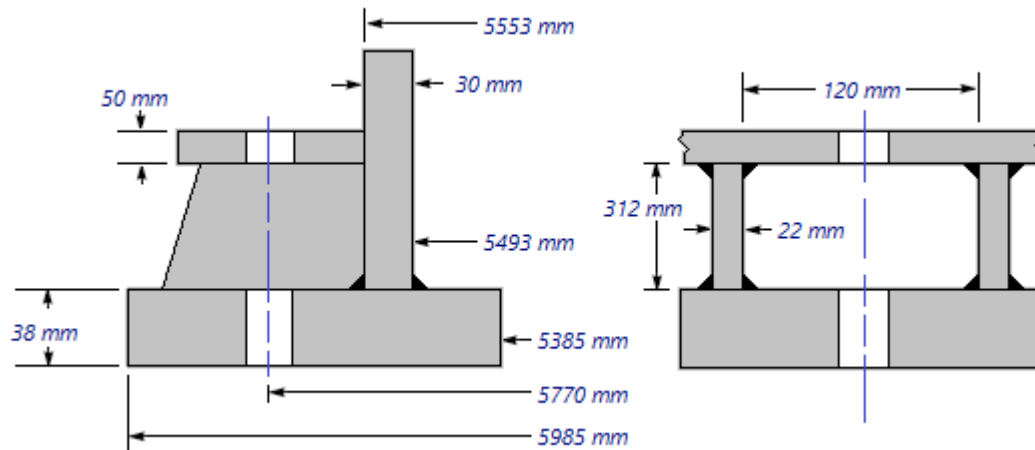
Abs Max of the all of the Stress Ratio's : 0.4027

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	38.0000	mm
Inside Diameter of Basing	5385.0000	mm
Outside Diameter of Basing	5985.0000	mm
Nominal Diameter of Bolts	64.0000	mm
Diameter of Bolt Circle	5770.0000	mm
Number of Bolts	40	

Thickness of Gusset Plates	22.0000	mm
Average Width of Gusset Plates	206.0000	mm
Height of Gussets	312.0000	mm
Distance between Gussets	120.0000	mm
Thickness of Top Plate or Ring	50.0000	mm
Radial Width of the Top Plate	206.0000	mm
Number of Gussets	80	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	150746.	N
Total Earthquake Shear on Support	550005.	N
Wind Moment on Support	2744061440.	N-mm
Earthquake Moment on Support	12295086080.	N-mm
Maximum Initial Bolt Load	128058.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	128612.6	kgm
Shop Test - Fabricated + water (full)	458497.1	kgm



Shipping	- Fab. + rem. intls.+ shipping app.	186785.5	kgm
Erected	- Fab. + rem. intls.+ insul. (etc)	188988.3	kgm
Empty	- Fab. + intls. + details + wghts.	188988.3	kgm
Operating	- Empty + operating liquid (no ca)	357708.1	kgm
Field Test	- Empty weight + test medium (full)	500225.6	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1 20"	1	OK	...	OK	8.33	OK	Passed
N5_2"	1	OK	...	OK	4.80	OK	Passed
N9 22"	1	OK	...	OK	8.33	OK	Passed
M4 24"	1	OK	...	...		OK	Passed
N6A 2"	1	OK	...	OK	4.80	OK	Passed
K4B 2"	1	OK	...	OK	4.80	OK	Passed
K3B 2"	1	OK	...	OK	4.80	OK	Passed
K2B 2"	1	OK	...	OK	4.80	OK	Passed
K1 B2"	1	OK	...	OK	4.80	OK	Passed
K1A 2"	1	OK	...	OK	4.80	OK	Passed
K2A 2"	1	OK	...	OK	4.80	OK	Passed
K3A 2"	1	OK	...	OK	4.80	OK	Passed
K4A 2"	1	OK	...	OK	4.80	OK	Passed
N2A 104"	...	OK	...	OK	8.16	OK	Passed
N2B 104"	...	OK	...	OK	8.16	OK	Passed
M3 24"	1	OK	...	...		OK	Passed
N6B 2"	1	OK	...	OK	4.80	OK	Passed
K5A_2"	1	OK	...	OK	4.80	OK	Passed
K6A_2"	1	OK	...	OK	4.80	OK	Passed
N10A_2"	1	OK	...	OK	4.80	OK	Passed
N10B_2"	1	OK	...	OK	4.80	OK	Passed
M2 24"	1	OK	...	...		OK	Passed
M1 24"	1	OK	...	...		OK	Passed
N3 20"	1	OK	...	OK	7.96	OK	Passed
N6 C 2"	1	OK	...	OK	4.80	OK	Passed
K6B 2"	1	OK	...	OK	4.80	OK	Passed
K5B 2"	1	OK	...	OK	4.80	OK	Passed
N7_2"	1	OK	...	OK	4.80	OK	Passed
N6 D 2"	1	OK	...	OK	4.80	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 0 Nozzle : N2B 104"
 Minimum MAWP Shells/Flanges : 0 Element : N4 Flange
 Minimum MAPnc Shells/Flanges : 0 Element : N4 Flange

Computed Vessel M.A.W.P. : 0 MPa

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
80	N1 20"	0.000	0.000	1016.000
80	N5_2"	0.000	180.000	114.328
90	N9 22"	1550.000	45.000	1117.600
90	M4 24"	1550.000	180.000	1219.200
90	N6A 2"	750.002	180.000	114.328
100	K4B 2"	5750.002	180.000	114.328
100	K3B 2"	5550.000	180.000	114.328
100	K2B 2"	5350.001	180.000	114.328
100	K1 B2"	5050.001	180.000	114.328
100	K1A 2"	5050.001	0.000	114.328
100	K2A 2"	4850.001	0.000	114.328
100	K3A 2"	4650.002	0.000	114.328
100	K4A 2"	4050.001	0.000	114.328
100	N2A 104"	3050.002	320.000	3900.000
100	N2B 104"	3050.002	225.000	3900.000
110	M3 24"	6950.001	0.000	1219.200



110	N6B 2"	6050.000	0.000	114.328
110	K5A 2"	8550.001	45.000	114.328
110	K6A 2"	8550.001	270.000	114.328
110	N10A 2"	6550.000	270.000	114.328
110	N10B 2"	7050.002	270.000	114.328
130	M2 24"	9050.003	180.000	1219.200
130	M1 24"	10700.003	90.000	1219.200
130	N3 20"	9950.004	0.000	952.000
130	N6 C 2"	9250.003	220.000	110.328
130	K6B 2"	9050.003	270.000	110.328
130	K5B 2"	9050.003	10.000	110.328
130	N7 2"	10050.005	45.000	110.328
150	N6 D 2"	11500.005	0.000	113.516

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N5_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	231.56	300
N6A 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.70	300
K4B 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.70	300
K3B 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.70	300
K2B 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	236.32	300
K1 B2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.70	300
K1A 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.70	300
K2A 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.70	300
K3A 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.70	300
K4A 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.70	300
N6B 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.70	300
K5A_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.70	300
K6A_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.70	300
N10A_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.70	300
N10B_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.70	300
N6 C 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	220.70	300
K6B 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	220.70	300
K5B 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	220.70	300
N7_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	220.70	300
N6 D 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	220.73	300
N1 20"	20.000 in	Actual	WNF	20.787	10.000	750.00	16.00	243.02	300
N3 20"	20.000 in	Actual	WNF	20.000	16.000	...	...	257.74	300
N9 22"	22.000 in	Actual	WNF	23.260	16.000	1258.80	16.00	251.89	150
M4 24"	24.000 in	Actual	WNF	25.575	20.000	1100.00	16.00	255.23	150
M3 24"	24.000 in	Actual	WNF	25.102	14.000	1149.60	12.00	254.52	150
M2 24"	24.000 in	Actual	WNF	25.575	20.000	1100.00	12.00	253.23	150
M1 24"	24.000 in	Actual	WNF	25.575	20.000	1100.00	14.00	253.23	150
N2A 104"	104.000 in	Actual	None	104.000	30.000	...	...	865.53	...
N2B 104"	104.000 in	Actual	None	104.000	30.000	...	...	865.53	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N5_2"	SA-105	10.000	10.000	...	...	...
N6A 2"	SA-105	14.000	10.000	...	...	...
K4B 2"	SA-105	10.000	10.000	...	...	...
K3B 2"	SA-105	10.000	10.000	...	...	...
K2B 2"	SA-105	10.000	10.000	...	...	...
K1 B2"	SA-105	10.000	10.000	...	...	...
K1A 2"	SA-105	10.000	10.000	...	...	...
K2A 2"	SA-105	10.000	10.000	...	...	...
K3A 2"	SA-105	10.000	10.000	...	...	...
K4A 2"	SA-105	10.000	10.000	...	...	...
N6B 2"	SA-105	12.000	10.000	...	...	...
K5A_2"	SA-105	12.000	10.000	...	...	...
K6A_2"	SA-105	12.000	10.000	...	...	...
N10A_2"	SA-105	12.000	10.000	...	...	...
N10B_2"	SA-105	12.000	10.000	...	...	...
N6 C 2"	SA-105	12.000	10.000	...	...	...
K6B 2"	SA-105	12.000	10.000	...	...	...

K5B 2"	SA-105	12.000	10.000	...	...	...
N7_2"	SA-105	12.000	10.000	...	...	...
N6 D 2"	SA-105	12.000	10.000	...	...	...
N1 20"	SA-516 70	12.000	10.000	16.000	16.000	...
N3 20"	SA-516 70	18.000	10.000	...	...	...
N9 22"	SA-516 70	10.000	10.000	14.000	16.000	...
M4 24"	SA-516 70	14.000	10.000	16.000	16.000	...
M3 24"	SA-516 70	12.000	10.000	12.000	12.000	...
M2 24"	SA-516 70	12.000	10.000	12.000	12.000	...
M1 24"	SA-516 70	12.000	10.000	14.000	14.000	...
N2A 104"	SA-516 70	10.000	10.000	...	...	...
N2B 104"	SA-516 70	10.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N5_2"	...	180.0	206.38	0.00	BTM DISH
N6A 2"	750.001	180.0	206.38	0.00	Shell-1
K4B 2"	5750.001	180.0	206.38	0.00	Shell-2
K3B 2"	5550.000	180.0	206.38	0.00	Shell-2
K2B 2"	5350.000	180.0	220.00	0.00	Shell-2
K1 B2"	5050.000	180.0	206.38	0.00	Shell-2
K1A 2"	5050.000	0.0	206.38	0.00	Shell-2
K2A 2"	4850.000	0.0	206.38	0.00	Shell-2
K3A 2"	4650.001	0.0	206.38	0.00	Shell-2
K4A 2"	4050.000	0.0	206.38	0.00	Shell-2
N6B 2"	6049.999	0.0	206.38	0.00	Shell-3
K5A 2"	8550.000	45.0	206.38	0.00	Shell-3
K6A 2"	8550.000	270.0	206.38	0.00	Shell-3
N10A 2"	6549.999	270.0	206.38	0.00	Shell-3
N10B 2"	7050.001	270.0	206.38	0.00	Shell-3
N6 C 2"	9250.001	220.0	206.38	0.00	Shell-4
K6B 2"	9050.002	270.0	206.38	0.00	Shell-4
K5B 2"	9050.002	10.0	206.38	0.00	Shell-4
N7_2"	10050.004	45.0	206.38	0.00	Shell-4
N6 D 2"	11500.003	0.0	206.38	0.00	Cone 1
N1 20"	...	0.0	220.00	0.00	BTM DISH
N3 20"	9950.003	0.0	220.00	0.00	Shell-4
N9 22"	1549.998	45.0	220.00	0.00	Shell-1
M4 24"	1549.998	180.0	220.00	0.00	Shell-1
M3 24"	6950.000	0.0	220.00	0.00	Shell-3
M2 24"	9050.002	180.0	220.00	0.00	Shell-4
M1 24"	10700.001	90.0	220.00	0.00	Shell-4
N2A 104"	3050.001	320.0	500.00	0.00	Shell-2
N2B 104"	3050.001	225.0	500.00	0.00	Shell-2



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SKIRT: 30.0mm THK X 5493.0mm ID X 1000.0mm	E250A
1	SKIRT: 20.0mm THK X 5503.0mm ID X 2700.0mm	E250A
3	SKIRT: 18.0mm THK X 5505.0mm ID X 3000.0mm	E250A
1	SKIRT: 16.0mm THK X 5508.0mm ID X 3000.0mm	E250A
1	SKIRT: 16.0mm THK X 5508.0mm ID X 1000.0mm	SA-516 60
1	ELLIPTICAL HEAD: 2.0 X 1, 20.0mm THK X 5506.0mm ID X 50.0mm	SA-516 70
2	CYLINDER: 16.0mm THK X 5506.0mm ID X 3000.0mm	SA-516 70
1	CYLINDER: 14.0mm THK X 5506.0mm ID X 1750.0mm	SA-516 70
1	CONCENTRIC CONE: 16.0mm THK X 5504.2mm X 4570.2mm ID X 1298.0 mm	SA-516 70
1	WELD NECK FLANGE: 260.0mm THK X 3960.0mm OD	SA-250 T1b
1	WELD NECK FLANGE: 60.0mm THK X 2889.0mm OD	SA-266 2
1	WELD NECK FLANGE: 60.0mm THK X 2889.1mm OD	SA-266 2
8	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 3700.0mm	...
5	LADDER: 4000.0mm	...
1	LADDER: 5000.0mm	...
2	LADDER: 2500.0mm	...
1	INSULATION: 1424mm X 40mm THK	Minral wool
1	CLADDING: 1425.000mm X 3.000mm THK	...
3	INSULATION: 3000mm X 40mm THK	Minral wool
3	CLADDING: 3000.000mm X 3.000mm THK	...
2	BAR RING STIFFENER: 5538.0mm ID X 6108.0mm OD X 16.0mm	SA-516 70
1	BAR RING STIFFENER: 5538.0mm ID X 6212.0mm OD X 14.0mm	SA-516 70
1	TRAYS # 1 SPACING 0. mm	
1	PACKING: 3000.000mm	...
1	BAR RING STIFFENER: 5538.0mm ID X 6112.0mm OD X 14.0mm	SA-516 70
1	INSULATION: 1749mm X 40mm THK	Minral wool
1	CLADDING: 1750.000mm X 3.000mm THK	...
1	BAR RING STIFFENER: 5534.0mm ID X 6134.0mm OD X 20.0mm	SA-516 70
1	INSULATION: 1298mm X 40mm THK	Minral wool
1	CLADDING: 1298.000mm X 3.000mm THK	...
1	INSULATION: 1200mm X 40mm THK	
1	CLADDING: 1200.000mm X 3.000mm THK	...



1	PLATFORM: 1000mm X 3100mm WIDE	
1	INSULATION: 400mm X 40mm THK	
1	CLADDING: 400.000mm X 3.000mm THK	...
1	BAR RING STIFFENER: 3760.0mm ID X 4360.0mm OD X 14.0mm	SA-516 70
1	INSULATION: 375mm X 40mm THK	
1	CLADDING: 282.000mm X 5.000mm THK	...
4	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 5985.0mm OD X 5385.0 ID X 38.0mm	E250A
40	BASERING BOLTS: 64.0mm DIA	SA-36
40	BASERING BOLT NUTS: 64.0mm DIA	...
1	BASERING TOP RING: 5965.0mm OD X 5553.0 ID X 50.0mm	...
80	BASERING GUSSET PLATES: 206.0mm X 206.0mm X 312.0mm X 22.0mm	...
1	GASKET: 3810.7mm OD X 3780.7mm ID	...
72	BODY FLANGE BOLTS: 38.1mm DIA	SA-193 B7
216	NUTS FOR BODY FLANGE BOLTS: 38.1mm DIA	...
2	CYLINDRICAL SEGMENT 16.0mm THK X 6870.9mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 3555.6mm X 3000.0mm	SA-516 70
7	CON. CONICAL SEGMENT 16.0mm THK X 1861.1mm X 1200.0mm X 1519.0mm	SA-516 70
1	CON. CONICAL SEGMENT 16.0mm THK X 13027.5mm X 1200.0mm X 10633.2	SA-516 70
4	CYLINDRICAL SEGMENT 12.0mm THK X 2114.6mm X 400.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 9622.2mm X 400.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 6343.8mm X 400.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
N4 Flange	[11]	B	-23	-48		0.468	12.000	1.00	No
N2A Flange	[11]	!	-1	-104		0.123	26.000	1.00	No
N2B Flange	[11]	!	-1	-104		0.123	26.000	1.00	No
BTM DISH	[10]	B	-14	-42	-29	0.534	16.000	1.00	No
BTM DISH	[7]	B	-7	-48	-29	0.422	20.000	1.00	No
Shell-1	[8]	B	-14	-44	-29	0.517	16.000	1.00	No
Shell-2	[8]	B	-14	-44	-29	0.519	16.000	1.00	No
Shell-3	[8]	B	-14	-46	-29	0.505	16.000	1.00	No
Shell-4	[8]	B	-18	-42	-29	0.583	14.000	1.00	No
Cone 1	[8]	B	-18	-39	-29	0.621	14.000	1.00	No
Cone 2	[8]	B	-18	-48	-29	0.514	14.000	1.00	No
Port shell	[8]	B	-23	-48	-29	0.469	12.000	1.00	No
Ring R1 Fr90	[23]	B	-14	-44	-29	0.517	16.000	1.00	No
Ring R2 Fr90	[23]	B	-14	-44	-29	0.517	16.000	1.00	No
Ring R1 Fr100	[23]	B	-18	-48	-29	0.519	14.000	1.00	No
Ring R1 Fr110	[23]	B	-18	-48	-29	0.505	14.000	1.00	No
R1	[23]	B	-18	-42	-29	0.583	14.000	1.00	No
R2	[23]	B	-23	-48	-29	0.469	12.000	1.00	No
N1 20"	[1]	B	-14	-48	-29	0.481	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N52"	[1]	A	6	-29		0.480	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N9 22"	[1]	B	-14	-47	-29	0.497	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M4 24"	[1]	B	-14	-47	-29	0.497	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N6A 2"	[1]	A	6	-26		0.507	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K4B 2"	[1]	A	6	-32		0.464	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K3B 2"	[1]	A	6	-32		0.464	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2B 2"	[1]	A	6	-32		0.464	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1 B2"	[1]	A	6	-32		0.464	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1A 2"	[1]	A	6	-32		0.464	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2A 2"	[1]	A	6	-32		0.464	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K3A 2"	[1]	A	6	-32		0.464	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K4A 2"	[1]	A	6	-32		0.464	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N2A 104"	[1]	B	2	-104		0.265	28.000	1.00	No
N2B 104"	[1]	B	2	-104		0.265	28.000	1.00	No
M3 24"	[1]	B	-23	-48	-29	0.464	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N6B 2"	[1]	A	6	-32		0.464	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K5A2"	[1]	A	6	-32		0.464	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K6A2"	[1]	A	6	-32		0.464	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N10A2"	[1]	A	6	-32		0.464	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N10B2"	[1]	A	6	-32		0.464	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M2 24"	[1]	B	-18	-46	-29	0.535	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M1 24"	[1]	B	-18	-46	-29	0.535	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N3 20"	[1]	B	6	-104		0.040	16.000	1.00	No



Nozzle Flg	[5]	A	-18	-104				
N6 C 2"	[1]	A	2	-26	0.535	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-104				
K6B 2"	[1]	A	2	-26	0.535	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-104				
K5B 2"	[1]	A	2	-26	0.535	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-104				
N72"	[1]	A	2	-26	0.535	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-104				
N6 D 2"	[1]	A	2	-28	0.517	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-104				
Noz1. Bolting	[22]		-48		1.000	64.000	1.00	No
Bolting	[21]		-48					

Warmest MDMT: 6 -26

Required Minimum Design Metal Temperature -1.0 °C

Warmest Computed Minimum Design Metal Temperature -26.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

- Impact test temps were not entered in and not considered in the analysis.
- UCS-66(i) applies to impact tested materials not by specification and
- UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
- The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note	70	90	There is a high jump in the Joint Eff.
Note	10	20	The wind diameter multiplier is very big.
Note	20	30	The wind diameter multiplier is very big.
Note	30	40	The wind diameter multiplier is very big.
Note	40	50	The wind diameter multiplier is very big.
Note	50	60	The wind diameter multiplier is very big.
Note	60	70	The wind diameter multiplier is very big.
Note	70	80	The wind diameter multiplier is very big.
Note	80	90	The wind diameter multiplier is very big.
Note	90	100	The wind diameter multiplier is very big.
Note	100	110	The wind diameter multiplier is very big.
Note	110	130	The wind diameter multiplier is very big.
Note	130	150	The wind diameter multiplier is very big.
Note	150	160	The wind diameter multiplier is very big.
Note	160	170	The wind diameter multiplier is very big.
Note	170	180	The wind diameter multiplier is very big.
Note	180	190	The wind diameter multiplier is very big.
Warn.	70	80	Skirt diameter at base is < at the top

Class From To: Check of Additional Element Data

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.





Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	0.3481	MPa
Design Internal Temperature	160.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-1.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	User Entered Pressure
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16		IP+VE+EW	
Load Case 17		NP+VO+OW	
Load Case 18		FS+BS+IP+OW	
Load Case 19		FS+BS+EP+OW	
Load Case 20		BL+IP+OW	
Wind Design Code		ASCE/SEI 7-16	
Wind Load Reduction Scale Factor		1.100	
Basic Wind Speed	[V]	98	km/hr
Surface Roughness Category		C: Open Terrain	
Type of Surface		Moderately Smooth	
Base Elevation		300	mm
Percent Wind for Hydrotest		50.0	
Using User defined Wind Press. Vs Elev.		N	
Height of Hill or Escarpment	H or Hh	0	mm
Distance Upwind of Crest	Lh	0	mm
Distance from Crest to the Vessel	x	0	mm
Type of Terrain (Hill, Escarpment)		Flat	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Site Elevation above Sea Level		0	mm

Seismic Design Code		ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor		1.000	
Importance Factor		1.250	
Table Value Fa		1.600	
Table Value Fv		2.400	
Max. Mapped Res. Acceleration	[Ss]	0.147	
Max. Eff. Ground Acceleration	[S]	0.065	
Force Modification Factor R		1.250	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	2.500	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Moment Reduction Factor Tau		1.000	

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	No
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/18
Element From Node	10



Element To Node	20	
Element Type	Skirt Support	
Description	Skirt-1	
Distance "FROM" to "TO"	1000	mm
Skirt Outside Diameter	5553	mm
Diameter of Skirt at Base	5553	mm
Skirt Thickness	30	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	30	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	160	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6	
Material Name	E250A	
Allowable Stress, Ambient	117	MPa
Allowable Stress, Operating	117	MPa
Allowable Stress, Hydrotest	207	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	202.2	MPa
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-2	
UNS Number	SA/IS 20	
Class / Thickness / Grade	:: t > 40	
Product Form	Plate; bar; shapes	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element From Node	10	
Detail Type	Weight	
Detail ID	BRACING. TAILIN	
Dist. from "FROM" Node / Offset dist	150	mm
Miscellaneous Weight	18631	N
Offset from Element Centerline	0	mm

Element#	2/18	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Skirt-2	
Distance "FROM" to "TO"	2700	mm
Skirt Outside Diameter	5543	mm
Diameter of Skirt at Base	5543	mm
Skirt Thickness	20	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	160	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6	
Material Name	E250A	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	2700	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	3604416	mm²
Platform Weight	15340	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm²
Platform Width	1500	mm
Platform Height	1000	mm



Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	3700	mm
Unit Weight of Ladder	0.2942	N/mm
Platform Length (top head platform)	0	mm

Element#	3/18	
Element From Node	30	
Element To Node	40	
Element Type	Skirt Support	
Description	Skirt-3	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	5541	mm
Diameter of Skirt at Base	5541	mm
Skirt Thickness	18	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	18	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	160	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6	
Material Name	E250A	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element#	4/18	
Element From Node	40	
Element To Node	50	
Element Type	Skirt Support	
Description	Skirt-4	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	5541	mm
Diameter of Skirt at Base	5541	mm
Skirt Thickness	18	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	18	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	160	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6	
Material Name	E250A	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	1000	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	3750720	mm ²
Platform Weight	15691	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.3628	N/mm



Platform Length (top head platform) 0 mm

Element# 5/18
 Element From Node 50
 Element To Node 60
 Element Type Skirt Support
 Description Skirt-5
 Distance "FROM" to "TO" 3000 mm
 Skirt Outside Diameter 5541 mm
 Diameter of Skirt at Base 5541 mm
 Skirt Thickness 18 mm
 Internal Corrosion Allowance 1 mm
 Nominal Thickness 18 mm
 External Corrosion Allowance 0 mm
 Design Temperature Internal Pressure 160 °C
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.6
 Material Name E250A
 Efficiency, Longitudinal Seam 0.7
 Efficiency, Head-to-Skirt or Circ. Seam 0.7
 Weld is pre-Heated No

Element From Node 50
 Detail Type Platform
 Detail ID PLAT:[1 OF 1]
 Dist. from "FROM" Node / Offset dist 2000 mm
 Platform Start Angle (degrees) 0.0
 Platform End Angle (degrees) 90.0
 Platform Wind Area 4238400 mm²
 Platform Weight 16054 N
 Platform Railing Weight 0 N/mm
 Platform Grating Weight 0.017 kg/cm²
 Platform Width 1500 mm
 Platform Height 1000 mm
 Platform Clearance or End Offset 85 mm
 Platform Force Coefficient 1.2
 Ladder Layout Angle 0.0
 Ladder Start Elevation 0 mm
 Ladder End Elevation 5000 mm
 Unit Weight of Ladder 0.3628 N/mm
 Platform Length (top head platform) 0 mm

Element# 6/18
 Element From Node 60
 Element To Node 70
 Element Type Skirt Support
 Description Skirt-6
 Distance "FROM" to "TO" 3000 mm
 Skirt Outside Diameter 5540 mm
 Diameter of Skirt at Base 5540 mm
 Skirt Thickness 16 mm
 Internal Corrosion Allowance 1 mm
 Nominal Thickness 16 mm
 External Corrosion Allowance 0 mm
 Design Temperature Internal Pressure 160 °C
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.6
 Material Name E250A
 Efficiency, Longitudinal Seam 0.7
 Efficiency, Head-to-Skirt or Circ. Seam 0.7
 Weld is pre-Heated No

Element From Node 60
 Detail Type Platform
 Detail ID PLAT:[1 OF 1]
 Dist. from "FROM" Node / Offset dist 1500 mm



Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	90.0	
Platform Wind Area	3019200	mm ²
Platform Weight	15137	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	2500	mm
Unit Weight of Ladder	0.3628	N/mm
Platform Length (top head platform)	0	mm

Element#	7/18	
Element From Node	70	
Element To Node	80	
Element Type	Skirt Support	
Description	Skirt-7	
Distance "FROM" to "TO"	1000	mm
Skirt Outside Diameter	5540	mm
Diameter of Skirt at Base	5540	mm
Skirt Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	160	°C
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6	
Material Name	SA-516 60	
Allowable Stress, Ambient	118	MPa
Allowable Stress, Operating	118	MPa
Allowable Stress, Hydrotest	198	MPa
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	193.8	MPa
UCS-66 Chart Curve Designation	C	
External Pressure Chart Name	CS-2	
UNS Number	K02100	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Platform	
Detail ID	BTM PLATFORM	
Dist. from "FROM" Node / Offset dist	1000	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	4819200	mm ²
Platform Weight	57827	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	2500	mm
Unit Weight of Ladder	0.3628	N/mm
Platform Length (top head platform)	0	mm

Element From Node	70
Detail Type	Weight



Detail ID	N1 pipe wt	
Dist. from "FROM" Node / Offset dist	200	mm
Miscellaneous Weight	42627	N
Offset from Element Centerline	4103	mm

Element#	8/18	
Element From Node	80	
Element To Node	90	
Element Type	Elliptical	
Description	BTM DISH	
Distance "FROM" to "TO"	50	mm
Inside Diameter	5506	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3481	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.4	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	-1375	mm
Height/Length of Liquid	1425	mm
Liquid Density	0.0009984	kg/cm³

Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	-1375	mm
Height/Length of Insulation	1425	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm³

Element From Node	80	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	-1375	mm
Height/Length of Lining	1425	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm³

Element From Node	80	
Detail Type	Nozzle	
Detail ID	N1 20"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	20	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	



Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	12116	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	
Element From Node	80	
Detail Type	Nozzle	
Detail ID	N52"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.09	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	80	
Detail Type	Weight	
Detail ID	VORTEX BREAKER	
Dist. from "FROM" Node / Offset dist	-1000	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	0	mm

Element#	9/18	
Element From Node	90	
Element To Node	100	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	5506	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3481	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	90	
Detail Type	Liquid	
Detail ID	Liquid 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.0009984	kg/cm³

Element From Node	90	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm³

Element From Node	90	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm³

Element From Node 90
Detail Type Ring
Detail ID Ring R1 Fr90
Dist. from "FROM" Node / Offset dist 200 mm
Inside Diameter of Ring 5538 mm
Thickness of Ring 16 mm
Outside Diameter of Ring 6108 mm
Material Name SA-516 70
Height of Section Ring 0 mm
Using Custom Stiffener Section No

Element From Node 90
Detail Type Ring
Detail ID Ring R2 Fr90
Dist. from "FROM" Node / Offset dist 2850 mm
Inside Diameter of Ring 5538 mm
Thickness of Ring 16 mm
Outside Diameter of Ring 6108 mm
Material Name SA-516 70
Height of Section Ring 0 mm
Using Custom Stiffener Section No

Element From Node 90
Detail Type Nozzle
Detail ID N9 22"
Dist. from "FROM" Node / Offset dist 1500 mm
Nozzle Diameter 22 in.
Nozzle Schedule None
Nozzle Class 150
Layout Angle 45.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 2902.9 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-516 70

Element From Node 90
Detail Type Nozzle
Detail ID M4 24"
Dist. from "FROM" Node / Offset dist 1500 mm
Nozzle Diameter 24 in.
Nozzle Schedule None
Nozzle Class 150
Layout Angle 180.0
Blind Flange (Y/N) Y
Weight of Nozzle (Used if > 0) 4532.4 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-516 70

Element From Node 90
Detail Type Nozzle
Detail ID N6A 2"
Dist. from "FROM" Node / Offset dist 700 mm
Nozzle Diameter 2 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 180.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 84.932 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-105

Element From Node 90
Detail Type Weight
Detail ID N9 pipe wt
Dist. from "FROM" Node / Offset dist 1500 mm
Miscellaneous Weight 27790 N
Offset from Element Centerline 4128 mm



Element#	10/18	
Element From Node	100	
Element To Node	110	
Element Type	Cylinder	
Description	Shell-2	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	5506	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3481	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	100	
Detail Type	Platform	
Detail ID	PLT-2	
Dist. from "FROM" Node / Offset dist	1000	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	5550719	mm ²
Platform Weight	29778	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.3628	N/mm
Platform Length (top head platform)	0	mm
Element From Node	100	
Detail Type	Liquid	
Detail ID	Liquid 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1021	mm
Liquid Density	0.0009984	kg/cm ³
Element From Node	100	
Detail Type	Insulation	
Detail ID	Ins: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³
Element From Node	100	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³
Element From Node	100	
Detail Type	Ring	
Detail ID	Ring R1 Fr100	
Dist. from "FROM" Node / Offset dist	2850	mm
Inside Diameter of Ring	5538	mm



Thickness of Ring	14	mm
Outside Diameter of Ring	6212	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	100	
Detail Type	Nozzle	
Detail ID	K4B 2"	
Dist. from "FROM" Node / Offset dist	2700	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	101.01	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	100	
Detail Type	Nozzle	
Detail ID	K3B 2"	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	101.01	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	100	
Detail Type	Nozzle	
Detail ID	K2B 2"	
Dist. from "FROM" Node / Offset dist	2300	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	104.69	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	100	
Detail Type	Nozzle	
Detail ID	K1 B2"	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	101.01	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	100	
Detail Type	Nozzle	
Detail ID	K1A 2"	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	101.01	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	



Element From Node 100
 Detail Type Nozzle
 Detail ID K2A 2"
 Dist. from "FROM" Node / Offset dist 1800 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 101.01 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105

Element From Node 100
 Detail Type Nozzle
 Detail ID K3A 2"
 Dist. from "FROM" Node / Offset dist 1600 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 101.01 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105

Element From Node 100
 Detail Type Nozzle
 Detail ID K4A 2"
 Dist. from "FROM" Node / Offset dist 1000 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 101.01 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105

Element From Node 100
 Detail Type Nozzle
 Detail ID N2A 104"
 Dist. from "FROM" Node / Offset dist 0.001454 mm
 Nozzle Diameter 104 in.
 Nozzle Schedule None
 Nozzle Class None
 Layout Angle 320.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 9652.2 N
 Grade of Attached Flange None
 Nozzle Matl SA-516 70

Element From Node 100
 Detail Type Nozzle
 Detail ID N2B 104"
 Dist. from "FROM" Node / Offset dist 0.001454 mm
 Nozzle Diameter 104 in.
 Nozzle Schedule None
 Nozzle Class None
 Layout Angle 225.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 9652.2 N
 Grade of Attached Flange None
 Nozzle Matl SA-516 70

Element From Node 100
 Detail Type Tray
 Detail ID CHIMNEY
 Dist. from "FROM" Node / Offset dist 2500 mm



Tray Spacing	0	mm
Number of Trays	1	
Tray Weight per Unit Area	0.03	kg/cm ²
Height of Liquid on Tray	250	mm
Density of Liquid on Tray	0.0009989	kg/cm ³
Support Ring and Bolting Bar Weight	2843.8	N

Element From Node	100	
Detail Type	Weight	
Detail ID	N2A INSERT PLT	
Dist. from "FROM" Node / Offset dist	0.001454	mm
Miscellaneous Weight	27947	N
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	N2B INSERT PLT	
Dist. from "FROM" Node / Offset dist	0.001454	mm
Miscellaneous Weight	27947	N
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	CT BEAM	
Dist. from "FROM" Node / Offset dist	2500	mm
Miscellaneous Weight	9806.2	N
Offset from Element Centerline	0	mm

Element#	11/18	
Element From Node	110	
Element To Node	130	
Element Type	Cylinder	
Description	Shell-3	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	5506	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3481	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	110	
Detail Type	Platform	
Detail ID	PLT-3	
Dist. from "FROM" Node / Offset dist	2000	mm
Platform Start Angle (degrees)	90.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	5550719	mm ²
Platform Weight	29778	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	110.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.3628	N/mm
Platform Length (top head platform)	0	mm



Element From Node	110	
Detail Type	Packing	
Detail ID	Structure Packi	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	3000	mm
Density	0.00025	kg/cm ³
Percent Volume Holdup	15.0	
Specific Gravity of Packing Liquid	1.0	

Element From Node	110	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³

Element From Node	110	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³

Element From Node	110	
Detail Type	Ring	
Detail ID	Ring R1 Fr110	
Dist. from "FROM" Node / Offset dist	2500	mm
Inside Diameter of Ring	5538	mm
Thickness of Ring	14	mm
Outside Diameter of Ring	6112	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	110	
Detail Type	Nozzle	
Detail ID	M3 24"	
Dist. from "FROM" Node / Offset dist	900	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	4218.4	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	110	
Detail Type	Nozzle	
Detail ID	N6B 2"	
Dist. from "FROM" Node / Offset dist	0.001161	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	101.01	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	110	
Detail Type	Nozzle	
Detail ID	K5A2"	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	



Layout Angle	45.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	111.68	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	110	
Detail Type	Nozzle	
Detail ID	K6A2"	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	111.68	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	110	
Detail Type	Nozzle	
Detail ID	N10A2"	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.09	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	110	
Detail Type	Nozzle	
Detail ID	N10B2"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.09	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	110	
Detail Type	Weight	
Detail ID	SUPPORT RING	
Dist. from "FROM" Node / Offset dist	2900	mm
Miscellaneous Weight	2843.7	N
Offset from Element Centerline	0	mm
Element From Node	110	
Detail Type	Weight	
Detail ID	MW DAVIT	
Dist. from "FROM" Node / Offset dist	500	mm
Miscellaneous Weight	4903.1	N
Offset from Element Centerline	0	mm
Element From Node	110	
Detail Type	Weight	
Detail ID	OTHER MISC	
Dist. from "FROM" Node / Offset dist	0.001161	mm
Miscellaneous Weight	9806.2	N
Offset from Element Centerline	0	mm
Element From Node	110	
Detail Type	Weight	
Detail ID	CLT 2	
Dist. from "FROM" Node / Offset dist	1500	mm



Miscellaneous Weight 26476 N
 Offset from Element Centerline 0 mm

Element From Node 110
 Detail Type Weight
 Detail ID 10T MISC
 Dist. from "FROM" Node / Offset dist 1500 mm
 Miscellaneous Weight 98062 N
 Offset from Element Centerline 0 mm

Element From Node 110
 Detail Type Weight
 Detail ID Support Grid
 Dist. from "FROM" Node / Offset dist 2990 mm
 Miscellaneous Weight 39447 N
 Offset from Element Centerline 0 mm

Element# 12/18
 Element From Node 130
 Element To Node 150
 Element Type Cylinder
 Description Shell-4
 Distance "FROM" to "TO" 1750 mm
 Inside Diameter 5506 mm
 Element Thickness 14 mm
 Internal Corrosion Allowance 1 mm
 Nominal Thickness 14 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 0.3481 MPa
 Design Temperature Internal Pressure 160 °C
 Design External Pressure 0.1034 MPa
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.6
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 130
 Detail Type Insulation
 Detail ID Ins: 90
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 1750 mm
 Thickness of Insulation 40 mm
 Density 0.000192 kg/cm³

Element From Node 130
 Detail Type Lining
 Detail ID Clad
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Lining 1750 mm
 Thickness of Lining 3 mm
 Density 0.008027 kg/cm³

Element From Node 130
 Detail Type Ring
 Detail ID R1
 Dist. from "FROM" Node / Offset dist 1705 mm
 Inside Diameter of Ring 5534 mm
 Thickness of Ring 20 mm
 Outside Diameter of Ring 6134 mm
 Material Name SA-516 70
 Height of Section Ring 0 mm
 Using Custom Stiffener Section No

Element From Node 130
 Detail Type Nozzle
 Detail ID M2 24"



Dist. from "FROM" Node / Offset dist	0.002908	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	4326.2	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	
Element From Node	130	
Detail Type	Nozzle	
Detail ID	M1 24"	
Dist. from "FROM" Node / Offset dist	1650	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	4432.3	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	
Element From Node	130	
Detail Type	Nozzle	
Detail ID	N3 20"	
Dist. from "FROM" Node / Offset dist	900	mm
Nozzle Diameter	20	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1969.9	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	
Element From Node	130	
Detail Type	Nozzle	
Detail ID	N6 C 2"	
Dist. from "FROM" Node / Offset dist	200	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	220.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	111.68	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	130	
Detail Type	Nozzle	
Detail ID	K6B 2"	
Dist. from "FROM" Node / Offset dist	0.002908	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	111.68	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	130	
Detail Type	Nozzle	
Detail ID	K5B 2"	
Dist. from "FROM" Node / Offset dist	0.002908	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	10.0	



Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	111.68	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	130	
Detail Type	Nozzle	
Detail ID	N72"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	45.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	101.55	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	130	
Detail Type	Weight	
Detail ID	BED LIMITER	
Dist. from "FROM" Node / Offset dist	0.002908	mm
Miscellaneous Weight	27810	N
Offset from Element Centerline	0	mm
Element From Node	130	
Detail Type	Weight	
Detail ID	DISTRIBUTOR	
Dist. from "FROM" Node / Offset dist	250	mm
Miscellaneous Weight	74378	N
Offset from Element Centerline	0	mm
Element From Node	130	
Detail Type	Weight	
Detail ID	SUPPORT RING	
Dist. from "FROM" Node / Offset dist	0.002908	mm
Miscellaneous Weight	2843.7	N
Offset from Element Centerline	0	mm
Element From Node	130	
Detail Type	Weight	
Detail ID	N3 pipe wt	
Dist. from "FROM" Node / Offset dist	900	mm
Miscellaneous Weight	52707	N
Offset from Element Centerline	4119	mm
Element From Node	130	
Detail Type	Weight	
Detail ID	SUPPORT RING	
Dist. from "FROM" Node / Offset dist	1650	mm
Miscellaneous Weight	5883.7	N
Offset from Element Centerline	0	mm
Element From Node	130	
Detail Type	Weight	
Detail ID	N3 INSERT PLT	
Dist. from "FROM" Node / Offset dist	900	mm
Miscellaneous Weight	1176.7	N
Offset from Element Centerline	0	mm
Element From Node	130	
Detail Type	Weight	
Detail ID	CLT-1	
Dist. from "FROM" Node / Offset dist	0.002908	mm
Miscellaneous Weight	26476	N
Offset from Element Centerline	0	mm
Element From Node	130	
Detail Type	Weight	
Detail ID	TRUNN + INSERT	



Dist. from "FROM" Node / Offset dist	1150	mm
Miscellaneous Weight	18631	N
Offset from Element Centerline	0	mm

Element From Node	130	
Detail Type	Weight	
Detail ID	N3 Feed Pipe	
Dist. from "FROM" Node / Offset dist	900	mm
Miscellaneous Weight	34811	N
Offset from Element Centerline	0	mm

Element#	13/18	
Element From Node	150	
Element To Node	160	
Element Type	Conical	
Description	Cone 1	
Distance "FROM" to "TO"	1298	mm
Element Outside Diameter	5534	mm
Element Thickness	14	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3481	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Cone Diameter at "To" End	4600	mm
Design Length of Cone	1298	mm
Half Apex Angle of Cone	19.787916	degrees
Toriconical (Y/N)	N	
Weld is pre-Heated	No	

Element From Node	150	
Detail Type	Platform	
Detail ID	PLT-5	
Dist. from "FROM" Node / Offset dist	850	mm
Platform Start Angle (degrees)	90.0	
Platform End Angle (degrees)	270.0	
Platform Wind Area	5550719	mm ²
Platform Weight	29762	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	140.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.3628	N/mm
Platform Length (top head platform)	0	mm

Element From Node	150	
Detail Type	Insulation	
Detail ID	Ins: 100	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1298	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³

Element From Node	150	
Detail Type	Lining	
Detail ID	Clading	
Dist. from "FROM" Node / Offset dist	0	mm



Height/Length of Lining	1298	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³

Element From Node	150	
Detail Type	Nozzle	
Detail ID	N6 D 2"	
Dist. from "FROM" Node / Offset dist	700	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	101.55	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	150	
Detail Type	Weight	
Detail ID	DEMISTER	
Dist. from "FROM" Node / Offset dist	200	mm
Miscellaneous Weight	20200	N
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	PIPE DAVIT	
Dist. from "FROM" Node / Offset dist	0.002326	mm
Miscellaneous Weight	12748	N
Offset from Element Centerline	3750	mm

Element#	14/18	
Element From Node	160	
Element To Node	170	
Element Type	Conical	
Description	Cone 2	
Distance "FROM" to "TO"	1200	mm
Element Outside Diameter	4600	mm
Element Thickness	14	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3481	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.6	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Cone Diameter at "To" End	3760	mm
Design Length of Cone	1200	mm
Half Apex Angle of Cone	19.290051	degrees
Toriconical (Y/N)	N	
Weld is pre-Heated	No	

Element From Node	160	
Detail Type	Insulation	
Detail ID	ins	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1200	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³

Element From Node	160	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm



Height/Length of Lining	1200	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³

Element#	15/18
Element From Node	170
Element To Node	180
Element Type	Cylinder
Description	Port shell
Distance "FROM" to "TO"	400 mm
Element Outside Diameter	3760 mm
Element Thickness	12 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3481 MPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.6
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	170
Detail Type	Platform
Detail ID	TOP
Dist. from "FROM" Node / Offset dist	0 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	360.0
Platform Wind Area	9390720 mm ²
Platform Weight	116061 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	3100 mm
Platform Height	1000 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.2
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.37 N/mm
Platform Length (top head platform)	0 mm

Element From Node	170
Detail Type	Insulation
Detail ID	ins
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	400 mm
Thickness of Insulation	40 mm
Density	0.000192 kg/cm ³

Element From Node	170
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	400 mm
Thickness of Lining	3 mm
Density	0.008027 kg/cm ³

Element From Node	170
Detail Type	Ring
Detail ID	R2
Dist. from "FROM" Node / Offset dist	20.001 mm
Inside Diameter of Ring	3760 mm
Thickness of Ring	14 mm
Outside Diameter of Ring	4360 mm



Material Name SA-516 70
 Height of Section Ring 0 mm
 Using Custom Stiffener Section No

Element# 16/18
 Element From Node 180
 Element To Node 190
 Element Type Flange
 Description N4 Flange
 Distance "FROM" to "TO" 270 mm
 Flange Inside Diameter 3736 mm
 Element Thickness 240 mm
 Internal Corrosion Allowance 1 mm
 Nominal Thickness 260 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 0.3481 MPa
 Design Temperature Internal Pressure 160 °C
 Design External Pressure 0.1034 MPa
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.6
 Material Name SA-250 T1b
 Allowable Stress, Ambient 88.898 MPa
 Allowable Stress, Operating 88.898 MPa
 Allowable Stress, Hydrotest 175.5 MPa
 Material Density 0.00775 kg/cm³
 P Number Thickness 18.999 mm
 Yield Stress, Operating 172.8 MPa
 UCS-66 Chart Curve Designation B
 External Pressure Chart Name CS-1
 UNS Number K11422
 Product Form Wld. tube
 Perform Flange Stress Calculation (Y/N) Y
 Weld is pre-Heated No

Element From Node 180
 Detail Type Insulation
 Detail ID ins
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 376 mm
 Thickness of Insulation 40 mm
 Density 0.000192 kg/cm³

Element From Node 180
 Detail Type Lining
 Detail ID clad
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Lining 282 mm
 Thickness of Lining 5 mm
 Density 0.008027 kg/cm³

Element# 17/18
 Element From Node 190
 Element To Node 200
 Element Type Flange
 Description N2A Flange
 Distance "FROM" to "TO" 112 mm
 Flange Inside Diameter 2642 mm
 Element Thickness 60 mm
 Internal Corrosion Allowance 1 mm
 Nominal Thickness 60 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 0.3481 MPa
 Design Temperature Internal Pressure 160 °C
 Design External Pressure 0.1034 MPa
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2



Material Name	SA-266 2 +IT
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	225 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	217.8 MPa
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	18/18
Element From Node	200
Element To Node	210
Element Type	Flange
Description	N2B Flange
Distance "FROM" to "TO"	112 mm
Flange Inside Diameter	2642 mm
Element Thickness	60 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	60 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3481 MPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
	Skirt-1	...	1000	...	1000
	Skirt-2	...	3700	...	2700
	Skirt-3	...	6700	...	3000
	Skirt-4	...	9700	...	3000
	Skirt-5	...	12700	...	3000
	Skirt-6	...	15700	...	3000
	Skirt-7	...	16700	...	1000
	BTM DISH	...	16750	...	50
	Shell-1	...	19750	...	3000
	Shell-2	...	22750	...	3000
	Shell-3	...	25750	...	3000
	Shell-4	...	27500	...	1750
	Cone 1	...	28798	...	1298
	Cone 2	...	29998	...	1200
	Port shell	...	30398	...	400
	N4 Flange	...	30671.2	...	270
	N2A Flange	...	19750	...	112
	N2B Flange	...	19750	...	112

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

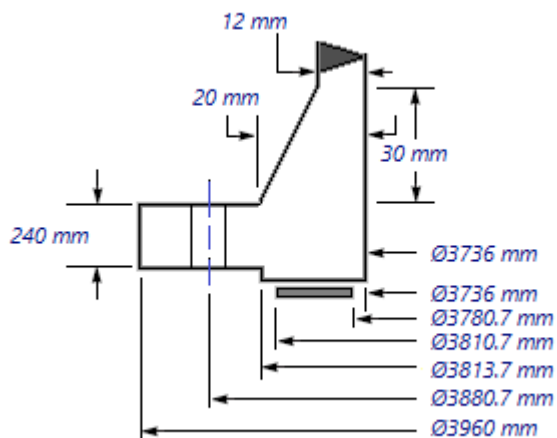
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Flange Input Data Values

Description: FLANGE :

N4 Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	0.35	MPa
Design Temperature		160	°C
Internal Corrosion Allowance	ci	1.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	3736.000	mm
Flange Outside Diameter	A	3960.000	mm
Flange Thickness	t	240.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	20.0000	mm
Length of Hub	h	30.0000	mm
Flange Material	SA-250 T1b		
Flange Material UNS number	K11422		
Flange Allowable Stress At Temperature	Sfo	88.90	MPa
Flange Allowable Stress At Ambient	Sfa	88.90	MPa
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	3880.700	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	72		
Flange Face Outside Diameter	Fod	3813.700	mm
Flange Face Inside Diameter	Fid	3736.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	3810.700	mm
Gasket Inside Diameter	Gi	3780.700	mm
Gasket Factor	m	4.2500	
Gasket Design Seating Stress	y	69.64	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		157399.00	N
Bending Moment on Flange		6181949952.00	N-mm



Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.35 \cdot (3736.0/2 + 1.0)) / (88.9 \cdot 1.0 - 0.6 \cdot 0.35) + Ca$$

$$= 8.3367 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (88.9 \cdot 1.0 \cdot 11.0) / (1869.0 + 0.6 \cdot 11.0)$$

$$= 0.521 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (88.9 \cdot 1.0 \cdot 12.0) / (1868.0 + 0.6 \cdot 12.0)$$

$$= 0.569 \text{ MPa}$$

Corroded Flange Thickness, $t_c = T - c_i$	239.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	3738.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	19.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	52.350 mm
Gasket Contact Width, $N = (G_o - G_i)/2$	15.000 mm
Basic Gasket Width, $b_o = N/2$	7.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.901 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	3796.898 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 3796.8977^2 \cdot 0.348$$

$$= 3941619.500 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.9011 \cdot \pi \cdot 3796.8977 \cdot 4.25 \cdot 0.35$$

$$= 243580.750 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 3738.0^2 \cdot 0.3481$$

$$= 3820282.250 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 3941620 - 3820282$$

$$= 121337.266 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(3941620 + 243581 + 0, 0)$$

$$= 4185200.500 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 69.64 \cdot 6.9011 \cdot \pi \cdot 3796.898 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 5732126.000 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((4185201 + 157399 + 4 \cdot 61819E+10 / 3796.9) / 172, 5732126 / 172)$$

$$= 63102.969 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 239.0 / (4.25 + 0.5)$$

$$= 378.095 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$



$$= 3880.7 \cdot \sin(\pi/72)$$

$$= 169.274 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{169.274/(2 \cdot 38.1 + 239.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	63102.969	65264.379	
Radial Distance between Hub and Bolts:	50.800	52.350	
Radial Distance between Bolts and Edge:	38.100	39.650	
Circ. Spacing between the Bolts:	82.550	169.274	378.095

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 65264.379 \cdot 172.0 / (69.64 \cdot \pi(3810.7 + 3780.7))$$

$$= 6.759 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 65264.3789$$

$$= 11224317.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 4185201 - 3941620$$

$$= 243580.81 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (3880.7 - 3796.8977) / 2$$

$$= 41.9011 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((3880.7 - 3738.0) / 2 + 41.9011)$$

$$= 56.6256 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (3880.7 - 3738.0 - 19.0) / 2$$

$$= 61.8501 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 111.0, Bb = 239.0, Cc = 30.0 \text{ and } Ddg = 15.0 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 111.0 \cdot 239.0^3 \cdot (1/3 - 0.210 \cdot (239.0/111.0) \cdot (1 - 1/12 \cdot (239.0/111.0)^4))$$

$$= 1047161280.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 30.0 \cdot 15.0^3 \cdot (1/3 - 0.105 \cdot (15.0/30.0) \cdot (1 - 1/192 \cdot (15.0/30.0)^4))$$

$$= 28436.082 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 63901.766 + 1.735$$

$$= 1047189760.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 111.0 \cdot 239.0^3 \left(\frac{1}{3} - 0.21 \left(\frac{239.0}{111.0} \right) \left(1 - \frac{1}{12} \left(\frac{239.0}{4.37} \right)^4 \right) \right)$$

$$= 1047161280.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(0.10472E+10, 0.10472E+10)$$

$$= 1047189760.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 7.453 \cdot 11.0^2 \cdot 202.776 \cdot 3738.0 / 0.419$$

$$= 142433568.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	3820282.	61.8501	1.0000	236380656.
Face Pressure, Mt	121337.	56.6256	1.0000	6873583.
Gasket Load, Mg	243581.	41.9011	1.0000	10210449.
Gasket Seating, Matm	11224317.	41.9011	1.0000	470502176.
Moment Load, Moe	...	...	...	116216088.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{I}{(0.384 \cdot IP + I)} \right] \cdot \left(\frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 61819E+10 \left[\frac{14243E+09}{(0.384 \cdot 10472E+10 + 14243E+09)} \right] \cdot (61.9 / (3880.7 - 2 \cdot 61.9)) +$$

$$157399 \cdot 61.85$$

$$= 116216088.000 \text{ N-mm}$$

Total Moment for Operation, Mo 369680768. N-mm

Total Moment for Gasket seating, Mg 470502176. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 202.776 mm

Hub Ratio, h/h0 = HL / H0 0.148

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.727

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.904

Factor V per 4.16.5 0.419

Factor f per 4.16.5 2.221

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 3960.0 / 3738.0$$

$$= 1.059$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.059^2 \cdot \log_{10}(1.059))) / (1.059^2 - 1) / (1.059 - 1)$$

$$= 33.387$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.059^2(1 + 8.55246 \cdot \log_{10}(1.059)) - 1) / ((1.0472 + 1.9448 \cdot 1.059^2) \cdot (1.059 - 1))$$

$$= 1.891$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.059^2(1 + 8.55246 \cdot \log_{10}(1.059)) - 1) / (1.36136(1.059^2 - 1) \cdot (1.059 - 1))$$

$$= 36.689$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.059^2 + 1) / (1.059^2 - 1)$$

= 17.352

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 36.689 \cdot 11.0^2 \cdot 202.776 / 0.419 \\ &= 2146169.000 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.904 / 202.776 \\ &= 0.004 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3 / d \\ &= (239.0 \cdot 0.004 + 1) / 1.891 + 239.0^3 / 2146169 \\ &= 7.453 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (2.2213 \cdot 36968E+09 / 3738.0) / (7.453 \cdot 19.0^2) \\ &= 81.62 \text{ MPa} \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (2.2213 \cdot 0.47050E+09 / 3738.0) / (7.453 \cdot 19.0^2) \\ &= 103.88 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.4204 \cdot 36968E+09 / 3738.0) / (7.453 \cdot 239.0^2) \\ &= 0.56 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.4204 \cdot 0.47050E+09 / 3738.0) / (7.453 \cdot 239.0^2) \\ &= 0.72 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (33.387 \cdot 36968E+09 / (239.0^2 \cdot 3738.0)) - 17.3522 \cdot 0.6 \\ &= 48.03 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (33.387 \cdot 0.47050E+09 / (239.0^2 \cdot 3738.0)) - 17.3522 \cdot 0.7 \\ &= 61.13 \text{ MPa} \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (82 + \max(0.6, 48)) / 2 \\ &= 64.83 \text{ MPa} \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (104 + \max(0.7, 61)) / 2 \\ &= 82.51 \text{ MPa} \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= (Wm1 + Fa + 4Me/G) / Ab \\ &= 10852593 / 65264.3789 \\ &= 166.30 \text{ MPa} \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm2 / Ab) \\ &= (5732126 / 65264.3789) \\ &= 87.84 \text{ MPa} \end{aligned}$$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	81.62	133.35	103.88	133.35
Radial Flange	0.56	88.90	0.72	88.90
Tangential Flange	48.03	88.90	61.13	88.90
Maximum Average	64.83	88.90	82.51	88.90
Bolting	166.30	172.00	87.84	172.00

Minimum Required Flange Thickness [Rigidity] 234.137 mm

Estimated M.A.W.P. (Operating) 0 MPa
 Estimated Finished Weight of Flange at given Thk. 2562.4 kgm
 Estimated Unfinished Weight of Forging at given Thk 2833.3 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.47050E+09 / 1.0 \cdot 1.0 \cdot 0.419 / (6.982 \cdot 199955 \cdot 11.0^2 \cdot 202.776 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.36968E+09 / 1.0 \cdot 1.0 \cdot 0.419 / (6.982 \cdot 192509 \cdot 11.0^2 \cdot 202.776 \cdot 0.3)$
 $= 0.816$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.47050E+09 / 1.0 \cdot 1.0 \cdot 0.419 / (7.453 \cdot 199955 \cdot 11.0^2 \cdot 202.776 \cdot 0.3)$
 $= 0.937$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.36968E+09 / 1.0 \cdot 1.0 \cdot 0.419 / (7.453 \cdot 192509 \cdot 11.0^2 \cdot 202.776 \cdot 0.3)$
 $= 0.765$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.468, Temperature Reduction per Fig. UCS 66.1 = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	82.35	133.35	102.91	133.35
Radial Flange	0.64	88.90	0.81	88.90
Tangential Flange	46.78	88.90	58.46	88.90
Maximum Average	64.56	88.90	80.68	88.90
Bolting	166.30	172.00	87.84	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 229.845 mm

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Warnings for Flange : FLANGE :

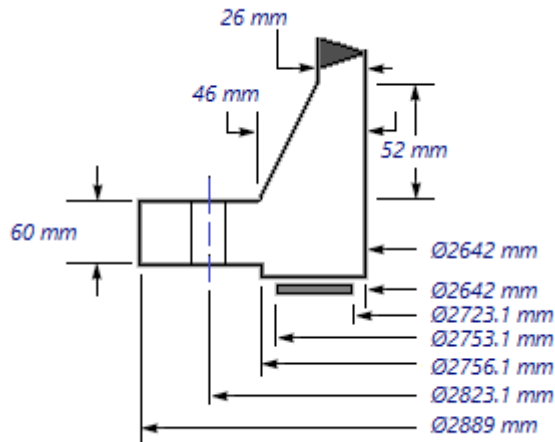
Note: Flange Hub Taper 0.3846 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by
 ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values

Description: FLANGE :

N2A Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	0.36	MPa
Design Temperature		160	°C
Internal Corrosion Allowance	ci	1.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2642.000	mm
Flange Outside Diameter	A	2889.000	mm
Flange Thickness	t	60.0000	mm
Thickness of Hub at Small End	go	26.0000	mm
Thickness of Hub at Large End	gl	46.0000	mm
Length of Hub	h	52.0000	mm
Flange Material		SA-266 2 +IT	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	2823.100	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		72	
Flange Face Outside Diameter	Fod	2756.100	mm
Flange Face Inside Diameter	Fid	2642.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2753.100	mm
Gasket Inside Diameter	Gi	2723.100	mm
Gasket Factor	m	4.2500	
Gasket Design Seating Stress	y	69.64	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		49030.85	N
Bending Moment on Flange		740066624.00	N-mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.36 \cdot (2642.0/2 + 1.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 0.36) + C_a$$

$$= 4.4365 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 25.0) / (1322.0 + 0.6 \cdot 25.0)$$

$$= 2.580 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 26.0) / (1321.0 + 0.6 \cdot 26.0)$$

$$= 2.684 \text{ MPa}$$

Corroded Flange Thickness, $t_c = T - c_i$	59.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2644.000 mm
Corroded Large Hub, $g_{lCor} = g_l - c_i$	45.000 mm
Corroded Small Hub, $g_{sCor} = g_s - c_i$	25.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g_{lCor}$	44.550 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	15.000 mm
Basic Gasket Width, $b_o = N/2$	7.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.901 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2739.298 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2739.2979^2 \cdot 0.358$$

$$= 2110592.500 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.9011 \cdot \pi \cdot 2739.2979 \cdot 4.25 \cdot 0.36$$

$$= 180785.547 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2644.0^2 \cdot 0.3582$$

$$= 1966295.375 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 2110593 - 1966295$$

$$= 144297.016 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(2110593 + 180786 + 0, 0)$$

$$= 2291378.000 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 69.64 \cdot 6.9011 \cdot \pi \cdot 2739.298 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 4135495.250 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((2291378 + 49031 + 4 \cdot 74007E+09 / 2739.3) / 172, 4135495 / 172)$$

$$= 24046.055 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 31.75 + 6 \cdot 59.0 / (4.25 + 0.5)$$

$$= 138.026 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2823.1 \cdot \sin(\pi / 72)$$

$$= 123.142 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ B_s / (2a + t) }, 1)$$

$$= \max(\sqrt{ 123.142 / (2 \cdot 31.75 + 59.0) }, 1)$$

$$= 1.0026$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	24046.055	43153.461	
Radial Distance between Hub and Bolts:	44.450	44.550	
Radial Distance between Bolts and Edge:	31.750	32.950	
Circ. Spacing between the Bolts:	71.450	123.142	138.026

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 43153.461 \cdot 172.0 / (69.64 \cdot \pi (2753.1 + 2723.1))$$

$$= 6.195 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 43153.4609$$

$$= 7421630.50 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 2291378 - 2110593$$

$$= 180785.62 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2823.1003 - 2739.2979) / 2$$

$$= 41.9012 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + h_g)$$

$$= 0.5 ((2823.1003 - 2644.0) / 2 + 41.9012)$$

$$= 65.7257 \text{ mm}$$



Distance to End Pressure Reaction [hd]:
 $= (C - B - g_1) / 2$
 $= (2823.1 - 2644.0 - 45.0) / 2$
 $= 67.0501 \text{ mm}$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:
 $Aa = 122.5$, $Bb = 59.0$, $Cc = 52.0$ and $Ddg = 35.0 \text{ mm}$

Compute the value of [Kab]:
 $= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$
 $= 122.5 \cdot 59.0^3 \cdot (1/3 - 0.210 \cdot (59.0/122.5) \cdot (1 - 1/12 \cdot (59.0/122.5)^4))$
 $= 5853068.000 \text{ mm}^4$

Compute the value of [Kcd]:
 $= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$
 $= 52.0 \cdot 35.0^3 \cdot (1/3 - 0.105 \cdot (35.0/52.0) \cdot (1 - 1/192 \cdot (35.0/52.0)^4))$
 $= 585769.000 \text{ mm}^4$

Flange Inertia Term [Ip,term1]:
 $= Kab + Kcd = 357.176 + 35.746$
 $= 6438837.500 \text{ mm}^4$

Flange Inertia Term [Ip,term2]:
 $= Ar \cdot t^3 \cdot (1/3 - 0.21 \cdot (t/AR) \cdot (1 - 1/12 \cdot (t/AR)^4))$
 $= 122.5 \cdot 59.0^3 \cdot (1/3 - 0.21 \cdot (59.0/122.5) \cdot (1 - 1/12 \cdot (59.0/122.5)^4))$
 $= 5853068.000 \text{ mm}^4$

Flange Inertia [Ip]:
 $= \max(Ip,term1, Ip,term2)$
 $= \max(6438838, 5853068)$
 $= 6438837.5 \text{ mm}^4$

Flange Moment of Inertia for Integral Types [I]:
 $= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$
 $= 0.0874 \cdot 0.661 \cdot 25.0^2 \cdot 257.099 \cdot 2644.0 / 0.377$
 $= 65167132.000 \text{ mm}^4$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1966295.	67.0501	1.0026	132238976.
Face Pressure, Mt	144297.	65.7257	1.0026	9512693.
Gasket Load, Mg	180786.	41.9012	1.0026	7598040.
Gasket Seating, Matm	7421630.	41.9012	1.0026	311915520.
Moment Load, Moe	...	...	...	74404816.

Moment Value, eq. 4.16.16, [Moe]:
 $= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 $= 4 \cdot 74007E+09 [65167132 / (0.384 \cdot 6438838 + 65167132)] \cdot (67.1 / (2823.1 - 2 \cdot 67.1)) +$
 $49031 \cdot 67.05$
 $= 74404816.000 \text{ N-mm}$

Total Moment for Operation, Mo 223754544. N-mm
 Total Moment for Gasket seating, Mg 311915520. N-mm

Effective Hub Length, $ho = \sqrt{Bcor \cdot goCor}$ 257.099 mm
 Hub Ratio, $h/h0 = HL / H0$ 0.202
 Thickness Ratio, $g1/g0 = (g1Cor/goCor)$ 1.800

Flange Factors for Integral Flange:
 Factor F per 4.16.5 0.897
 Factor V per 4.16.5 0.377
 Factor f per 4.16.5 2.136

Stress factor from Table 4.16.4 [K]:

$$\begin{aligned} &= A / B \\ &= 2889.0/2644.0 \\ &= 1.093 \end{aligned}$$

Stress factor from Table 4.16.4 [Y]:

$$\begin{aligned} &= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1) \\ &= (0.66845 + (5.7169(1.093^2 \cdot \log_{10}(1.093))) / (1.093^2 - 1)) / (1.093 - 1) \\ &= 21.833 \end{aligned}$$

Stress factor from Table 4.16.4 [T]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1)) \\ &= (1.093^2(1 + 8.55246 \cdot \log_{10}(1.093)) - 1) / ((1.0472 + 1.9448 \cdot 1.093^2) \cdot (1.093 - 1)) \\ &= 1.880 \end{aligned}$$

Stress factor from Table 4.16.4 [U]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1)) \\ &= (1.093^2(1 + 8.55246 \cdot \log_{10}(1.093)) - 1) / (1.36136(1.093^2 - 1) \cdot (1.093 - 1)) \\ &= 23.992 \end{aligned}$$

Stress factor from Table 4.16.4 [Z]:

$$\begin{aligned} &= (K^2 + 1) / (K^2 - 1) \\ &= (1.093^2 + 1) / (1.093^2 - 1) \\ &= 11.314 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 23.992 \cdot 25.0^2 \cdot 257.099 / 0.377 \\ &= 10228841.000 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.897 / 257.099 \\ &= 0.003 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / (T + t^3/d) \\ &= (59.0 \cdot 0.003 + 1) / (1.88 + 59.0^3 / 10228841) \\ &= 0.661 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (2.1363 \cdot 2.2375E+09 / 2644.0) / (0.6615 \cdot 45.0^2) \\ &= 134.93 \text{ MPa} \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (2.1363 \cdot 0.31192E+09 / 2644.0) / (0.6615 \cdot 45.0^2) \\ &= 188.09 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (1.2743 \cdot 2.2375E+09 / 2644.0) / (0.6615 \cdot 59.0^2) \\ &= 46.82 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (1.2743 \cdot 0.31192E+09 / 2644.0) / (0.6615 \cdot 59.0^2) \\ &= 65.27 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (21.8332 \cdot 2.2375E+09 / (59.0^2 \cdot 2644.0)) - 11.314 \cdot 47 \\ &= 0.89 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (21.8332 \cdot 0.31192E+09 / (59.0^2 \cdot 2644.0)) - 11.314 \cdot 65 \\ &= 1.24 \text{ MPa} \end{aligned}$$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (135 + \max(47, 0.9)) / 2$
 $= 90.87 \text{ MPa}$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (188 + \max(65, 1)) / 2$
 $= 126.68 \text{ MPa}$

Bolt Stress, Operating [BSo]:
 $= (Wm1 + Fa + 4Me/G) / Ab$
 $= 3420637 / 43153.4609$
 $= 79.27 \text{ MPa}$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (4135495 / 43153.4609)$
 $= 95.84 \text{ MPa}$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	134.93	207.00	188.09	207.00
Radial Flange	46.82	138.00	65.27	138.00
Tangential Flange	0.89	138.00	1.24	138.00
Maximum Average	90.87	138.00	126.68	138.00
Bolting	79.27	172.00	95.84	172.00

Minimum Required Flange Thickness 55.016 mm

Estimated M.A.W.P. (Operating) 1 MPa
 Estimated Finished Weight of Flange at given Thk. 621.0 kgm
 Estimated Unfinished Weight of Forging at given Thk 931.4 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.31846E+09 / 1.0237 \cdot 1 \cdot 0.377 / (0.648 \cdot 202713 \cdot 25.0^2 \cdot 257.099 \cdot 0.3)$
 $= 0.965 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.22845E+09 / 1.0237 \cdot 1 \cdot 0.377 / (0.648 \cdot 194578 \cdot 25.0^2 \cdot 257.099 \cdot 0.3)$
 $= 0.721 \text{ (should be } \leq 1)$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.31192E+09 / 1.0026 \cdot 1 \cdot 0.377 / (0.661 \cdot 202713 \cdot 25.0^2 \cdot 257.099 \cdot 0.3)$
 $= 0.945 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 0.22375E+09 / 1.0026 \cdot 1 \cdot 0.377 / (0.661 \cdot 194578 \cdot 25.0^2 \cdot 257.099 \cdot 0.3)$
 $= 0.706 \text{ (should be } \leq 1)$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:



Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	126.85	207.00	175.34	207.00
Radial Flange	45.60	138.00	63.03	138.00
Tangential Flange	0.62	138.00	0.86	138.00
Maximum Average	86.22	138.00	119.18	138.00
Bolting	79.27	172.00	95.84	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 51.664 mm

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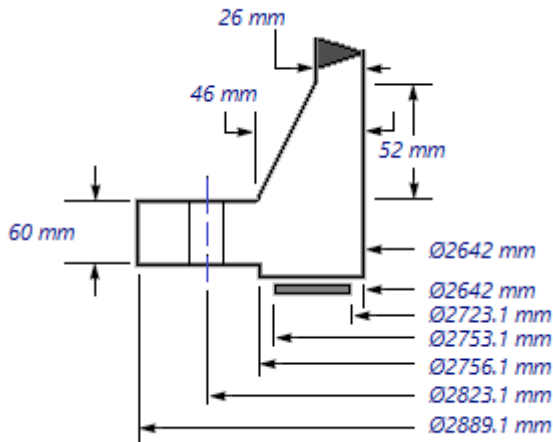
Warnings for Flange : FLANGE :

Note: Flange Hub Taper 0.3846 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
 This warning should not be considered for Standard Flanges as permitted by
 ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values Description: FLANGE :

N2B Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	0.36	MPa
Design Temperature		160	°C
Internal Corrosion Allowance	ci	1.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2642.000	mm
Flange Outside Diameter	A	2889.100	mm
Flange Thickness	t	60.0000	mm
Thickness of Hub at Small End	go	26.0000	mm
Thickness of Hub at Large End	gl	46.0000	mm
Length of Hub	h	52.0000	mm
Flange Material		SA-266 2 +IT	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	2823.100	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		72	
Flange Face Outside Diameter	Fod	2756.100	mm
Flange Face Inside Diameter	Fid	2642.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2753.100	mm
Gasket Inside Diameter	Gi	2723.100	mm
Gasket Factor	m	4.2500	
Gasket Design Seating Stress	y	69.64	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		49030.80	N
Bending Moment on Flange		1356739968.00	N-mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.36 \cdot (2642.0/2 + 1.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 0.36) + C_a$$

$$= 4.4365 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 25.0) / (1322.0 + 0.6 \cdot 25.0)$$

$$= 2.580 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 26.0) / (1321.0 + 0.6 \cdot 26.0)$$

$$= 2.684 \text{ MPa}$$

Corroded Flange Thickness, $t_c = T - c_i$	59.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2644.000 mm
Corroded Large Hub, $g_{lCor} = g_l - c_i$	45.000 mm
Corroded Small Hub, $g_{sCor} = g_s - c_i$	25.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g_{lCor}$	44.550 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	15.000 mm
Basic Gasket Width, $b_o = N/2$	7.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.901 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2739.298 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2739.2979^2 \cdot 0.358$$

$$= 2110592.500 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.9011 \cdot \pi \cdot 2739.2979 \cdot 4.25 \cdot 0.36$$

$$= 180785.547 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2644.0^2 \cdot 0.3582$$

$$= 1966295.375 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 2110593 - 1966295$$

$$= 144297.016 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(2110593 + 180786 + 0, 0)$$

$$= 2291378.000 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 69.64 \cdot 6.9011 \cdot \pi \cdot 2739.298 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 4135495.250 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((2291378 + 49031 + 4 \cdot 13567E+10 / 2739.3) / 172, 4135495 / 172)$$

$$= 25123.264 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 31.75 + 6 \cdot 59.0 / (4.25 + 0.5)$$

$$= 138.026 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2823.1 \cdot \sin(\pi / 72)$$

$$= 123.142 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ B_s / (2a + t) }, 1)$$

$$= \max(\sqrt{ 123.142 / (2 \cdot 31.75 + 59.0) }, 1)$$

$$= 1.0026$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	25123.264	43153.461	
Radial Distance between Hub and Bolts:	44.450	44.550	
Radial Distance between Bolts and Edge:	31.750	33.000	
Circ. Spacing between the Bolts:	71.450	123.142	138.026

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 43153.461 \cdot 172.0 / (69.64 \cdot \pi (2753.1 + 2723.1))$$

$$= 6.195 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 43153.4609$$

$$= 7421630.50 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 2291378 - 2110593$$

$$= 180785.62 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2823.1003 - 2739.2979) / 2$$

$$= 41.9012 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + h_g)$$

$$= 0.5 ((2823.1003 - 2644.0) / 2 + 41.9012)$$

$$= 65.7257 \text{ mm}$$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (2823.1 - 2644.0 - 45.0) / 2$
 $= 67.0501 \text{ mm}$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:
 Aa = 122.55, Bb = 59.0, Cc = 52.0 and Ddg = 35.0 mm

Compute the value of [Kab]:
 $= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$
 $= 122.55 \cdot 59.0^3 \cdot (1/3 - 0.210 \cdot (59.0/122.55) \cdot (1 - 1/12 \cdot (59.0/122.55)^4))$
 $= 5856479.000 \text{ mm}^4$

Compute the value of [Kcd]:
 $= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$
 $= 52.0 \cdot 35.0^3 \cdot (1/3 - 0.105 \cdot (35.0/52.0) \cdot (1 - 1/192 \cdot (35.0/52.0)^4))$
 $= 585769.000 \text{ mm}^4$

Flange Inertia Term [Ip,term1]:
 $= Kab + Kcd = 357.385 + 35.746$
 $= 6442248.000 \text{ mm}^4$

Flange Inertia Term [Ip,term2]:
 $= Ar \cdot t^3 \cdot (1/3 - 0.21 \cdot (t/AR) \cdot (1 - 1/12 \cdot (t/AR)^4))$
 $= 122.55 \cdot 59.0^3 \cdot (1/3 - 0.21 \cdot (59.0/122.55) \cdot (1 - 1/12 \cdot (59.0/122.55)^4))$
 $= 5856479.000 \text{ mm}^4$

Flange Inertia [Ip]:
 $= \max(Ip,term1, Ip,term2)$
 $= \max(6442248, 5856479)$
 $= 6442248.0 \text{ mm}^4$

Flange Moment of Inertia for Integral Types [I]:
 $= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$
 $= 0.0874 \cdot 0.661 \cdot 25.0^2 \cdot 257.099 \cdot 2644.0 / 0.377$
 $= 65168376.000 \text{ mm}^4$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1966295.	67.0501	1.0026	132238976.
Face Pressure, Mt	144297.	65.7257	1.0026	9512693.
Gasket Load, Mg	180786.	41.9012	1.0026	7598040.
Gasket Seating, Matm	7421630.	41.9012	1.0026	311915520.
Moment Load, Moe	...	...	...	133661000.

Moment Value, eq. 4.16.16, [Moe]:
 $= 4Me [I / (0.384 \cdot IP + I)] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$
 $= 4 \cdot 13567E+10 [65168376 / (0.384 \cdot 6442248 + 65168376)] \cdot (67.1 / (2823.1 - 2 \cdot 67.1)) + 49031 \cdot 67.05$
 $= 133661000.000 \text{ N-mm}$

Total Moment for Operation, Mo 283010720. N-mm
 Total Moment for Gasket seating, Mg 311915520. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 257.099 mm
 Hub Ratio, h/h0 = HL / H0 0.202
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.800

Flange Factors for Integral Flange:
 Factor F per 4.16.5 0.897
 Factor V per 4.16.5 0.377
 Factor f per 4.16.5 2.136

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2889.1/2644.0$$

$$= 1.093$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.093^2 \cdot \log_{10}(1.093))) / (1.093^2 - 1)) / (1.093 - 1)$$

$$= 21.825$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.093^2(1 + 8.55246 \cdot \log_{10}(1.093)) - 1) / ((1.0472 + 1.9448 \cdot 1.093^2) \cdot (1.093 - 1))$$

$$= 1.880$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.093^2(1 + 8.55246 \cdot \log_{10}(1.093)) - 1) / (1.36136(1.093^2 - 1) \cdot (1.093 - 1))$$

$$= 23.983$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.093^2 + 1) / (1.093^2 - 1)$$

$$= 11.310$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 23.983 \cdot 25.0^2 \cdot 257.099 / 0.377$$

$$= 10224891.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.897 / 257.099$$

$$= 0.003 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / (T + t^3/d)$$

$$= (59.0 \cdot 0.003 + 1) / (1.88 + 59.0^3 / 10224891)$$

$$= 0.661$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (2.1363 \cdot 28301E+09 / 2644.0) / (0.6615 \cdot 45.0^2)$$

$$= 170.66 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (2.1363 \cdot 0.31192E+09 / 2644.0) / (0.6615 \cdot 45.0^2)$$

$$= 188.09 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.2743 \cdot 28301E+09 / 2644.0) / (0.6615 \cdot 59.0^2)$$

$$= 59.22 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.2743 \cdot 0.31192E+09 / 2644.0) / (0.6615 \cdot 59.0^2)$$

$$= 65.27 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (21.8247 \cdot 28301E+09 / (59.0^2 \cdot 2644.0)) - 11.3096 \cdot 59$$

$$= 1.14 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (21.8247 \cdot 0.31192E+09 / (59.0^2 \cdot 2644.0)) - 11.3096 \cdot 65$$

$$= 1.26 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:
 $= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$
 $= (171 + \max(59, 1)) / 2$
 $= 114.94 \text{ MPa}$

Average Flange Stress, Seating [SAa]:
 $= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$
 $= (188 + \max(65, 1)) / 2$
 $= 126.68 \text{ MPa}$

Bolt Stress, Operating [BSo]:
 $= (W_{m1} + F_a + 4M_e/G) / A_b$
 $= 4320756 / 43153.4609$
 $= 100.13 \text{ MPa}$

Bolt Stress, Seating [BSa]:
 $= (W_{m2} / A_b)$
 $= (4135495 / 43153.4609)$
 $= 95.84 \text{ MPa}$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	170.66	207.00	188.09	207.00
Radial Flange	59.22	138.00	65.27	138.00
Tangential Flange	1.14	138.00	1.26	138.00
Maximum Average	114.94	138.00	126.68	138.00
Bolting	100.13	172.00	95.84	172.00

Minimum Required Flange Thickness 55.016 mm

Estimated M.A.W.P. (Operating) 0 MPa
 Estimated Finished Weight of Flange at given Thk. 621.2 kgm
 Estimated Unfinished Weight of Forging at given Thk 931.8 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.31846E+09 / 1.0237 \cdot 1 \cdot 0.377 / (0.648 \cdot 202713 \cdot 25.0^2 \cdot 257.099 \cdot 0.3)$
 $= 0.965 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.28895E+09 / 1.0237 \cdot 1 \cdot 0.377 / (0.648 \cdot 194578 \cdot 25.0^2 \cdot 257.099 \cdot 0.3)$
 $= 0.913 \text{ (should be } \leq 1)$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.31192E+09 / 1.0026 \cdot 1 \cdot 0.377 / (0.661 \cdot 202713 \cdot 25.0^2 \cdot 257.099 \cdot 0.3)$
 $= 0.945 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 0.28301E+09 / 1.0026 \cdot 1 \cdot 0.377 / (0.661 \cdot 194578 \cdot 25.0^2 \cdot 257.099 \cdot 0.3)$
 $= 0.893 \text{ (should be } \leq 1)$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:



Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	160.54	207.00	175.34	207.00
Radial Flange	57.71	138.00	63.03	138.00
Tangential Flange	0.80	138.00	0.88	138.00
Maximum Average	109.12	138.00	119.18	138.00
Bolting	100.13	172.00	95.84	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 51.664 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt-1	...	...	30	1	5553	...	1
Skirt-2	...	...	20	1	5543	...	2
Skirt-3	...	...	18	1	5541	...	3
Skirt-4	...	...	18	1	5541	...	4
Skirt-5	...	...	18	1	5541	...	5
Skirt-6	...	...	16	1	5540	...	6
Skirt-7	...	...	16	1	5540	...	7
BTM DISH	0.4015	20	1	5506	138		8
Shell-1	0.3876	16	1	5506	138		9
Shell-2	0.3582	16	1	5506	138		10
Shell-3	0.3481	16	1	5506	138		11
Shell-4	0.3481	14	1	5506	138		12
Cone 1	0.3481	16	1	5534	138		13
Cone 2	0.3481	16	1	4600	138		14
Port shell	0.3481	12	1	3760	138		15
N4 Flange	0.3481	260	1	3736	88.898		16
N2A Flange	0.3582	60	1	2642	138		17
N2B Flange	0.3582	60	1	2642	138		18

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-1	...		No Calc	No Calc	30	No Calc	1
Skirt-2	...		No Calc	No Calc	20	No Calc	2
Skirt-3	...		No Calc	No Calc	18	No Calc	3
Skirt-4	...		No Calc	No Calc	18	No Calc	4
Skirt-5	...		No Calc	No Calc	18	No Calc	5
Skirt-6	...		No Calc	No Calc	16	No Calc	6
Skirt-7	...		No Calc	No Calc	16	No Calc	7
BTM DISH	0.34815	0.69819	0.80155	16	9.0118		8
Shell-1	0.34815	0.70976	0.79923	16	8.74763		9
Shell-2	0.34815	0.73916	0.79923	16	8.15884		10
Shell-3	0.34815	0.74917	0.79923	16	7.95848		11
Shell-4	0.34815	0.64956	0.69963	14	7.95848		12
Cone 1	0.34815	0.61114	0.65824	14	8.41133		13
Cone 2	0.34815	0.73777	0.79465	14	7.14152		14
Port shell	0.34815	0.80933	0.88309	12	5.73819		15
N4 Flange	0.34815	0.37913	0.37913	240	234.137		16
N2A Flange	0.34815	0.57097	0.67435	60	55.0164		17
N2B Flange	0.34815	0.42885	0.53165	60	55.0164		18

Minimum 0 0

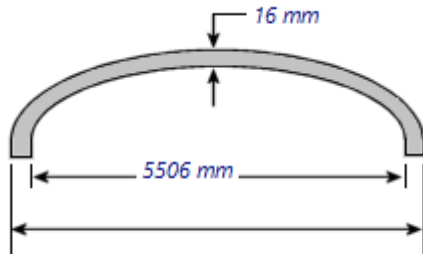
MAWP: 0 MPa, limited by: N4 Flange.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: BTM DISH

Elliptical Head Design Information:							
Design Pressure	P	0.402 MPa		Int. Design Temperature	160 °C		
Material		SA-516 70		Ext. Design Temperature	0 °C		
Length		50.000 mm		External Pressure Chart	CS-2		
				UNS Number	K02700		
Int. Corr. All.	c	1.0000	mm	Allowable Stress (ope)	S	138.0	MPa
Ext. Corr. All.	ce	0.0000	mm	Allowable Stress (amb)	Sa	138.0	MPa
Inside Diameter	Di	5506.000	mm	Actual Stress		73.7	MPa
Spec. Min. Thk	t,e, or sa	16.0000	mm	Spec. Nominal Thk.		20.000	mm
Surface Area		34212504.0	mm²	SG of Contents		1.000	
Element Volume		23044.4	ltr	Weight of Contents		225110.9	N
Empty Weight		56873.3	N	Operating Weight		281984.1	N
				Comp. All. Ope. Stress		80.7	MPa
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 68.825 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 34.413 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.402 \cdot 5508.0 \cdot 1.) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.402)$$

$$= 8.0118 + 1.0000 = 9.0118 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.053 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 15.0) / (1. \cdot 5508.0 + 0.2 \cdot 15.0)$$

$$= 1 - 0.1 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 16.0) / (1.0 \cdot 5506.0 + 0.2 \cdot 16.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.402 \cdot (1. \cdot 5508.0 + 0.2 \cdot 15.0)) / (2 \cdot 1.0 \cdot 15.0)$$

$$= 73.726 \text{ MPa}$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (0.402 \cdot 2754.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.402) + 1.0$
 $= 9.027 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.040 MPa
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138.0 \cdot 1.0 \cdot 19.0) / (2754.0 + 0.6 \cdot 19.0)$
 $= 1 - 0 = 1 \text{ MPa}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (5508.0 / (2 \cdot 1377.5))^2) / 6$
 $= 0.999516$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.586 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 16.0, tr = 8.012, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.534$, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 20.0, tr = 8.027, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.422$, Temp. Reduction = 45 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.402 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	20.000 mm	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.422	Temperature Difference	41 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 68.825 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 34.413 mm

EXTERNAL PRESSURE RESULTS for: BTM DISH

Elliptical Head From 80 to 90 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
15.000	5538.00	369.20	0.0003762	37.61

MAEP = $B / (K \cdot Do / t) = 37.6104 / (0.9 \cdot 369.2) = 0.1132 \text{ MPa}$

PV Elite 28 U2 Licensee: SL User
FileName : C-1901_R0_Ope
Element: BTM DISH

Step: 7 2:08pm

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HEXAGON

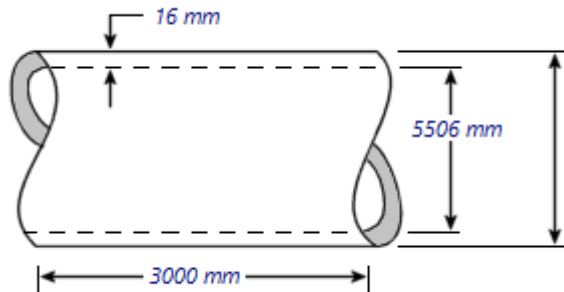
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
14.339	5538.00	386.23	0.0003596	35.95

$$MAEP = B/(K0*Do/t) = 35.9522/(0.9 *386.2289) = 0.1034 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	0.388 MPa	Int. Design Temperature		160 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	5506.000 mm	Actual Stress		71.4 MPa
Spec. Min. Thk t,e, or sa		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		52194420.0 mm²	SG of Contents		1.000
Element Volume		71443.3 ltr	Weight of Contents		698620.1 N
Empty Weight		63287.0 N	Operating Weight		761907.1 N
			Comp. All. Ope. Stress		80.7 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.388 \cdot 2754.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.388)$$

$$= 7.7476 + 1.0000 = 8.7476 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.039 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (2754.0 + 0.6 \cdot 15.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (2753.0 + 0.6 \cdot 16.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.388 \cdot (2754.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 71.389 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.290 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 16.0, tr = 7.748, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.517$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.388 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.517	Temperature Difference	30 °C
UCS-66(c) red.	No	Computed Min. Temp.	-44 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 90 to Ring R1 Fr90 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	5538.00	708.84	369.20	0.1280	0.0017830	101.09

$$MAEP = (4*B) / (3*(Do/t)) = (4*101.0931) / (3*369.2) = 0.3651 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.410	5538.00	708.84	747.39	0.1280	0.0005799	57.97

$$MAEP = (4*B) / (3*(Do/t)) = (4*57.9743) / (3*747.3942) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	5538.00	3709.59	369.20	0.6698	0.0002866	28.66

$$MAEP = (4*B) / (3*(Do/t)) = (4*28.6558) / (3*369.2) = 0.1035 \text{ MPa}$$

Cylindrical Shell From Ring R1 Fr90 to Ring R2 Fr90 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	5538.00	2650.00	369.20	0.4785	0.0004084	40.83

$$MAEP = (4*B) / (3*(Do/t)) = (4*40.8308) / (3*369.2) = 0.1475 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.081	5538.00	2650.00	423.37	0.4785	0.0003285	32.84

$$MAEP = (4*B) / (3*(Do/t)) = (4*32.8408) / (3*423.3743) = 0.1034 \text{ MPa}$$



Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	5538.00	3708.65	369.20	0.6697	0.0002867	28.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.6634)/(3*369.2) = 0.1035 \text{ MPa}$$

Cylindrical Shell From Ring R2 Fr90 to 100 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	5538.00	3000.00	369.20	0.5417	0.0003583	35.82

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.8193)/(3*369.2) = 0.1294 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.818	5538.00	3000.00	400.79	0.5417	0.0003110	31.09

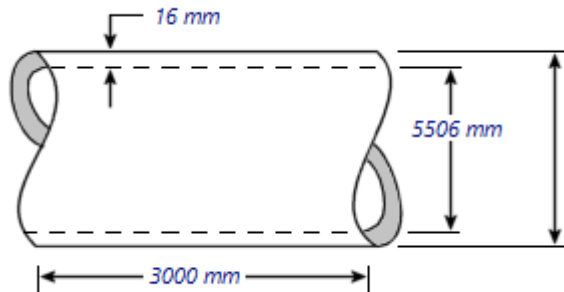
$$MAEP = (4*B)/(3*(Do/t)) = (4*31.0891)/(3*400.7923) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	5538.00	3709.85	369.20	0.6699	0.0002866	28.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.6537)/(3*369.2) = 0.1035 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	0.358 MPa	Int. Design Temperature		160 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	5506.000 mm	Actual Stress		66.0 MPa
Spec. Min. Thk t,e, or sa		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		52194420.0 mm²	SG of Contents		1.000
Element Volume		71443.3 ltr	Weight of Contents		237763.7 N
Empty Weight		63287.0 N	Operating Weight		301050.7 N
			Comp. All. Ope. Stress		80.7 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.358 \cdot 2754.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.358)$$

$$= 7.1588 + 1.0000 = 8.1588 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.010 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (2754.0 + 0.6 \cdot 15.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (2753.0 + 0.6 \cdot 16.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.358 \cdot (2754.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 65.972 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.290 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 16.0, tr = 7.779, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr * E^*/(tg - c) = 0.519$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.389 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.519	Temperature Difference	30 °C
UCS-66(c) red.	No	Computed Min. Temp.	-44 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 100 to Ring R1 Fr100 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	5538.00	3000.00	369.20	0.5417	0.0003583	35.82

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.8193)/(3*369.2) = 0.1294 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.818	5538.00	3000.00	400.79	0.5417	0.0003110	31.09

$$MAEP = (4*B)/(3*(Do/t)) = (4*31.0891)/(3*400.7923) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	5538.00	3709.85	369.20	0.6699	0.0002866	28.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.6537)/(3*369.2) = 0.1035 \text{ MPa}$$

Cylindrical Shell From Ring R1 Fr100 to 110 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	5538.00	2650.00	369.20	0.4785	0.0004084	40.83

$$MAEP = (4*B)/(3*(Do/t)) = (4*40.8308)/(3*369.2) = 0.1475 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.081	5538.00	2650.00	423.37	0.4785	0.0003285	32.84

$$MAEP = (4*B)/(3*(Do/t)) = (4*32.8408)/(3*423.3745) = 0.1034 \text{ MPa}$$

PV Elite 28 U2 Licensee: SL User
FileName : C-1901_R0_Ope
Element: Shell-2

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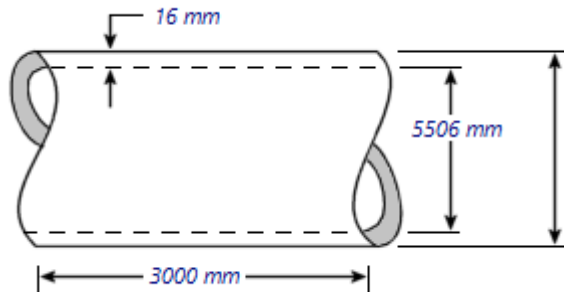
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	5538.00	3708.65	369.20	0.6697	0.0002867	28.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.6634)/(3*369.2) = 0.1035 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	0.348 MPa	Int. Design Temperature		160 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	5506.000 mm	Actual Stress		64.1 MPa
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		52194420.0 mm ²	SG of Contents		0.000
Element Volume		71443.3 ltr	Weight of Contents		0.0 N
Empty Weight		63287.0 N	Operating Weight		63287.0 N
			Comp. All. Ope. Stress		80.7 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.348 \cdot 2754.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.348)$$

$$= 6.9585 + 1.0000 = 7.9585 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (2754.0 + 0.6 \cdot 15.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (2753.0 + 0.6 \cdot 16.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.348 \cdot (2754.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 64.129 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.290 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 16.0, tr = 7.579, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.505, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -46 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.379 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.505	Temperature Difference	32 °C
UCS-66(c) red.	No	Computed Min. Temp.	-46 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 110 to Ring R1 Fr110 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	5538.00	2650.00	369.20	0.4785	0.0004084	40.83

$$MAEP = (4*B)/(3*(Do/t)) = (4*40.8308)/(3*369.2) = 0.1475 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.081	5538.00	2650.00	423.37	0.4785	0.0003285	32.84

$$MAEP = (4*B)/(3*(Do/t)) = (4*32.8408)/(3*423.3745) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	5538.00	3708.65	369.20	0.6697	0.0002867	28.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.6634)/(3*369.2) = 0.1035 \text{ MPa}$$

Cylindrical Shell From Ring R1 Fr110 to 130 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	5538.00	2205.00	369.20	0.3982	0.0004960	49.58

$$MAEP = (4*B)/(3*(Do/t)) = (4*49.5841)/(3*369.2) = 0.1791 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.100	5538.00	2205.00	457.68	0.3982	0.0003551	35.50

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.5013)/(3*457.6779) = 0.1034 \text{ MPa}$$

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Element: Shell-3

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Results for Maximum Stiffened Length

Slen

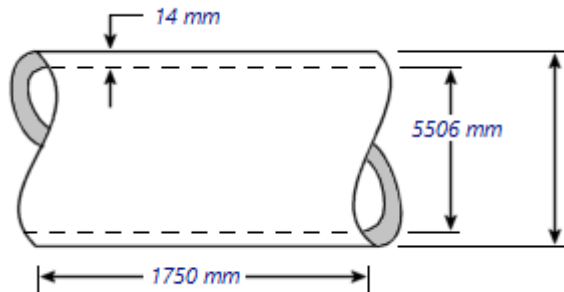
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	5538.00	3709.53	369.20	0.6698	0.0002866	28.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.6563)/(3*369.2) = 0.1035 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	0.348 MPa	Int. Design Temperature		160 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1750.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	5506.000 mm	Actual Stress		74.0 MPa
Spec. Min. Thk t _e , or s _a		14.0000 mm	Spec. Nominal Thk.		14.000 mm
Surface Area		30424752.0 mm ²	SG of Contents		0.000
Element Volume		41675.2 ltr	Weight of Contents		0.0 N
Empty Weight		32290.5 N	Operating Weight		32290.5 N
			Comp. All. Ope. Stress		70.0 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.348 \cdot 2754.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.348)$$

$$= 6.9585 + 1.0000 = 7.9585 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (2754.0 + 0.6 \cdot 13.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (2753.0 + 0.6 \cdot 14.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.348 \cdot (2754.0 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0)$$

$$= 73.963 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.254 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14.0, tr = 7.579, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.583, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.379 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.583	Temperature Difference	24 °C
UCS-66(c) red.	No	Computed Min. Temp.	-42 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-4

Cylindrical Shell From 130 to R1 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	5534.00	2205.00	425.69	0.3984	0.0003953	39.52

$$MAEP = (4*B)/(3*(Do/t)) = (4*39.518)/(3*425.6922) = 0.1238 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.095	5534.00	2205.00	457.54	0.3984	0.0003550	35.49

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.4904)/(3*457.5375) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	5534.00	2614.02	425.69	0.4724	0.0003304	33.03

$$MAEP = (4*B)/(3*(Do/t)) = (4*33.0335)/(3*425.6922) = 0.1035 \text{ MPa}$$

Cylindrical Shell From R1 to 150 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	5534.00	2563.00	425.69	0.4631	0.0003373	33.73

$$MAEP = (4*B)/(3*(Do/t)) = (4*33.726)/(3*425.6922) = 0.1056 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.888	5534.00	2563.00	429.40	0.4631	0.0003332	33.31

$$MAEP = (4*B)/(3*(Do/t)) = (4*33.3086)/(3*429.401) = 0.1034 \text{ MPa}$$

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FileName : C-1901_R0_Ope
Element: Shell-4

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Results for Maximum Stiffened Length

Slen

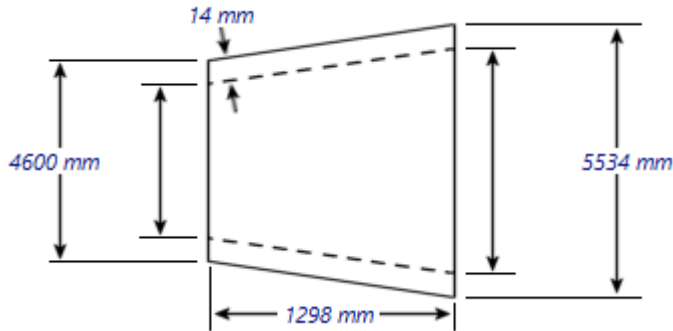
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	5534.00	2613.89	425.69	0.4723	0.0003304	33.04

$$MAEP = (4*B)/(3*(Do/t)) = (4*33.0352)/(3*425.6922) = 0.1035 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Cone 1

Conical Head/Section Design Information:					
Design Pressure	P	0.348 MPa	Int. Design Temperature		160 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1298.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Outside Diameter	Do	5534.000 mm	Actual Stress		78.6 MPa
Spec. Min. Thk t,e, or sa		14.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		21977188.0 mm²	SG of Contents		0.000
Element Volume		25946.0 ltr	Weight of Contents		0.0 N
Empty Weight		26635.1 N	Operating Weight		26635.1 N
			Comp. All. Ope. Stress		74.4 MPa
			To End Diameter		4600.000 mm
Line of Support	Neither End				

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot Do) / (2 \cdot \cos(a) \cdot (S \cdot E + 0.4 \cdot P))$$
 per Appendix 1-4 (e)

$$= (0.348 \cdot 5534.0) / (2 \cdot 0.941 \cdot (138.0 \cdot 1.0 + 0.4 \cdot 0.35))$$

$$= 7.4113 + 1.0000 = 8.4113 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D - 0.8 \cdot t \cdot \cos(a))$$
 per Appendix 1-4 (e)(5)

$$= (2 \cdot 138.0 \cdot 1.0 \cdot (13.0) \cdot 0.941) / (5534.0 - 0.8 \cdot (13.0) \cdot 0.941)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D - 0.8 \cdot t \cdot \cos(a))$$
 per Appendix 1-4 (e)

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 14.0 \cdot 0.941) / (5534.0 - 0.8 \cdot 14.0 \cdot 0.941)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (D - 0.8 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot t \cdot \cos(a))$$

$$= (0.348 \cdot (5534.0 - 0.8 \cdot (13.0) \cdot 0.941)) / (2 \cdot 1.0 \cdot (13.0) \cdot 0.941)$$

$$= 78.613 \text{ MPa}$$

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 0.349 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14.0, tr = 8.07, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.621$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.379 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.621	Temperature Difference	21 °C
UCS-66(c) red.	No	Computed Min. Temp.	-39 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Cone 1

Cylindrical Shell From 150 to 160 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	5532.24	2563.00	425.56	0.4633	0.0003374	33.73

$$MAEP = (4*B) / (3*(Do/t)) = (4*33.73) / (3*425.557) = 0.1057 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.885	5532.24	2563.00	429.34	0.4633	0.0003331	33.30

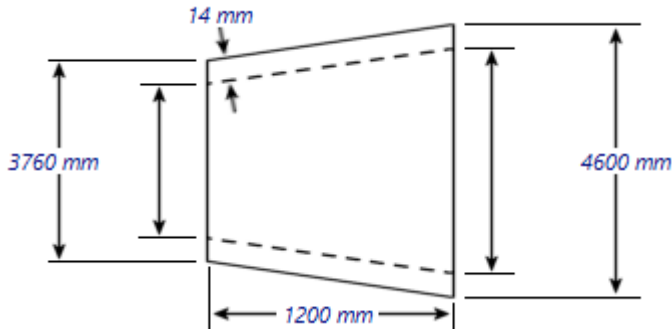
$$MAEP = (4*B) / (3*(Do/t)) = (4*33.3041) / (3*429.3419) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	5532.24	2614.95	425.56	0.4727	0.0003303	33.03

$$MAEP = (4*B) / (3*(Do/t)) = (4*33.0251) / (3*425.557) = 0.1035 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Cone 2

Conical Head/Section Design Information:					
Design Pressure	P	0.348 MPa	Int. Design Temperature		160 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1200.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Outside Diameter	Do	4600.000 mm	Actual Stress		65.1 MPa
Spec. Min. Thk t _e , or s _a		14.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		16712472.0 mm²	SG of Contents		0.000
Element Volume		16292.8 ltr	Weight of Contents		0.0 N
Empty Weight		20240.3 N	Operating Weight		20240.3 N
			Comp. All. Ope. Stress		89.2 MPa
			To End Diameter		3760.000 mm
Line of Support	Neither End				

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D_o) / (2 \cdot \cos(a) \cdot (S \cdot E + 0.4 \cdot P))$$
 per Appendix 1-4 (e)

$$= (0.348 \cdot 4600.0) / (2 \cdot 0.9439 \cdot (138.0 \cdot 1.0 + 0.4 \cdot 0.35))$$

$$= 6.1415 + 1.0000 = 7.1415 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D - 0.8 \cdot t \cdot \cos(a))$$
 per Appendix 1-4 (e)(5)

$$= (2 \cdot 138.0 \cdot 1.0 \cdot (13.0) \cdot 0.9439) / (4600.0 - 0.8 \cdot (13.0) \cdot 0.9439)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D - 0.8 \cdot t \cdot \cos(a))$$
 per Appendix 1-4 (e)

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 14.0 \cdot 0.9439) / (4600.0 - 0.8 \cdot 14.0 \cdot 0.9439)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (D - 0.8 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot t \cdot \cos(a))$$

$$= (0.348 \cdot (4600.0 - 0.8 \cdot (13.0) \cdot 0.9439)) / (2 \cdot 1.0 \cdot (13.0) \cdot 0.9439)$$

$$= 65.120 \text{ MPa}$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 0.427 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14.0, tr = 6.688, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr * E*/(tg - c) = 0.514$, Temp. Reduction = 31 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.379 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.514	Temperature Difference	30 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Cone 2

Cylindrical Shell From 160 to 170 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	4598.33	2563.00	353.72	0.5574	0.0003720	37.19

$$MAEP = (4*B)/(3*(Do/t)) = (4*37.1873)/(3*353.718) = 0.1402 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.599	4598.33	2563.00	396.45	0.5574	0.0003076	30.75

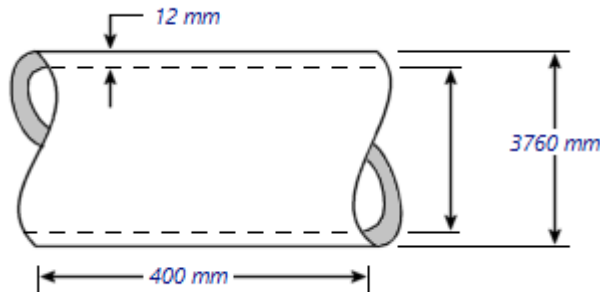
$$MAEP = (4*B)/(3*(Do/t)) = (4*30.7509)/(3*396.4495) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	4598.33	3423.03	353.72	0.7444	0.0002746	27.46

$$MAEP = (4*B)/(3*(Do/t)) = (4*27.4583)/(3*353.718) = 0.1035 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Port shell

Cylindrical Shell Design Information:					
Design Pressure	P	0.348 MPa	Int. Design Temperature		160 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		400.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Outside Diameter	Do	3760.000 mm	Actual Stress		59.4 MPa
Spec. Min. Thk t _e , or s _a		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		4724955.0 mm ²	SG of Contents		0.000
Element Volume		4385.7 ltr	Weight of Contents		0.0 N
Empty Weight		4295.5 N	Operating Weight		4295.5 N
			Comp. All. Ope. Stress		87.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (0.348 \cdot 1880.0) / (138.0 \cdot 1.0 + 0.4 \cdot 0.348)$$

$$= 4.7382 + 1.0000 = 5.7382 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 11.0) / (1880.0 - 0.4 \cdot 11.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (1880.0 - 0.4 \cdot 12.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

$$= (0.348 \cdot (1880.0 - 0.4 \cdot 11.0)) / (1.0 \cdot 11.0)$$

$$= 59.362 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.320 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12.0, tr = 5.159, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.469, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	0.379 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.469	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Port shell

Cylindrical Shell From 170 to R2 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	3760.00	2563.00	341.82	0.6816	0.0003170	31.69

$$MAEP = (4*B)/(3*(Do/t)) = (4*31.6924)/(3*341.8182) = 0.1236 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.253	3760.00	2563.00	366.72	0.6816	0.0002845	28.45

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.4454)/(3*366.7192) = 0.1034 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	3760.00	3039.48	341.82	0.8084	0.0002653	26.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*26.5262)/(3*341.8182) = 0.1035 \text{ MPa}$$

Cylindrical Shell From R2 to 180 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	3760.00	380.00	341.82	0.1011	0.0027169	110.09

$$MAEP = (4*B)/(3*(Do/t)) = (4*110.0943)/(3*341.8182) = 0.4294 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.543	3760.00	380.00	827.64	0.1011	0.0006421	64.20

$$MAEP = (4*B)/(3*(Do/t)) = (4*64.1987)/(3*827.639) = 0.1034 \text{ MPa}$$

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PV Elite 28 U2 Licensee: SL User
FileName : C-1901_R0_Ope
Element: Port shell

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Results for Maximum Stiffened Length

Slen

Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	3760.00	3039.48	341.82	0.8084	0.0002653	26.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*26.5262)/(3*341.8182) = 0.1035 \text{ MPa}$$

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Hydrostatic Test Pressure Results:

UG99(b) = 1.30 * M.A.W.P. * Sa/S 0.493 MPa
 UG99b(b)(35) = 1.30 * Design Pres * Sa/S 0.453 MPa
 UG99(c) = 1.30 * M.A.P. - Head(Hyd) 0.439 MPa
 UG100 = 1.10 * M.A.W.P. * Sa/S 0.417 MPa
 PED = max(1.43*DP, 1.25*DP*ratio) 0.498 MPa
 PED&UG99(b)(35) = max(1.43*DP, 1.3*DP*ratio) 0.498 MPa
 App 27-4 = M.A.W.P. 0.379 MPa

User Defined Hydrostatic Test Pressure at High Point 0.588 MPa

Horizontal Test performed per: User Hydro Pressure

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
BTM DISH	110.6	234.0	0.473	0.64
Shell-1	110.9	234.0	0.474	0.64
Shell-2	110.9	234.0	0.474	0.64
Shell-3	110.9	234.0	0.474	0.64
Shell-4	126.7	234.0	0.541	0.64
Cone 1	134.7	234.0	0.576	0.64
Cone 2	110.0	234.0	0.470	0.63
Port shell	97.7	234.0	0.418	0.63

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1 20"	Nozzle	138.00	138.00	1.000
N1 20"	Pad	138.00	138.00	1.000
N5_2"	Nozzle	138.00	138.00	1.000
N9 22"	Nozzle	138.00	138.00	1.000
N9 22"	Pad	138.00	138.00	1.000
M4 24"	Nozzle	138.00	138.00	1.000
M4 24"	Pad	138.00	138.00	1.000
N6A 2"	Nozzle	138.00	138.00	1.000
K4B 2"	Nozzle	138.00	138.00	1.000
K3B 2"	Nozzle	138.00	138.00	1.000
K2B 2"	Nozzle	138.00	138.00	1.000
K1 B2"	Nozzle	138.00	138.00	1.000
K1A 2"	Nozzle	138.00	138.00	1.000
K2A 2"	Nozzle	138.00	138.00	1.000
K3A 2"	Nozzle	138.00	138.00	1.000
K4A 2"	Nozzle	138.00	138.00	1.000
N2A 104"	Nozzle	138.00	138.00	1.000
N2B 104"	Nozzle	138.00	138.00	1.000
M3 24"	Nozzle	138.00	138.00	1.000
M3 24"	Pad	138.00	138.00	1.000
N6B 2"	Nozzle	138.00	138.00	1.000
K5A_2"	Nozzle	138.00	138.00	1.000
K6A_2"	Nozzle	138.00	138.00	1.000
N10A_2"	Nozzle	138.00	138.00	1.000
N10B_2"	Nozzle	138.00	138.00	1.000
M2 24"	Nozzle	138.00	138.00	1.000
M2 24"	Pad	138.00	138.00	1.000
M1 24"	Nozzle	138.00	138.00	1.000
M1 24"	Pad	138.00	138.00	1.000
N3 20"	Nozzle	138.00	138.00	1.000
N6 C 2"	Nozzle	138.00	138.00	1.000
K6B 2"	Nozzle	138.00	138.00	1.000
K5B 2"	Nozzle	138.00	138.00	1.000



N7 2"	Nozzle	138.00	138.00	1.000
N6 D 2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
Ring R1 Fr90	138.00	138.00	1.000
Ring R2 Fr90	138.00	138.00	1.000
Ring R1 Fr100	138.00	138.00	1.000
Ring R1 Fr110	138.00	138.00	1.000
R1	138.00	138.00	1.000
R2	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
BTM DISH	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Cone 1	138.00	138.00	1.000
Cone 2	138.00	138.00	1.000
Port shell	138.00	138.00	1.000
N4 Flange	88.90	88.90	1.000
N2A Flange	138.00	138.00	1.000
N2B Flange	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1 20"	16.70	234.00	0.071
N5 2"	1.36	225.00	0.006
N9 22"	11.60	234.00	0.050
M4 24"	10.18	234.00	0.043
N6A 2"	1.36	225.00	0.006
K4B 2"	1.36	225.00	0.006
K3B 2"	1.36	225.00	0.006
K2B 2"	1.36	225.00	0.006
K1 B2"	1.36	225.00	0.006
K1A 2"	1.36	225.00	0.006
K2A 2"	1.36	225.00	0.006
K3A 2"	1.36	225.00	0.006
K4A 2"	1.36	225.00	0.006
N2A 104"	28.03	234.00	0.120
N2B 104"	28.03	234.00	0.120
M3 24"	14.37	234.00	0.061
N6B 2"	1.36	225.00	0.006
K5A 2"	1.36	225.00	0.006
K6A 2"	1.36	225.00	0.006
N10A 2"	1.36	225.00	0.006
N10B 2"	1.36	225.00	0.006
M2 24"	10.18	234.00	0.043
M1 24"	10.18	234.00	0.043
N3 20"	9.94	234.00	0.042
N6 C 2"	1.36	225.00	0.006
K6B 2"	1.36	225.00	0.006
K5B 2"	1.36	225.00	0.006
N7 2"	1.36	225.00	0.006
N6 D 2"	1.36	225.00	0.006



Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.493 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.453 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	0.490 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.417 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.498 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.498 MPa
App 27-4	= M.A.W.P.	0.379 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 0.379 • 1.0
 = 0.493 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
BTM DISH	118.1	234.0	0.505	0.64
Shell-1	115.9	234.0	0.495	0.63
Shell-2	110.5	234.0	0.472	0.60
Shell-3	105.1	234.0	0.449	0.57
Shell-4	115.0	234.0	0.491	0.54
Cone 1	118.3	234.0	0.506	0.52
Cone 2	95.6	234.0	0.409	0.51
Port shell	85.2	234.0	0.364	0.50

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1 20"	Nozzle	138.00	138.00	1.000
N1 20"	Pad	138.00	138.00	1.000
N5_2"	Nozzle	138.00	138.00	1.000
N9 22"	Nozzle	138.00	138.00	1.000
N9 22"	Pad	138.00	138.00	1.000
M4 24"	Nozzle	138.00	138.00	1.000
M4 24"	Pad	138.00	138.00	1.000
N6A 2"	Nozzle	138.00	138.00	1.000
K4B 2"	Nozzle	138.00	138.00	1.000
K3B 2"	Nozzle	138.00	138.00	1.000
K2B 2"	Nozzle	138.00	138.00	1.000
K1 B2"	Nozzle	138.00	138.00	1.000
K1A 2"	Nozzle	138.00	138.00	1.000
K2A 2"	Nozzle	138.00	138.00	1.000
K3A 2"	Nozzle	138.00	138.00	1.000
K4A 2"	Nozzle	138.00	138.00	1.000
N2A 104"	Nozzle	138.00	138.00	1.000
N2B 104"	Nozzle	138.00	138.00	1.000
M3 24"	Nozzle	138.00	138.00	1.000
M3 24"	Pad	138.00	138.00	1.000
N6B 2"	Nozzle	138.00	138.00	1.000
K5A_2"	Nozzle	138.00	138.00	1.000
K6A_2"	Nozzle	138.00	138.00	1.000
N10A_2"	Nozzle	138.00	138.00	1.000
N10B_2"	Nozzle	138.00	138.00	1.000
M2 24"	Nozzle	138.00	138.00	1.000
M2 24"	Pad	138.00	138.00	1.000
M1 24"	Nozzle	138.00	138.00	1.000
M1 24"	Pad	138.00	138.00	1.000
N3 20"	Nozzle	138.00	138.00	1.000

N6 C 2"	Nozzle	138.00	138.00	1.000
K6B 2"	Nozzle	138.00	138.00	1.000
K5B 2"	Nozzle	138.00	138.00	1.000
N7 2"	Nozzle	138.00	138.00	1.000
N6 D 2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
Ring R1 Fr90	138.00	138.00	1.000
Ring R2 Fr90	138.00	138.00	1.000
Ring R1 Fr100	138.00	138.00	1.000
Ring R1 Fr110	138.00	138.00	1.000
R1	138.00	138.00	1.000
R2	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
BTM DISH	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Cone 1	138.00	138.00	1.000
Cone 2	138.00	138.00	1.000
Port shell	138.00	138.00	1.000
N4 Flange	88.90	88.90	1.000
N2A Flange	138.00	138.00	1.000
N2B Flange	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1 20"	16.73	234.00	0.071
N5 2"	1.36	225.00	0.006
N9 22"	11.37	234.00	0.049
M4 24"	9.97	234.00	0.043
N6A 2"	1.33	225.00	0.006
K4B 2"	1.27	225.00	0.006
K3B 2"	1.27	225.00	0.006
K2B 2"	1.27	225.00	0.006
K1 B2"	1.27	225.00	0.006
K1A 2"	1.27	225.00	0.006
K2A 2"	1.27	225.00	0.006
K3A 2"	1.27	225.00	0.006
K4A 2"	1.27	225.00	0.006
N2A 104"	26.17	234.00	0.112
N2B 104"	26.17	234.00	0.112
M3 24"	12.76	234.00	0.055
N6B 2"	1.21	225.00	0.005
K5A 2"	1.21	225.00	0.005
K6A 2"	1.21	225.00	0.005
N10A 2"	1.21	225.00	0.005
N10B 2"	1.21	225.00	0.005
M2 24"	8.57	234.00	0.037
M1 24"	8.57	234.00	0.037
N3 20"	8.37	234.00	0.036
N6 C 2"	1.14	225.00	0.005
K6B 2"	1.14	225.00	0.005
K5B 2"	1.14	225.00	0.005

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N7 2"	1.14	225.00	0.005
N6 D 2"	1.11	225.00	0.005

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

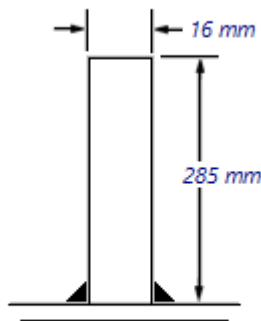
MDMT Calculations for: Ring R1 Fr90, Shell to Ring Junction, Curve: B

Govrn. thk, $t_g = 16.0$, $t_r = 7.748$, $c = 1.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.517$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: Ring R1 Fr90, SA-516 70 , Bar Ring: 285.0 x 16.0 mm

Material treatment options:



Effective Length of Shell: 317 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	4755.608	7.5000	35667.062
Ring :	4560.000	157.5000	718200.188
Total:	9315.608		753867.250

Centroid of Ring plus Shell: 81 mm

	Inertia	Distance	A*Dist ²
Shell:	89167.570	73.4252	25638702.000
Ring :	30865486.000	-76.5749	26738510.000
Total:	30954652.000		52377212.000

Available Moment of Inertia, Ring + Shell: 0.833E+08 mm⁴

Required Stress in Ring plus Shell Breq 24.25 MPa
 Required Strain in Ring plus Shell Areq 0.0002420

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5538.0^2 \cdot 1679.418(15.0 + 4560.0005/1679.418)0.000242) / 10.9$$

$$= 20258088 \text{ mm}^4$$

MDMT Calculations for: Ring R2 Fr90, Shell to Ring Junction, Curve: B

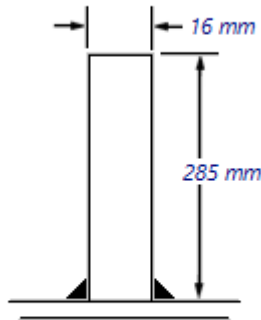
Govrn. thk, $t_g = 16.0$, $t_r = 7.748$, $c = 1.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.517$, Temp. Reduction = 30 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: Ring R2 Fr90, SA-516 70 , Bar Ring: 285.0 x 16.0 mm

Material treatment options:



Effective Length of Shell: 317 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	4755.608	7.5000	35667.062
Ring :	4560.000	157.5000	718200.188
Total:	9315.608		753867.250

Centroid of Ring plus Shell: 81 mm

	Inertia	Distance	A*Dist ²
Shell:	89167.570	73.4252	25638702.000
Ring :	30865486.000	-76.5749	26738510.000
Total:	30954652.000		52377212.000

Available Moment of Inertia, Ring + Shell: 0.833E+08 mm⁴

Required Stress in Ring plus Shell Breq 25.85 MPa
 Required Strain in Ring plus Shell Areq 0.0002580

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5538.0^2 \cdot 2825.0005(15.0 + 4560.0005/2825.0005)0.000258) / 10.9$$

$$= 34071744 \text{ mm}^4$$

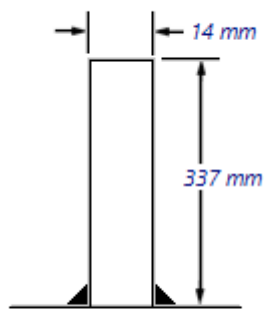
MDMT Calculations for: Ring R1 Fr100, Shell to Ring Junction, Curve: B

Govern. thk, $t_g = 14.0$, $c = 1.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.519$, Temp. Reduction = 30 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: Ring R1 Fr100, SA-516 70 , Bar Ring: 337.0 x 14.0 mm

Material treatment options:



Effective Length of Shell: 317 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	4755.608	7.5000	35667.062
Ring :	4718.002	183.5001	865753.812
Total:	9473.610		901420.875

Centroid of Ring plus Shell: 95 mm

	Inertia	Distance	A*Dist ²
Shell:	89167.570	87.6507	36535672.000
Ring :	44651568.000	-88.3494	36826884.000
Total:	44740736.000		73362560.000

Available Moment of Inertia, Ring + Shell: 0.118E+09 mm⁴

Required Stress in Ring plus Shell Breq 25.77 MPa
 Required Strain in Ring plus Shell Areq 0.0002570

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5538.0^2 \cdot 2825.0(15.0 + 4718.0024/2825.0)0.000257) / 10.9$$

$$= 34053920 \text{ mm}^4$$

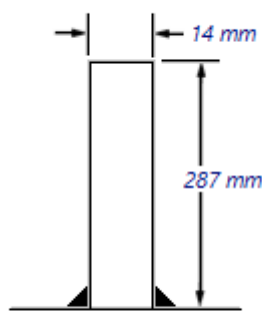
MDMT Calculations for: Ring R1 Fr110, Shell to Ring Junction, Curve: B

Govern. thk, tg = 14.0, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.505, Temp. Reduction = 32 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: Ring R1 Fr110, SA-516 70 , Bar Ring: 287.0 x 14.0 mm

Material treatment options:



Effective Length of Shell: 317 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	4755.608	7.5000	35667.062
Ring :	4018.002	158.5001	636853.500
Total:	8773.609		672520.562

Centroid of Ring plus Shell: 77 mm

	Inertia	Distance	A*Dist ²
Shell:	89167.570	69.1527	22741748.000
Ring :	27579892.000	-81.8474	26916580.000
Total:	27669060.000		49658328.000

Available Moment of Inertia, Ring + Shell: 0.773E+08 mm⁴

Required Stress in Ring plus Shell Breq 25.79 MPa
 Required Strain in Ring plus Shell Areq 0.0002570

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (5538.0^2 \cdot 2427.4988(15.0 + 4018.0017/2427.4988)0.000257) / 10.9$$

$$= 29236124 \text{ mm}^4$$

MDMT Calculations for: R1, Shell to Ring Junction, Curve: B

Govrn. thk, tg = 14.0, tr = 7.579, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.583, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations for: R2, Shell to Ring Junction, Curve: B

Govrn. thk, tg = 12.0, tr = 5.159, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.469, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
------	----	----------------------	------------------------	--------------------------	----------	-----------------



10	20	16700	...	...	No Calc	No Calc
20	30	16700	...	...	No Calc	No Calc
30	40	16700	...	...	No Calc	No Calc
40	50	16700	...	...	No Calc	No Calc
50	60	16700	...	...	No Calc	No Calc
60	70	16700	...	...	No Calc	No Calc
70	80	16700	...	...	No Calc	No Calc
80	90	No Calc	5538	15	0.00037619	37.6104
90	Ring	708.835	5538	15	0.001783	101.093
Ring	Ring	2650	5538	15	0.0004084	40.8308
Ring	100	3000	5538	15	0.00035827	35.8193
100	Ring	3000	5538	15	0.00035827	35.8193
Ring	110	2650	5538	15	0.0004084	40.8308
110	Ring	2650	5538	15	0.0004084	40.8308
Ring	130	2205	5538	15	0.00049595	49.5841
130	Ring	2205	5534	13	0.00039527	39.518
Ring	150	2563	5534	13	0.00033734	33.726
150	160	2563	5532.24	13	0.00033738	33.73
160	170	2563	4598.33	13	0.00037196	37.1873
170	Ring	2563	3760	11	0.000317	31.6924
Ring	180	379.999	3760	11	0.0027169	110.094
180	190	No Calc	...	239	No Calc	No Calc
190	200	No Calc	...	59	No Calc	No Calc
200	210	No Calc	...	59	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	...	No Calc	...	No Calc
50	60	...	No Calc	...	No Calc
60	70	...	No Calc	...	No Calc
70	80	...	No Calc	...	No Calc
80	90	16	15.3386	0.10342	0.11319
90	Ring	16	8.40974	0.10342	0.36509
Ring	Ring	16	14.0806	0.10342	0.14746
Ring	100	16	14.8176	0.10342	0.12936
100	Ring	16	14.8176	0.10342	0.12936
Ring	110	16	14.0806	0.10342	0.14746
110	Ring	16	14.0806	0.10342	0.14746
Ring	130	16	13.1002	0.10342	0.17907
130	Ring	14	13.0952	0.10342	0.12378
Ring	150	14	13.8877	0.10342	0.10563
150	160	14	13.8854	0.10342	0.10568
160	170	14	12.5988	0.10342	0.14018
170	Ring	12	11.2531	0.10342	0.12362
Ring	180	12	5.54304	0.10342	0.42945
180	190	240	234.137	0.10342	No Calc
190	200	60	55.0164	0.10342	No Calc
200	210	60	55.0164	0.10342	No Calc

Minimum

0

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	16700	No Calc	No Calc	No Calc
20	30	16700	No Calc	No Calc	No Calc
30	40	16700	No Calc	No Calc	No Calc
40	50	16700	No Calc	No Calc	No Calc



50	60	16700	No Calc	No Calc	No Calc
60	70	16700	No Calc	No Calc	No Calc
70	80	16700	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc
90	Ring	708.835	3709.59	No Calc	No Calc
Ring	Ring	2650	3708.65	20258088	83331816
Ring	100	3000	3709.85	34071744	83331816
100	Ring	3000	3709.85	No Calc	No Calc
Ring	110	2650	3708.65	34053920	118103224
110	Ring	2650	3708.65	No Calc	No Calc
Ring	130	2205	3709.53	29236124	77327344
130	Ring	2205	2614.02	No Calc	No Calc
Ring	150	2563	2613.89	No Calc	No Calc
150	160	2563	2614.95	No Calc	No Calc
160	170	2563	3423.03	No Calc	No Calc
170	Ring	2563	3039.48	No Calc	No Calc
Ring	180	379.999	3039.48	No Calc	No Calc
180	190	No Calc	No Calc	No Calc	No Calc
190	200	No Calc	No Calc	No Calc	No Calc
200	210	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	7933.43	...	7799.67	...	...
20	30	7261.87	...	6900.06	...	...
30	40	7261.87	...	6859.7	...	...
40	50	7261.87	...	6859.7	...	...
50	60	7261.87	...	6859.7	...	...
60	70	6456.13	...	6053.77	...	...
70	80	2152.07	...	2084.99	...	...
80	90	5799.84	23044.4	5509.85	23069.1	...
90	100	6453.91	71443.3	6051.61	71495.2	...
100	110	6453.91	71443.3	6051.61	71495.2	...
110	130	6453.91	71443.3	6051.61	71495.2	...
130	150	3292.94	41675.2	3058.32	41705.5	...
150	160	2716.21	25946	2546.99	25967.8	...
160	170	2064.07	16292.8	1935.57	16309.4	...
170	180	438.051	4385.72	401.655	4390.42	...
180	190	2772.24	3182.7	2745.86	3183.06	...
190	200	621.034	614.616	613.825	615.048	...
200	210	621.245	614.616	614.037	615.048	...
Total		83276	330085.97	78998	330340.97	0

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Wght	1900	...	150	...	BRACING. TAILIN
20	Plat	1564.35	2316.52	2700	-2316.52	PLAT:[1 OF 1]
40	Plat	1600.15	2317.15	1000	-2317.15	PLAT:[1 OF 1]
50	Plat	1637.15	2317.15	2000	-2317.15	PLAT:[1 OF 1]
60	Plat	1543.65	2318.09	1500	-2318.09	PLAT:[1 OF 1]
70	Plat	5897.1	0.32242E-04	1000	...	BTM PLATFORM
70	Wght	4347	...	200.001	-4103	N1 pipe wt
80	Liqd	22956.4	...	-687.5	...	Liquid: 40
80	Insl	363.26	...	-662.5	...	Ins: 40
80	Lini	894.658	...	-662.5	...	clad
80	Nozl	1235.62	...	-1528.49	...	N1 20"
80	Nozl	10.4111	-1000	-4342.19	...	N5_2"
80	Wght	50	...	-1000	...	VORTEX BREAKER
90	Liqd	71244.2	...	1500	...	Liquid 50
90	Insl	404.221	...	1500	...	Ins: 50
90	Lini	1250.48	...	1500	...	clad
90	Ring	646.501	...	200.002	...	Ring R1 Fr90
90	Ring	646.501	...	2850	...	Ring R2 Fr90
90	Nozl	296.036	2024.44	1500	-2024.44	N9 22"
90	Nozl	462.209	-2862.99	1500	...	M4 24"
90	Nozl	8.66123	-2856.19	700.003	...	N6A 2"
90	Wght	2834	2918.94	1500	-2918.94	N9 pipe wt
100	Plat	3036.67	...	1000	-2317.46	PLT-2
100	Liqd	24246.8	...	510.5	...	Liquid 60
100	Insl	404.221	...	1500	...	Ins: 60
100	Lini	1250.48	...	1500	...	clad
100	Ring	674.875	...	2850	...	Ring R1 Fr100
100	Nozl	10.3008	-2856.19	2700	...	K4B 2"
100	Nozl	10.3008	-2856.19	2500	...	K3B 2"
100	Nozl	10.6765	-2862.99	2300	...	K2B 2"
100	Nozl	10.3008	-2856.19	2000	...	K1 B2"
100	Nozl	10.3008	2856.19	2000	...	K1A 2"
100	Nozl	10.3008	2856.19	1800	...	K2A 2"
100	Nozl	10.3008	2856.19	1600	...	K3A 2"
100	Nozl	10.3008	2856.19	1000	...	K4A 2"
100	Nozl	1605.35	2300.43	0.0014539	1930.29	N2A 104"



100	Nozl	1605.56	-2123.44	0.0014539	2123.44
100	Tray	13389.4	...	2500	...
100	Wght	2850	...	0.0014539	...
100	Wght	2850	...	0.0014539	...
100	Wght	1000.02	...	2500	...
110	Plat	3036.67	-2317.46	2000	...
110	Pack	17839.6	...	1500	...
110	Insl	404.221	...	1500	...
110	Lini	1250.48	...	1500	...
110	Ring	569.854	...	2500	...
110	Nozl	430.19	2862.99	900.001	...
110	Nozl	10.3008	2856.19	0.0011613	...
110	Nozl	11.3894	2019.63	2500	-2019.63
110	Nozl	11.3894	...	2500	2856.19
110	Nozl	10.4111	...	500.001	2856.19
110	Nozl	10.4111	...	1000	2856.19
110	Wght	290	...	2900	...
110	Wght	500.009	...	500.001	...
110	Wght	1000.02	...	0.0011613	...
110	Wght	2700	...	1500	...
110	Wght	10000.2	...	1500	...
110	Wght	4022.7	...	2990	...
130	Insl	235.626	...	875	...
130	Lini	729.446	...	875	...
130	Ring	852.266	...	1705	...
130	Nozl	441.178	-2862.99	0.0029078	...
130	Nozl	452.002	...	1650	-2862.99
130	Nozl	200.884	2862.99	900.003	...
130	Nozl	11.3894	-2187.97	200.003	1835.92
130	Nozl	11.3894	...	0.0029078	2856.19
130	Nozl	11.3894	2812.8	0.0029078	-495.972
130	Nozl	10.3559	2019.63	1000	-2019.63
130	Wght	2836	...	0.0029078	...
130	Wght	7585	...	250.003	...
130	Wght	290	...	0.0029078	...
130	Wght	5375	4119	900.003	...
130	Wght	600.011	...	1650	...
130	Wght	120	...	900.003	...
130	Wght	2700	...	0.0029078	...
130	Wght	1900	...	1150	...
130	Wght	3550	...	900.003	...
150	Plat	3035.07	-2316.91	850	...
150	Insl	194.359	...	649	...
150	Lini	494.917	...	649	...
150	Nozl	10.3559	2551.86	700.003	...
150	Wght	2060	...	200.003	...
150	Wght	1300.02	651.181	0.0023256	-3693.03
160	Insl	147.986	...	600	...
160	Lini	376.945	...	600	...
170	Plat	11835.7	0.33043E-04	...	...
170	Insl	36.7167	...	200	...
170	Lini	113.096	...	200	...
170	Ring	415.177	...	20.0011	...
180	Insl	12.2663	...	188	...
180	Lini	132.818	...	141	...
110	Pliq	10709.95	...	1500.000	...

N2B 104"
 CHIMNEY
 N2A INSERT PLT
 N2B INSERT PLT
 CT BEAM
 PLT-3
 Structure Packi
 Ins: 70
 clad
 Ring R1 Fr110
 M3 24"
 N6B 2"
 K5A_2"
 K6A_2"
 N10A_2"
 N10B_2"
 SUPPORT RING
 MW DAVIT
 OTHER MISC
 CLT 2
 10T MISC
 Support Grid
 Ins: 90
 Clad
 R1
 M2 24"
 M1 24"
 N3 20"
 N6 C 2"
 K6B 2"
 K5B 2"
 N7_2"
 BED LIMITER
 DISTRIBUTOR
 SUPPORT RING
 N3 pipe wt
 SUPPORT RING
 N3 INSERT PLT
 CLT-1
 TRUNN + INSERT
 N3 Feed Pipe
 PLT-5
 Ins: 100
 Clading
 N6 D 2"
 DEMISTER
 PIPE DAVIT
 ins
 clad
 TOP
 ins
 clad
 R2
 ins
 clad

Note: The individual tray liquid weights are listed below, but these weights are included in the tray weights above.

From	Tliq	Weight	X Offset	Y Offset
100	Tliq	5952.70	0.000	2500.000

Total Weight of Each Detail Type:

Platforms	33186.5
Packing	17839.6
Packing Liquid	10710.0



Liquid	118447.4
Insulation	2202.9
Cladding & Overlay	6493.3
Stiffeners	3805.2
Nozzles	6939.7
Trays	7146.7
Tray Bolting Weight	290.0
Liquid on Trays	5952.7
Weights	62660.0

Sum of the Detail Weights 275673.9 kgm

Note:

Because some nozzles were attached to a head or flange, their weights include the attached head or flange.

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	82034.2	82034.2	82034.2	82034.2	82034.2	82034.2
Cladding	6493.3	6493.3	6493.3	6493.3	6493.3	6493.3
Stif. Rings	3805.2	3805.2	3805.2	3805.2	3805.2	3805.2
Nozzles	6939.7	6939.7	6939.7	6939.7	6939.7	6939.7
Tray Bar	290.0	290.0	290.0	290.0	290.0	290.0
Wld Weights	29050.3	29050.3	29050.3	29050.3	29050.3	29050.3
Platforms	...	...	33186.5	33186.5	33186.5	33186.5
Packing	...	...	17839.6	17839.6	17839.6	17839.6
Insulation	...	...	...	2202.9	2202.9	2202.9
Trays	...	...	7146.7	7146.7	7146.7	7146.7
Ope Weights	...	...	...	...	...	33609.7
Ope. Liquid	...	...	...	...	...	118447.4
Tray Liquid	...	...	...	...	...	5952.7
Packing Liq	...	...	...	...	...	10710.0
Test Liquid	...	329884.5	...	...	...	...
Totals	128612.6	458497.1	186785.5	188988.3	188988.3	357708.1

Field Installation Options:

Insulation installed before lifting.

Lining installed before lifting.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	128612.6 kgm
Shop Test Wt.	- Fabricated weight + water (full)	458497.1 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	186785.5 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	188988.3 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	188988.3 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	357708.1 kgm
Field Test Wt.	- Empty weight + test medium (full)	500225.6 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	149473.5 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	318646.8 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	99987.8 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	17445264	17256770
20	30	47017388	46678096
30	40	52222696	51883404
40	50	52222696	51883404
50	60	52222696	51883404
60	70	52213272	51911680



70	80	17404426	17303894
80	90	34212504	33727016
90	100	52194420	51892828
100	110	52194420	51892828
110	130	52194420	51892828
130	150	30424752	30270812
150	160	21977188	21829810
160	170	16712472	16577057
170	180	4724955	4694795
180	190	4944416	3403727
190	200	2064175	929610
200	210	2064648	929610

Total 564456768.0 556841536.0 mm²
Total 564.5 556.8 m²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	9833.43	9833.43	9833.43	...	9833.43
20	30	8826.23	8826.23	8826.23	35549940	8826.23
30	40	7261.87	7261.87	7261.87	...	7261.87
40	50	8862.03	8862.03	8862.03	36373304	8862.03
50	60	8899.03	8899.03	8899.03	37214372	8899.03
60	70	7999.78	7999.78	7999.78	35103184	7999.78
70	80	8049.17	12396.2	8049.17	174968128	12396.2
80	90	7095.88	31310.2	31384.2	...	8353.79
90	100	8513.82	84246.7	81568.2	87029896	13002.5
100	110	26595.9	59450.1	99650.2	5192009	29250.6
110	130	42874.3	59261.7	98089.1	676888	48551.8
130	150	10893.8	31204.9	53508.7	12431224	31204.9
150	160	7061.66	9810.93	33681.1	47097796	9810.93
160	170	2064.07	2589	18871.8	...	2589
170	180	12688.9	12838.8	17221.8	...	12838.8
180	190	2772.24	2917.33	6098.09	...	2917.33

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	222598	357708	499805
20	30	212765	347875	489971
30	40	203938	339048	481145
40	50	196677	331787	473883
50	60	187815	322925	465021
60	70	178916	314026	456122
70	80	170916	306026	448122
80	90	158520	293630	440073
90	100	150166	262319	408689
100	110	137163	178073	327121
110	130	107913	118623	227471
130	150	59360.9	59360.9	129382
150	160	28156	28156	75872.8
160	170	18345.1	18345.1	42191.7
170	180	15756.1	15756.1	23319.9
180	190	2917.33	2917.33	6098.09

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

		Cumulative	Cumulative	Cumulative
--	--	------------	------------	------------



From	To	Empty Mom. N-mm	Oper. Mom. N-mm	Hydro. Mom. N-mm
10	20	471636736	471636736	471636736
20	30	471636736	471636736	471636736
30	40	436086752	436086752	436086752
40	50	436086752	436086752	436086752
50	60	399713504	399713504	399713504
60	70	362499072	362499072	362499072
70	80	327395904	327395904	327395904
80	90	152427808	152427808	152427808
90	100	152427808	152427808	152427808
100	110	65397916	65397916	65397916
110	130	60205904	60205904	60205904
130	150	59529016	59529016	59529016
150	160	47097796	47097796	47097796
160	170	...	...	...
170	180	...	...	...
180	190	...	...	...



Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N1 20"	4.48	5.10	160	300	GR 1.1	1.658	4.484	4.484	4.484
N5_2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
N9 22"	1.54	1.96	160	150	GR 1.1	...	...	...	...
M4 24"	1.54	1.96	160	150	GR 1.1	...	...	...	...
N6A 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K4B 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K3B 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K2B 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K1 B2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K1A 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K2A 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K3A 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K4A 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
M3 24"	1.54	1.96	160	150	GR 1.1	...	...	...	...
N6B 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K5A_2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K6A_2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
N10A_2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
N10B_2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
M2 24"	1.54	1.96	160	150	GR 1.1	...	...	...	...
M1 24"	1.54	1.96	160	150	GR 1.1	...	...	...	...
N3 20"	4.48	5.10	160	300	GR 1.1	1.658	4.484	4.484	4.484
N6 C 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K6B 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K5B 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
N7_2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
N6 D 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
Min Rating	1.540	1.960, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N1 20"	551.688	0.50	49497480	37500	0.379	4.484	Pass
N3 20"	551.688	0.50	49497480	37500	0.379	4.484	Pass

Evaluating Flange on Nozzle [N1 20"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(49497480) + 4 \cdot \text{abs}(37500) \cdot 551.688 \leq \pi \cdot 551.688^3 \cdot [(4.484 - 0.379) + 500 \cdot 4.484]$$

$$= 0.87475E+09 \leq 0.33493E+10 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1 20" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.484, 0.500 \cdot 4.484 + 4.484 - 874746368.0 / (\pi \cdot 551.688^3))$$

$$= \min(4.484, 5.069)$$

$$= 4.484 \text{ MPa}$$

Evaluating Flange on Nozzle [N3 20"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$



$$= 16(49497480) + 4 \cdot \text{abs}(37500) 551.688 \leq \pi \cdot 551.688^3 [(4.484 - 0.379) + .500 \cdot 4.484]$$
$$= 0.87475\text{E}+09 \leq 0.33493\text{E}+10 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3 20" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.484, 0.500 \cdot 4.484 + 4.484 - 874746368.0 / (\pi \cdot 551.688^3))$$

$$= \min(4.484, 5.069)$$

$$= 4.484 \text{ MPa}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 2.5518 Hz.

The Natural Frequency for the Vessel (Ope...) is 2.16029 Hz.

The Natural Frequency for the Vessel (Filled) is 1.68014 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	98 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.181 \cdot 0.904) / (1 + 1.7 \cdot 3.4 \cdot 0.181)))$
 $= \min(0.85, 0.88)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	2.160 Hz
Natural Frequency of Vessel (Empty)	2.552 Hz
Natural Frequency of Vessel (Test)	1.680 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.577
Structure Height to Diameter ratio	5.623
Height to top of Structure	30671.174 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Zg = 274320.0$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 500.0 + 300.0 = 800.0$ mm
 $= 2.625$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
Because z (2.625 ft.) < 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:



K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.896^2)$$

$$= 16.0 \text{ psf } [0.766] \text{ kPa}$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.577 \cdot 95.635$$

$$= 750.5 \text{ lbs. } [3338.4] \text{ N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-1	800.0	0.000	0.000	0.000	0.849	1.000	0.766
Skirt-2	2650.0	0.000	0.000	0.000	0.849	1.000	0.766
Skirt-3	5500.0	0.000	0.000	0.000	0.883	1.000	0.766
Skirt-4	8500.0	0.000	0.000	0.000	0.967	1.000	0.766
Skirt-5	11500.0	0.000	0.000	0.000	1.031	1.000	0.766
Skirt-6	14500.0	0.000	0.000	0.000	1.082	1.000	0.766
Skirt-7	16500.0	0.000	0.000	0.000	1.112	1.000	0.766
BTM DISH	17025.0	0.000	0.000	0.000	1.120	1.000	0.766
Shell-1	18550.0	0.000	0.000	0.000	1.140	1.000	0.766
Shell-2	21550.0	0.000	0.000	0.000	1.177	1.000	0.766
Shell-3	24550.0	0.000	0.000	0.000	1.209	1.000	0.766
Shell-4	26925.0	0.000	0.000	0.000	1.233	1.000	0.766
Cone 1	28428.9	0.000	0.000	0.000	1.247	1.000	0.766
Cone 2	29677.8	0.000	0.000	0.000	1.259	1.000	0.766
Port shell	30498.0	0.000	0.000	0.000	1.266	1.000	0.766
N4 Flange	30836.2	0.000	0.000	0.000	1.269	1.000	0.766

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm
- Lc - Total length of conical section(s) of vessel, mm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, N
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
- Z - Maximum amplitude of vibration at top of vessel, mm
- DL - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.000003 (High); 0.000003 < 0.000004 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 2498.0/30671.174 = 0.081

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[17.92/100.63^2] \cdot 10^4 = 17.702$ [Geometry Violation]

Compute the vibration possibility. If $V_p > 0.000004$ no chance. $[V_p]$:
 $= W / (L \cdot Dr^2)$
 $= 3477917 / (30671.17 \cdot 5333.292^2)$
 $= 0.39866E-05$

Since V_p is > 0.000004 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PLAT:[1 OF 1]	3604416.25	4000.00	0.77	2761.15	1.20
PLAT:[1 OF 1]	3750720.00	8000.00	0.77	2873.22	1.20
PLAT:[1 OF 1]	4238400.00	12000.00	0.77	3246.81	1.20
PLAT:[1 OF 1]	3019200.00	14500.00	0.77	2312.85	1.20
BTM PLATFORM	4819200.00	17000.00	0.77	3691.73	1.20
PLT-2	5550719.50	21050.00	0.77	4252.11	1.20
PLT-3	5550719.50	25050.00	0.77	4252.11	1.20
PLT-5	5550719.50	28650.00	0.77	4252.11	1.20
TOP	9390720.00	30298.00	0.77	7193.72	1.20

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
BRACING. TAIL	0.00	450.00	0.77	0.00
N1 pipe wt	0.00	16200.00	0.77	0.00
VORTEX BREAKER	0.00	16000.00	0.77	0.00
N9 pipe wt	0.00	18550.00	0.77	0.00
N2A INSERT PL	0.00	20050.00	0.77	0.00
N2B INSERT PL	0.00	20050.00	0.77	0.00
CT BEAM	0.00	22550.00	0.77	0.00
SUPPORT RING	0.00	25950.00	0.77	0.00
MW DAVIT	0.00	23550.00	0.77	0.00
OTHER MISC	0.00	23050.00	0.77	0.00
CLT 2	0.00	24550.00	0.77	0.00
10T MISC	0.00	24550.00	0.77	0.00
Support Grid	0.00	26040.00	0.77	0.00
BED LIMITER	0.00	26050.00	0.77	0.00
DISTRIBUTOR	0.00	26300.00	0.77	0.00
SUPPORT RING	0.00	26050.00	0.77	0.00
N3 pipe wt	0.00	26950.00	0.77	0.00
SUPPORT RING	0.00	27700.00	0.77	0.00
N3 INSERT PLT	0.00	26950.00	0.77	0.00
CLT-1	0.00	26050.00	0.77	0.00
TRUNN + INSERT	0.00	27200.00	0.77	0.00
N3 Feed Pipe	0.00	26950.00	0.77	0.00
DEMISTER	0.00	28000.00	0.77	0.00
PIPE DAVIT	0.00	27800.00	0.77	0.00

The Natural Frequency for the Vessel (Ope...) is 2.16029 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	800	8884.8	8884800	0.76606	3672.24
20	30	2650	8868.8	23945760	0.76606	14481.8
30	40	5500	8865.6	26596800	0.76606	10992.9
40	50	8500	8865.6	26596800	0.76606	13967.6
50	60	11500	8865.6	26596800	0.76606	14719.7
60	70	14500	8864	26592002	0.76606	13535.1
70	80	16500	8864	8864000	0.76606	7847.61
80	90	17025	8988.8	449440	0.76606	185.761



90	100	18550	8988.8	26966400	0.76606	11145.7
100	110	21550	8988.8	26966400	0.76606	15714.5
110	130	24550	8988.8	26966400	0.76606	15918.3
130	150	26925	8982.4	15719199	0.76606	6497.02
150	160	28428.9	8232.39	10685640	0.76606	9130.25
160	170	29677.8	6813.33	8176002	0.76606	3379.29
170	180	30498	6144	2457600	0.76606	8876.97
180	190	30836.2	6105.6	1648512	0.76606	681.359

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104\end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= $0.020(100.62720.75)$
= 0.635 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/2.16)
= 0.463

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of (1.7 • 0.635, 0.463) per 12.8.2
= 0.463

Seismic Response Coefficient per equation 12.8-2 [Cs]:
= $Sds / (R / I)$
= $0.157 / (1.25 / 1.25)$
= 0.157

Maximum value of C_s per equation 12.8-3 [Cs]:
= $Sd1 / (T(R / I_e))$
= $0.104 / (0.463(1.25 / 1.25))$
= 0.225

Minimum value of C_s per equation 15.4-1 [Cs]:
= $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
= $\max((0.044 \cdot 0.1568 \cdot 1.25), 0.030)$
= 0.030

Total Base Shear [V]:
= $C_s \cdot W$ (Equation 12.8-1):
= $0.1568 \cdot 3507685$
= 550005.000 N

Vertical load per 12.4-4, [Ev]:
= $0.2 \cdot Sds \cdot W$
= $0.2 \cdot 0.157 \cdot 3507685$



= 110001.0 N

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 2.16029 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	500	96426.5	387.888	403.725	110001
20	30	2350	86550	1636.34	1703.15	106977
30	40	5200	71209.9	2979.09	3100.72	104263
40	50	8200	86901	5732.95	5967.02	102030
50	60	11200	87263.8	7863.06	8184.1	99304.5
60	70	14200	78445.9	8961.85	9327.76	96567.9
70	80	16200	121557	15842.9	10707.2	94107.8
80	90	16725	307028	41312.7	9745.02	90295.8
90	100	18250	826123	121296	12758.4	80667.4
100	110	21250	582967	99665	46407.1	54760.2
110	130	24250	581120	113375	85372.9	36478.4
130	150	26625	305995	65545.6	23816.6	18254.4
150	160	28149	96206	21787.4	16322.3	8658.43
160	170	29398	25387.7	6004.56	4982.56	5641.41
170	180	30198	125897	30586.7	31464	4845.25
180	190	30536.2	28607.3	7027.99	6951.15	897.125



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	500	150746	550005	2.437E+09	11.99E+09
20	30	2350	147074	549617	2.288E+09	11.44E+09
30	40	5200	132592	547981	1910744960	9.956E+09
40	50	8200	121599	545002	1529304320	8.316E+09
50	60	11200	107631	539269	1185319040	6.689E+09
60	70	14200	92911.8	531406	884382400	5.082E+09
70	80	16200	79376.7	522444	625844928	3.501E+09
80	90	16725	71529.1	506601	550361472	2.986E+09
90	100	18250	71343.3	465288	546788224	2.962E+09
100	110	21250	60197.6	343992	349396800	1747318272
110	130	24250	44483.2	244327	192311920	864481152
130	150	26625	28564.9	130952	82695392	301333792
150	160	28149	22067.9	65406.6	38373768	129450200
160	170	29398	12937.6	43619.2	15646001	58663748
170	180	30198	9558.33	37614.7	2142946	9903598
180	190	30536.2	681.359	7027.99	92019.9	949154

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	150746	0.00079357	0.0031415	0.62177E-05
20	30	147074	0.0196	0.051265	0.28752E-04
30	40	132592	0.1036	0.17307	0.51607E-04
40	50	121599	0.25782	0.35607	0.69625E-04
50	60	107631	0.46616	0.58653	0.83349E-04
60	70	92911.8	0.71628	0.85448	0.94638E-04
70	80	79376.7	0.90218	0.95062	0.97579E-04
80	90	71529.1	0.95306	0.9555	0.97721E-04
90	100	71343.3	1.10507	1.25981	0.00010464
100	110	60197.6	1.41863	1.58055	0.00010878
110	130	44483.2	1.74472	1.91043	0.00011086
130	150	28564.9	2.0076	2.10505	0.00011148
150	160	22067.9	2.17745	2.24994	0.00011174
160	170	12937.6	2.31702	2.38414	0.00011189
170	180	9558.33	2.40652	2.4289	0.00011189
180	190	681.359	2.44401	2.45911	0.00011189

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	344.556	2153.48
20	30	343.936	2149.6
30	40	343.812	2148.82
40	50	343.812	2148.82
50	60	343.812	2148.82
60	70	343.75	2148.44
70	80	343.75	2148.44
80	90	348.589	2178.68
90	100	348.589	2178.68
100	110	348.589	2178.68
110	130	348.589	2178.68
130	150	348.341	2177.13
150	160	319.255	1995.35
160	170	264.224	1651.4
170	180	238.267	1489.17
180	190	236.778	1479.86

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 153.356 Actual deflection : 2.459 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	520529	503269	714858112	691404160
20	30	347019	329730	477425600	453800960
30	40	312320	295022	429838880	406181184
40	50	312320	295022	429838880	406181184
50	60	312320	295022	429838880	406181184
60	70	277666	260361	382352736	358651136
70	80	277666	260361	382352736	358651136
80	90	277567	260268	382075264	358391520
90	100	277567	260268	382075264	358391520
100	110	277567	260268	382075264	358391520
110	130	277567	260268	382075264	358391520
130	150	242782	225483	334191776	310490912
150	160	257977	239597	354996640	329830784
160	170	213663	198448	244132144	226852112
170	180	141297	129558	131973896	121073280
180	190	141297	129558	131973896	121073280



Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	98.2782	173.877	-109.248	-111.691
20	30	98.2782	173.877	-94.8099	-96.4939
30	40	98.2782	173.877	-90.6283	-92.0719
40	50	98.2782	173.877	-90.6283	-92.0719
50	60	98.2782	173.877	-90.6283	-92.0719
60	70	98.2782	173.877	-80.6498	-81.2092
70	80	99.1182	166.317	-80.6498	-81.2092
80	90	165.597	280.795	-80.6789	-81.2385
90	100	165.597	280.795	-80.6789	-81.2385
100	110	165.597	280.795	-80.6789	-81.2385
110	130	165.597	280.795	-80.6789	-81.2385
130	150	165.597	280.795	-69.9723	-70.4576
150	160	165.597	280.795	-74.3633	-74.879
160	170	165.597	280.795	-89.1868	-89.8054
170	180	165.597	280.795	-87.1418	-87.7462
180	190	106.678	210.596	-81.0673	-82.8749



Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	...	...	...
50	60	...	...	...
60	70	...	...	...
70	80	...	...	...
80	90	31.8904	-9.57141	45.1472
90	100	31.8904	-9.57141	45.1472
100	110	31.8904	-9.57141	45.1472
110	130	31.8904	-9.57141	45.1472
130	150	36.8073	-11.032	52.1081
150	160	36.792	-11.0336	52.0865
160	170	30.5395	-9.17597	43.2348
170	180	29.5071	-8.86344	41.7733
180	190	...	...	...

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-4.3376	-6.97039	-9.73932	0.68192	0.68192
20	30	-6.32805	-10.3465	-14.5727	1.03897	1.03897
30	40	-6.77911	-11.2703	-15.9937	1.07328	1.07328
40	50	-6.53772	-11.0289	-15.7523	1.07328	1.07328
50	60	-6.24313	-10.7343	-15.4577	0.98376	0.98376
60	70	-6.73909	-11.8282	-17.1805	1.01041	1.01041
70	80	-6.43777	-11.5269	-16.8791	0.91256	0.91256
80	90	-5.97299	-6.60083	-5.97299	0.42517	0.42517
90	100	-5.65822	-6.28606	-5.65822	0.42517	0.42517
100	110	-5.16828	-5.79613	-5.16828	0.18242	0.18242
110	130	-4.06613	-4.46968	-4.06613	0.16794	0.16794
130	150	-2.58176	-2.58176	-2.58176	0.19166	0.19166
150	160	-1.15244	-1.15244	-1.15244	0.14275	0.14275
160	170	-0.90657	-0.90657	-0.90657	...	...
170	180	-1.19265	-1.19265	-1.19265	...	...
180	190	-0.22083	-0.22083	-0.22083	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	0.68192	3.52418	17.3337	1.76209	...
20	30	1.03897	5.04123	25.1977	2.52061	...
30	40	1.07328	4.70266	24.5035	2.35133	...
40	50	1.07328	3.76387	20.4668	1.88194	...
50	60	0.98376	2.91727	16.4624	1.45863	...
60	70	1.01041	2.46507	14.1657	1.23253	...
70	80	0.91256	1.74444	9.7578	0.87222	...



80	90	0.42517	1.53515	8.32911	0.76758	...
90	100	0.42517	1.52518	8.26131	0.76259	...
100	110	0.18242	0.97459	4.87388	0.4873	...
110	130	0.16794	0.53643	2.41134	0.26821	...
130	150	0.19166	0.26625	0.9702	0.13313	...
150	160	0.14275	0.11631	0.39235	0.058153	...
160	170	...	0.068948	0.25852	0.034474	...
170	180	...	0.017694	0.081772	0.008847	...
180	190	...	0.00075979	0.007837	0.0003799	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	9.14385
20	30	...	...	...	13.321
30	40	...	...	...	13.0481
40	50	...	...	...	11.0275
50	60	...	...	...	9.04039
60	70	...	...	...	8.04718
70	80	...	...	...	5.92914
80	90	...	...	...	5.25486
90	100	...	...	...	5.22236
100	110	...	...	...	3.36645
110	130	...	...	...	1.75818
130	150	...	...	...	0.80974
150	160	...	...	...	0.38218
160	170	...	...	...	0.26064
170	180	...	...	...	0.08288
180	190	...	...	...	0.0077513

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	...	0.21859
20	30	...	0.32447
30	40	...	0.35344
40	50	...	0.34587
50	60	...	0.33663
60	70	...	0.37093
70	80	...	0.36148
80	90	...	0.34696
90	100	...	0.30997
100	110	...	0.21042
110	130	...	0.14017
130	150	...	0.080964
150	160	...	0.03614
160	170	...	0.02843
170	180	...	0.037402
180	190	...	0.0069251

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...



60	70	...	...	...	...
70	80	...	...	...	...
80	90	...	...	...	...
90	100	...	...	...	...
100	110	...	...	...	...
110	130	...	...	...	...
130	150	...	...	...	...
150	160	...	...	...	...
160	170	...	...	...	...
170	180	...	...	...	...
180	190	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	130	...
130	150	...
150	160	...
160	170	...
170	180	...
180	190	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-8.54	109.25		0.0782
20		98.28	-12.41	94.81		0.1309
30		98.28	-12.56	90.63		0.1385
40		98.28	-11.37	90.63		0.1255
50		98.28	-10.14	90.63		0.1119
60		98.28	-10.21	80.65		0.1267
70		99.12	-9.09	80.65		0.1128
80		165.60	-7.93	80.68		0.0983
90		165.60	-7.61	80.68		0.0943
100		165.60	-6.33	80.68		0.0784
110		165.60	-4.77	80.68		0.0591
130		165.60	-3.04	69.97		0.0434
150		165.60	-1.41	74.36		0.0190
160		165.60	-0.98	89.19		0.0109
170		165.60	-1.21	87.14		0.0139

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	5.71	98.28	-13.94	109.25	0.0581	0.1276
20	8.36	98.28	-20.36	94.81	0.0850	0.2148
30	7.70	98.28	-20.55	90.63	0.0783	0.2267
40	5.91	98.28	-18.29	90.63	0.0601	0.2018



50	4.12	98.28	-15.93	90.63	0.0419	0.1758
60	2.69	98.28	-15.43	80.65	0.0274	0.1913
70	0.77	99.12	-12.92	80.65	0.0077	0.1602
80	0.05	165.60	-11.31	80.68	0.0003	0.1401
90	0.30	165.60	-11.00	80.68	0.0018	0.1363
100		165.60	-8.51	80.68		0.1054
110		165.60	-5.85	80.68		0.0725
130		165.60	-3.50	69.97		0.0501
150		165.60	-1.64	74.36		0.0221
160		165.60	-1.14	89.19		0.0128
170		165.60	-1.24	87.14		0.0142

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-11.18	109.25		0.1023
20		98.28	-16.43	94.81		0.1733
30		98.28	-17.05	90.63		0.1881
40		98.28	-15.87	90.63		0.1751
50		98.28	-14.64	90.63		0.1615
60		98.28	-15.30	80.65		0.1898
70		99.12	-14.18	80.65		0.1759
80		165.60	-8.56	80.68		0.1061
90		165.60	-8.24	80.68		0.1021
100		165.60	-6.95	80.68		0.0862
110		165.60	-5.17	80.68		0.0641
130		165.60	-3.04	69.97		0.0434
150		165.60	-1.41	74.36		0.0190
160		165.60	-0.98	89.19		0.0109
170		165.60	-1.21	87.14		0.0139

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	11.26	98.28	-24.77	109.25	0.1146	0.2267
20	16.21	98.28	-36.26	94.81	0.1650	0.3824
30	14.66	98.28	-36.49	90.63	0.1492	0.4027
40	10.86	98.28	-32.22	90.63	0.1105	0.3556
50	7.05	98.28	-27.84	90.63	0.0717	0.3072
60	3.72	98.28	-26.63	80.65	0.0378	0.3302
70		99.12	-21.84	80.65		0.2707
80	2.50	165.60	-15.01	80.68	0.0151	0.1860
90	2.71	165.60	-14.66	80.68	0.0164	0.1817
100		165.60	-10.64	80.68		0.1319
110		165.60	-6.91	80.68		0.0856
130		165.60	-3.66	69.97		0.0523
150		165.60	-1.65	74.36		0.0222
160		165.60	-1.14	89.19		0.0127
170		165.60	-1.24	87.14		0.0142

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		173.88	-12.18	111.69		0.1091
20		173.88	-18.13	96.49		0.1879
30		173.88	-19.42	92.07		0.2109
40		173.88	-18.71	92.07		0.2032
50		173.88	-17.90	92.07		0.1944
60		173.88	-19.42	81.21		0.2392
70		166.32	-18.66	81.21		0.2298
80		280.79	-7.17	81.24		0.0882
90		280.79	-6.85	81.24		0.0843
100		280.79	-5.84	81.24		0.0719
110		280.79	-4.50	81.24		0.0554
130		280.79	-2.91	70.46		0.0413
150		280.79	-1.35	74.88		0.0181
160		280.79	-0.94	89.81		0.0105



170 280.79 -1.20 87.75 0.0137

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		173.88	-10.42	111.69		0.0933
20		173.88	-15.61	96.49		0.1618
30		173.88	-17.07	92.07		0.1854
40		173.88	-16.83	92.07		0.1827
50		173.88	-16.44	92.07		0.1786
60		173.88	-18.19	81.21		0.2240
70		166.32	-17.79	81.21		0.2191
80		280.79	-6.40	81.24		0.0788
90		280.79	-6.08	81.24		0.0749
100		280.79	-5.35	81.24		0.0659
110		280.79	-4.23	81.24		0.0521
130		280.79	-2.77	70.46		0.0394
150		280.79	-1.30	74.88		0.0173
160		280.79	-0.91	89.81		0.0101
170		280.79	-1.19	87.75		0.0136

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-11.18	109.25		0.1023
20		98.28	-16.43	94.81		0.1733
30		98.28	-17.05	90.63		0.1881
40		98.28	-15.87	90.63		0.1751
50		98.28	-14.64	90.63		0.1615
60		98.28	-15.30	80.65		0.1898
70		99.12	-14.18	80.65		0.1759
80	27.25	165.60		80.68	0.1646	
90	27.55	165.60		80.68	0.1664	
100	27.25	165.60		80.68	0.1646	
110	28.13	165.60		80.68	0.1698	
130	34.68	165.60		69.97	0.2094	
150	35.90	165.60		69.99	0.2168	
160	29.70	165.60		84.21	0.1794	
170	28.33	165.60		87.14	0.1711	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	11.26	98.28	-24.77	109.25	0.1146	0.2267
20	16.21	98.28	-36.26	94.81	0.1650	0.3824
30	14.66	98.28	-36.49	90.63	0.1492	0.4027
40	10.86	98.28	-32.22	90.63	0.1105	0.3556
50	7.05	98.28	-27.84	90.63	0.0717	0.3072
60	3.72	98.28	-26.63	80.65	0.0378	0.3302
70		99.12	-21.84	80.65		0.2707
80	34.39	165.60		80.68	0.2077	
90	34.60	165.60		80.68	0.2089	
100	31.36	165.60		80.68	0.1894	
110	30.14	165.60		80.68	0.1820	
130	35.47	165.60		69.97	0.2142	
150	36.21	165.60		69.99	0.2187	
160	29.92	165.60		84.21	0.1807	
170	28.43	165.60		87.14	0.1717	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-11.18	111.69		0.1001
20		98.28	-16.43	96.49		0.1702
30		98.28	-17.05	92.07		0.1851
40		98.28	-15.87	92.07		0.1723



50	98.28	-14.64	92.07	0.1590
60	98.28	-15.30	81.21	0.1884
70	99.12	-14.18	81.21	0.1747
80	165.60	-18.13	81.24	0.2232
90	165.60	-17.81	81.24	0.2192
100	165.60	-16.52	81.24	0.2034
110	165.60	-14.75	81.24	0.1815
130	165.60	-14.07	70.46	0.1997
150	165.60	-12.45	70.48	0.1766
160	165.60	-10.15	84.79	0.1197
170	165.60	-10.07	87.75	0.1148

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	11.26	98.28	-24.77	111.69	0.1146	0.2217
20	16.21	98.28	-36.26	96.49	0.1650	0.3758
30	14.66	98.28	-36.49	92.07	0.1492	0.3964
40	10.86	98.28	-32.22	92.07	0.1105	0.3500
50	7.05	98.28	-27.84	92.07	0.0717	0.3024
60	3.72	98.28	-26.63	81.21	0.0378	0.3280
70		99.12	-21.84	81.21		0.2689
80		165.60	-24.58	81.24		0.3026
90		165.60	-24.23	81.24		0.2983
100		165.60	-20.21	81.24		0.2488
110		165.60	-16.48	81.24		0.2029
130		165.60	-14.69	70.46		0.2086
150		165.60	-12.69	70.48		0.1800
160		165.60	-10.31	84.79		0.1216
170		165.60	-10.10	87.75		0.1151

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		173.88	-12.18	111.69		0.1091
20		173.88	-18.13	96.49		0.1879
30		173.88	-19.42	92.07		0.2109
40		173.88	-18.71	92.07		0.2032
50		173.88	-17.90	92.07		0.1944
60		173.88	-19.42	81.21		0.2392
70		166.32	-18.66	81.21		0.2298
80	40.37	280.79		81.24	0.1438	
90	40.68	280.79		81.24	0.1449	
100	40.65	280.79		81.24	0.1448	
110	41.52	280.79		81.24	0.1479	
130	49.85	280.79		70.46	0.1775	
150	51.13	280.79		74.88	0.1821	
160	42.36	280.79		89.81	0.1509	
170	40.59	280.79		87.75	0.1446	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		173.88	-10.42	111.69		0.0933
20		173.88	-15.61	96.49		0.1618
30		173.88	-17.07	92.07		0.1854
40		173.88	-16.83	92.07		0.1827
50		173.88	-16.44	92.07		0.1786
60		173.88	-18.19	81.21		0.2240
70		166.32	-17.79	81.21		0.2191
80	39.60	280.79		81.24	0.1410	
90	39.91	280.79		81.24	0.1421	
100	40.16	280.79		81.24	0.1430	
110	41.25	280.79		81.24	0.1469	
130	49.72	280.79		70.46	0.1771	
150	51.08	280.79		74.88	0.1819	
160	42.33	280.79		89.81	0.1507	



170 40.58 280.79 87.75 0.1445

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-5.02	109.25		0.0459
20		98.28	-7.37	94.81		0.0777
30		98.28	-7.85	90.63		0.0866
40		98.28	-7.61	90.63		0.0840
50		98.28	-7.23	90.63		0.0797
60		98.28	-7.75	80.65		0.0961
70		99.12	-7.35	80.65		0.0911
80	26.34	165.60		80.68	0.1591	
90	26.66	165.60		80.68	0.1610	
100	26.90	165.60		80.68	0.1625	
110	27.99	165.60		80.68	0.1690	
130	34.42	165.60		69.97	0.2078	
150	35.78	165.60		69.99	0.2161	
160	29.63	165.60		84.21	0.1789	
170	28.31	165.60		87.14	0.1710	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-6.74	109.25		0.0617
20		98.28	-9.83	94.81		0.1037
30		98.28	-10.65	90.63		0.1175
40		98.28	-10.42	90.63		0.1150
50		98.28	-10.14	90.63		0.1119
60		98.28	-11.09	80.65		0.1375
70		99.12	-10.81	80.65		0.1340
80	25.70	165.60		80.68	0.1552	
90	26.00	165.60		80.68	0.1570	
100	26.46	165.60		80.68	0.1598	
110	27.70	165.60		80.68	0.1673	
130	34.41	165.60		69.97	0.2078	
150	35.72	165.60		69.99	0.2157	
160	29.70	165.60		84.21	0.1793	
170	28.41	165.60		87.14	0.1716	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-7.65	109.25		0.0700
20		98.28	-11.39	94.81		0.1201
30		98.28	-12.34	90.63		0.1362
40		98.28	-12.10	90.63		0.1335
50		98.28	-11.72	90.63		0.1293
60		98.28	-12.84	80.65		0.1592
70		99.12	-12.44	80.65		0.1542
80	25.71	165.60		80.68	0.1553	
90	26.03	165.60		80.68	0.1572	
100	26.28	165.60		80.68	0.1587	
110	27.59	165.60		80.68	0.1666	
130	34.42	165.60		69.97	0.2078	
150	35.78	165.60		69.99	0.2161	
160	29.63	165.60		84.21	0.1789	
170	28.31	165.60		87.14	0.1710	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-5.02	109.25		0.0459
20		98.28	-7.37	94.81		0.0777
30		98.28	-7.85	90.63		0.0866
40		98.28	-7.61	90.63		0.0840



50		98.28	-7.23	90.63	0.0797
60		98.28	-7.75	80.65	0.0961
70		99.12	-7.35	80.65	0.0911
80	26.34	165.60		80.68	0.1591
90	26.66	165.60		80.68	0.1610
100	26.90	165.60		80.68	0.1625
110	27.99	165.60		80.68	0.1690
130	34.42	165.60		69.97	0.2078
150	35.78	165.60		69.99	0.2161
160	29.63	165.60		84.21	0.1789
170	28.31	165.60		87.14	0.1710

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-7.65	109.25		0.0700
20		98.28	-11.39	94.81		0.1201
30		98.28	-12.34	90.63		0.1362
40		98.28	-12.10	90.63		0.1335
50		98.28	-11.72	90.63		0.1293
60		98.28	-12.84	80.65		0.1592
70		99.12	-12.44	80.65		0.1542
80		165.60	-7.03	80.68		0.0871
90		165.60	-6.71	80.68		0.0832
100		165.60	-5.98	80.68		0.0741
110		165.60	-4.64	80.68		0.0575
130		165.60	-2.77	69.97		0.0396
150		165.60	-1.30	74.36		0.0174
160		165.60	-0.91	89.19		0.0102
170		165.60	-1.19	87.14		0.0137

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		81.90	-7.43	91.04		0.0817
20		81.90	-11.06	79.01		0.1400
30		81.90	-11.99	75.52		0.1588
40		81.90	-11.76	75.52		0.1557
50		81.90	-11.38	75.52		0.1507
60		81.90	-12.47	67.21		0.1855
70		82.60	-12.08	67.21		0.1797
80	26.06	138.00		67.23	0.1889	
90	26.34	138.00		67.23	0.1909	
100	26.49	138.00		67.23	0.1919	
110	27.73	138.00		67.23	0.2009	
130	34.50	138.00		58.31	0.2500	
150	35.82	138.00		58.33	0.2596	
160	29.66	138.00		70.18	0.2149	
170	28.35	138.00		72.62	0.2055	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		81.90	-7.43	93.08		0.0799
20		81.90	-11.06	80.41		0.1376
30		81.90	-11.99	76.73		0.1563
40		81.90	-11.76	76.73		0.1532
50		81.90	-11.38	76.73		0.1483
60		81.90	-12.47	67.67		0.1842
70		82.60	-12.08	67.67		0.1785
80		138.00	-16.25	67.70		0.2400
90		138.00	-15.97	67.70		0.2359
100		138.00	-15.34	67.70		0.2266
110		138.00	-14.07	67.70		0.2078
130		138.00	-13.72	58.71		0.2337
150		138.00	-12.29	58.73		0.2093
160		138.00	-10.05	70.66		0.1423

170 138.00 -10.02 73.12 0.1370

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-7.65	109.25		0.0700
20		98.28	-11.39	94.81		0.1201
30		98.28	-12.34	90.63		0.1362
40		98.28	-12.10	90.63		0.1335
50		98.28	-11.72	90.63		0.1293
60		98.28	-12.84	80.65		0.1592
70		99.12	-12.44	80.65		0.1542
80	25.71	165.60		80.68	0.1553	
90	26.03	165.60		80.68	0.1572	
100	26.28	165.60		80.68	0.1587	
110	27.59	165.60		80.68	0.1666	
130	34.42	165.60		69.97	0.2078	
150	35.78	165.60		69.99	0.2161	
160	29.63	165.60		84.21	0.1789	
170	28.31	165.60		87.14	0.1710	

Absolute Maximum of the all of the Stress Ratio's 0.4027

Governing Element: Skirt-3

Governing Load Case 4 : NP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-1	...	-6.8	0.7	17.3	11.3	+ Tens
Skirt-1	...	-6.8	-0.7	-17.3	-24.8	- Comp
Skirt-2	...	-10.0	1.0	25.2	16.2	+ Tens
Skirt-2	...	-10.0	-1.0	-25.2	-36.3	- Comp
Skirt-3	...	-10.9	1.1	24.5	14.7	+ Tens
Skirt-3	...	-10.9	-1.1	-24.5	-36.5	- Comp
Skirt-4	...	-10.7	1.1	20.5	10.9	+ Tens
Skirt-4	...	-10.7	-1.1	-20.5	-32.2	- Comp
Skirt-5	...	-10.4	1.0	16.5	7.0	+ Tens
Skirt-5	...	-10.4	-1.0	-16.5	-27.8	- Comp
Skirt-6	...	-11.5	1.0	14.2	3.7	+ Tens
Skirt-6	...	-11.5	-1.0	-14.2	-26.6	- Comp
Skirt-7	...	-11.2	0.9	9.8	...	+ Tens
Skirt-7	...	-11.2	-0.9	-9.8	-21.8	- Comp
BTM DISH	...	-6.3	0.4	8.3	2.5	+ Tens
BTM DISH	...	-6.3	-0.4	-8.3	-15.0	- Comp
Shell-1	...	-6.0	0.4	8.3	2.7	+ Tens
Shell-1	...	-6.0	-0.4	-8.3	-14.7	- Comp
Shell-2	...	-5.6	0.2	4.9	...	+ Tens
Shell-2	...	-5.6	-0.2	-4.9	-10.6	- Comp
Shell-3	...	-4.3	0.2	2.4	...	+ Tens
Shell-3	...	-4.3	-0.2	-2.4	-6.9	- Comp
Shell-4	...	-2.5	0.2	1.0	...	+ Tens
Shell-4	...	-2.5	-0.2	-1.0	-3.7	- Comp
Cone 1	...	-1.1	0.1	0.4	...	+ Tens
Cone 1	...	-1.1	-0.1	-0.4	-1.7	- Comp



Cone 2	...	-0.9	...	0.3	...	+ Tens
Cone 2	...	-0.9	...	-0.3	-1.1	- Comp
Port shell	...	-1.2	...	0.1	...	+ Tens
Port shell	...	-1.2	...	-0.1	-1.2	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-7

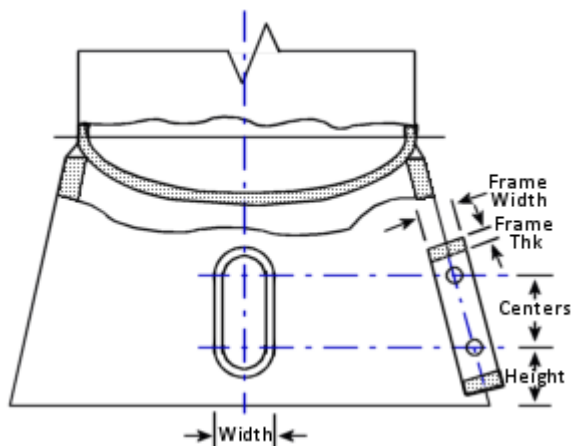
Total vessel (empty) weight	Fg	1840770.875 N
Head Weight below section 2-2	DeltaFg	81917.266 N
Weight of vessel contents (liquid)	Ff	1219866.750 N
Vessel Gross Weight	Fg + Ff	3060637.750 N
Skirt Mean Diameter	Dz	5525.000 mm
Skirt Analysis Thickness (Corroded)	ez	15.000 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	0.000 N
Moment at section 1-1	M1	2986037760.000 N-mm
Applied Moment - Bottom of Skirt	Mb	3500768768.000 N-mm
Weight at Bottom of Skirt	W	3000887.750 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	5523.000 mm
Head Corroded Thickness	eb	15.000 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 1377.5 \text{ mm} \\
 K &= D_i / (2 * h_i) = 5508.0 / (2 * 1377.5) = 2.0 \\
 R &= D_i (0.44 * K + 0.02) = 5508.0 * (0.44 * 2.0 + 0.02) = 4955.4399 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 5508.0 * ((0.5 / 2.0) - 0.08) = 936.8602 \text{ mm}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	993453670400.000 mm ⁴
Moment of Inertia	Iyy =	993453670400.000 mm ⁴
Section Modulus	W =	359621568.000 mm ³
Section Metal Area	A =	260359.516 mm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned}
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(0.99345E+12, 0.99345E+12) / 260359.516)} \\
 &= 1953.382 \text{ mm}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:



$$\begin{aligned} &= r_g / L \\ &= 1953.382 / 1000.0 \\ &= 1.953 \end{aligned}$$

Correct the Moment [M4]:

top of skirt is the critical end
Controlling Moment [M4] = 0.29860E+10 N-mm

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$\begin{aligned} &= -M / Z - W / A \\ &= -0.29860E+10 / .14385E+10 - 3000888 / 260359.51562 \\ &= -13.602 \text{ MPa} \end{aligned}$$

Upwind Side:

$$\begin{aligned} &= +(M) / Z - W / A \\ &= +.29860E+10 / .14385E+10 - 3000888 / 260359.51562 \\ &= -9.452 \text{ MPa} \end{aligned}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$\begin{aligned} &= F_1 - F_g - F_f + 4 \cdot M_1 / D_z \\ &= 0.0 - 1840771 - 1219867 + 4 \cdot 0.29860E+10 / 5525.0 \\ &= -899676.8 \text{ N} \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned} &= F_1 - F_g - F_f - 4 \cdot M_1 / D_z \\ &= 0.0 - 1840771 - 1219867 - 4 \cdot 0.29860E+10 / 5525.0 \\ &= -5221598.5 \text{ N} \end{aligned}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$\begin{aligned} &= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (-0.9E+06 + 81917.266 + 1219867) / (\pi \cdot 5523.0 \cdot 15.0) + 0.348 \cdot 5523.0 / (4 \cdot 15.0) \\ &= 33.59 \text{ MPa} \end{aligned}$$

Stress [σ_{1qm}]:

$$\begin{aligned} &= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (-0.5E+07 + 81917.266 + 1219867) / (\pi \cdot 5523.0 \cdot 15.0) + 0.348 \cdot 5523.0 / (4 \cdot 15.0) \\ &= 16.98 \text{ MPa} \end{aligned}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$\begin{aligned} &= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (1219867 + 81917.266) / (\pi \cdot 5523.0 \cdot 15.0) + 0.348 \cdot 5523.0 / (4 \cdot 15.0) \\ &= 37.05 \text{ MPa} \end{aligned}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$\begin{aligned} &= F_{zp} / (\pi \cdot D_z \cdot e_z) \\ &= -899676.812 / (\pi \cdot 5525.0 \cdot 15.0) \\ &= -3.46 \text{ MPa} \end{aligned}$$

Stress [σ_{3qm}]:

$$\begin{aligned} &= F_{zq} / (\pi \cdot D_z \cdot e_z) \\ &= -5221598.500 / (\pi \cdot 5525.0 \cdot 15.0) \\ &= -20.06 \text{ MPa} \end{aligned}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:

$$\begin{aligned} &= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b) \\ &= 1 - 0.5(5523.0 + 15.0 - 5525.0 + 15.0) / (936.86 + 15.0) \\ &= 0.9853 \end{aligned}$$



$\gamma = \arccos(\cos(\gamma)) = \arccos(0.985) = 9.839 \text{ deg}$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(15.0^2 + 15.0^2 + 2 \cdot 15.0 \cdot 15.0 \cdot 0.985)}$$

$$= 14.945 \text{ mm}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 14.94 \cdot -899676.8 = -13.4509 \times 10^6 \text{ N-mm}$$

$$M_q = a \cdot F_{zq} = 14.94 \cdot -5221598.5 = -78.0671 \times 10^6 \text{ N-mm}$$

Correction factor [C]:

$$= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2$$

$$= 0.63 - 0.057 \left(\frac{15.0}{15.0} \right)^2$$

$$= 0.573$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.573 \cdot 6 \cdot -0.1E+08 / (\pi \cdot 5523.0 \cdot 15.0^2)$$

$$= -11.84 \text{ MPa}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.573 \cdot 6 \cdot -0.8E+08 / (\pi \cdot 5523.0 \cdot 15.0^2)$$

$$= -68.73 \text{ MPa}$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.573 \cdot 6 \cdot -0.1E+08 / (\pi \cdot 5525.0 \cdot 15.0^2)$$

$$= -11.84 \text{ MPa}$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.573 \cdot 6 \cdot -0.8E+08 / (\pi \cdot 5525.0 \cdot 15.0^2)$$

$$= -68.70 \text{ MPa}$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 0.591/217.441 = 0.0027$$

$$y = 125 \cdot eb/D_b = 125 \cdot 15.0/5523.0 = 0.3395$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994$$

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.339 + 3.994 = 4.3548$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b \left(\left(\frac{y}{y_A} \right) \cdot \alpha - 1 \right) / (4 \cdot eb)$$

$$= (0.348 + 0.0) \cdot 5523.0 \left(\left(\frac{9.84}{40.0} \right) \cdot 4.355 - 1 \right) / (4 \cdot 15.0)$$

$$= 2.28 \text{ MPa}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 33.592 - -11.842 + 2.28$$

$$= 47.71 \text{ MPa}$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 33.592 + -11.842 - 2.28$$

$$= 19.47 \text{ MPa}$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$



$$= 16.985 - -68.727 + 2.28$$

$$= 87.99 \text{ MPa}$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 16.985 + -68.727 - 2.28$$

$$= -54.02 \text{ MPa}$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 37.049 + -11.842 + 2.28$$

$$= 27.49 \text{ MPa}$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 37.049 - -11.842 - 2.28$$

$$= 46.61 \text{ MPa}$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 37.049 + -68.727 + 2.28$$

$$= -29.40 \text{ MPa}$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 37.049 - -68.727 - 2.28$$

$$= 103.50 \text{ MPa}$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= -3.456 - -11.837$$

$$= 8.38 \text{ MPa}$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= -3.456 + -11.837$$

$$= -15.29 \text{ MPa}$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -20.057 - -68.702$$

$$= 48.65 \text{ MPa}$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -20.057 + -68.702$$

$$= -88.76 \text{ MPa}$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 137.997(3 - (1 / 1.5)) \cdot (33.592 / 117.998)^2$$

$$= 406.54 \text{ MPa}$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 137.997(3 - (1 / 1.5)) \cdot (16.985 / 117.998)^2$$

$$= 412.09 \text{ MPa}$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 137.997(3 - (1 / 1.5)) \cdot (37.049 / 117.998)^2$$

$$= 404.92 \text{ MPa}$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$$

$$= 137.997(3 - (1 / 1.5)) \cdot (37.049 / 117.998)^2$$

$$= 404.92 \text{ MPa}$$



For $\sigma_{3pi\text{tot}}$ & $\sigma_{3po\text{tot}}$, the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 137.997(3 - (1/1.5) \cdot (-3.456/117.998)^2)$$

$$= 413.91 \text{ MPa}$$

For $\sigma_{3qi\text{tot}}$ & $\sigma_{3qo\text{tot}}$, the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 137.997(3 - (1/1.5) \cdot (-20.057/117.998)^2)$$

$$= 411.33 \text{ MPa}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	47.71	406.54	Pass	46
Stot1po	19.47	406.54	Pass	47
Stot1qi	87.99	412.09	Pass	48
Stot1qo	-54.02	412.09	Pass	49
Stot2pi	27.49	404.92	Pass	50
Stot2po	46.61	404.92	Pass	51
Stot2qi	-29.40	404.92	Pass	52
Stot2qo	103.50	404.92	Pass	53
Stot3pi	8.38	413.91	Pass	54
Stot3po	-15.29	413.91	Pass	55
Stot3qi	48.65	411.33	Pass	56
Stot3qo	-88.76	411.33	Pass	57

The stresses are satisfactory in the region without openings.

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**Shop/Field Installation Options :**

Platform(s) installed in the Shop.

Packing is installed in the Shop.

Trays are installed in the Shop.

Insulation is installed in the Field before being lifted.

Lining is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	22205.566 mm
Center of Gravity of the Packing	24249.998 mm
Center of Gravity of the Liquid	18227.904 mm
Center of Gravity of the Insulation	22322.580 mm
Center of Gravity of the Lining	22480.385 mm
Center of Gravity of the Stiffening Rings	23423.988 mm
Center of Gravity of the Nozzles	20188.551 mm
Center of Gravity of the Trays	22250.000 mm
Center of Gravity of the Added Weights (Operating)	23279.824 mm
Center of Gravity of the Added Weights (Empty)	22269.350 mm
Center of Gravity of Bare Shell New and Cold	14137.588 mm
Center of Gravity of Bare Shell Corroded	14082.163 mm
Vessel CG in the Operating Condition	19252.283 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	18665.859 mm
Vessel CG in the Test Condition	21270.877 mm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	5553.0000	mm
Skirt Thickness	STHK	30.0000	mm
Skirt Internal Corrosion Allowance	SCA	1.0000	mm
Skirt External Corrosion Allowance		0.0000	mm
Skirt Material		E250A	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	38.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		E250A	
Basing Ope. All. Stress	BASOPE, f4	117.00	MPa
Basing Yield Stress		250.00	MPa
Inside Diameter of Basing	DI, D4	5385.0000	mm
Outside Diameter of Basing	DOU, D5	5985.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	64.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Root Area of a Single Bolt	Area	2151.0000	mm ²
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	114.00	MPa
Number of Bolts	RN, nb	40	
Diameter of Bolt Circle	DC, DBC	5770.0000	mm
Bolt Allowable Shear Stress		82.800	MPa
Ultimate Comp. Strength of Concrete	FPC	40.0	MPa
Allowable Comp. Strength of Concrete	FC	14.0	MPa
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	22.0000	mm
Width of Gussets at Top Plate	TWDT	206.0000	mm
Width of Gussets at Base Plate	BWDT	206.0000	mm
Gusset Plate Elastic Modulus	E	201746.0	MPa
Gusset Plate Yield Stress	SY	250.0	MPa
Height of Chair	h1	400.0000	mm
Distance between Gussets	RG, b6	120.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	60.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	50.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	206.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	90.0000	mm
External Corrosion Allowance	CA	1.5000	mm
Dead Weight of Vessel	DW	1853219.0	N
Operating Weight of Vessel	ROW	3507685.0	N
Earthquake Moment on Basing	EQMOM	12295086080.0	N-mm
Wind Moment on Basing	WIMOM	2744061440.0	N-mm
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.12295E+11/5770.0) - 3397684) / 40$$

$$= 128057.9 \text{ N}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.27441E+10/5770.0) - 1853219) / 40$$

$$= 1207.6 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/5770.0) - 3507685) / 40$$

$$= -87692.1 \text{ N}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 2151.0000 mm²
 Area Available in all the Bolts Abss * RN : 86040.0000 mm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.261	0.391	1.400	2.528	0.446	0.779	788201.1	4185885.2	14691.3
4	0.456	0.424	1.899	2.099	0.403	0.785	969420.9	4367104.5	21756.7
6	0.440	0.432	1.860	2.136	0.407	0.785	953519.2	4351203.0	21035.4
8	0.428	0.426	1.831	2.163	0.409	0.785	941699.5	4339383.5	20514.0
10	0.427	0.427	1.829	2.165	0.410	0.785	940718.4	4338402.5	20471.2
12	0.427	0.427	1.829	2.164	0.410	0.785	940963.6	4338647.5	20481.9

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 940964 / (4.747 \cdot 5770.0 \cdot 2.164)$$

$$= 31.75 \text{ MPa, Should be } \leq 114.0 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 40 \cdot 2151.0 / (\pi \cdot 5770.0)$$

$$= 4.747 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc / [(2kd + t3) / (2kd)]$$

$$= 2.404 / [(2 \cdot 0.4268 \cdot 5770.0 + 300.0) / (2 \cdot 0.4268 \cdot 5770.0)]$$

$$= 2.551 \text{ MPa, Should be } \leq 14.0 \text{ MPa}$$

Values for table 10.3, l = 216.0, b = 226.587, l/b = 0.953275

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(11968.582, 15962.533) = 15962.533 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 15962.533 / 165.0)} + 1.5$$

$$= 25.594 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 206.0, b = 120.0, b/l = 0.582524

As b/l (0.583) is less than 1, inverting b/l = 1.717.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 245188.72 / (4 \cdot \pi) [1.3(\ln((2 \cdot 206.0 \cdot \sin(\pi \cdot 108.5/206.0)) / (\pi \cdot 53.975))) + 1] - [(0.7 - 0.048) \cdot 245188.72 / (4 \cdot \pi)]$$

$$= 29214.2383 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(Mo) / Fb)} + Ca$$

$$= \sqrt{(6 \cdot \text{Abs}(29214.24) / 166.66)} + 1.5$$

= 33.9318 mm

Required Gusset Plate Thickness [tg]:
 $= P / (\text{Stress Term} \cdot l_{avg}) + C_a$
 $= 245188.72 / (124.11 \cdot 206.0) + 1.5$
 $= 11.091 \text{ (not less than } 9.525 + 1.5 \text{) mm}$

Bolt spacing [m]:
 $= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$
 $= \pi \cdot 5770.0 / 40 = 453.175 \text{ mm}$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:
 $= 1.76(P \cdot a / (m \cdot (h + t_{ba}) \cdot 1.5 \cdot S_{ktope}))^{2/3} \cdot r^{1/3} + C_a$
 $= 1.76(245189 \cdot 108.5 / (453.17 \cdot 350.0 \cdot 175))^{2/3} \cdot 2776.51^{1/3} + C_a$
 $= 24.002 + 1.0 = 25.002 \text{ mm}$

Shear Stress in a Single Bolt [tb]:
 $= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 550005 / (2151.0 \cdot 40)$
 $= 6.4 \text{ MPa. Must be less than } 82.8 \text{ MPa.}$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	25.5939	mm
Actual Basing Thickness as entered by user	38.0000	mm
Required Thickness of Chair Cap	33.9318	mm
Actual Top Ring Thickness as entered by user	50.0000	mm
Required Gusset thickness, + CA	11.0910	mm
Actual Gusset Thickness as entered by user	22.0000	mm
Required Thickness of Skirt for Local Stress	25.0024	mm
Given Thickness of Skirt	30.0000	mm
Required Gusset Height to meet local stress	232.5095	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:
 $= \text{SkirtStress} \cdot (\text{SkirtThickness} - C_A)$
 $= 24.767 \cdot (30.0 - 1.0)$
 $= 718.19 \text{ N/mm}$

Results for Computed Minimum Basing Weld Size [BWeld]:
 $= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 718 / [(0.4 \cdot 202) \cdot 2 \cdot 0.707]$
 $= 6.28 \text{ mm}$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:
 $= \text{Bolt Load} / (C_{mwth} + 2 \cdot (H_g + T_{ta}))$
 $= 245188.7 / (210.0 + 2 \cdot (312.0 + 50.0))$
 $= 262.515 \text{ N/mm}$

Horizontal Plate Load [Wh]:
 $= \text{Bolt Load} \cdot e / (C_{mwth} \cdot (H_g + T_{ta}) + 0.6667 \cdot (H_g + T_{ta})^2)$
 $= 245188.7 \cdot 108.5 / (210.0 \cdot (362.0) + 0.6667 \cdot (362.0)^2)$
 $= 162.826 \text{ N/mm}$

Resultant Weld Load [Wr]:
 $= \sqrt{ W_v^2 + W_h^2 }$
 $= \sqrt{ 262.51^2 + 162.83^2 }$
 $= 308.911 \text{ N/mm}$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:
 $= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 308.91 / [(0.4 \cdot 202) \cdot 2 \cdot 0.707]$
 $= 2.701 \text{ mm}$



Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:
= Bolt Load / (2 • TopWth)
= 245188.7 / (2 • 206.0)
= 595.118 N/mm

Weld Load [Wh]:
= Bolt Load • e / (2 • Hgt • TopWth)
= 245188.7 • 108.5 / (2 • 362.0 • 206.0)
= 178.371 N/mm

Resultant Weld Load [Wr]:
= $\sqrt{Wv^2 + Wh^2}$
= $\sqrt{595.12^2 + 178.37^2}$
= 621.274 N/mm

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
= Wr / [(0.4 • Yield) • 2 • 0.707]
= 621.27 / [(0.4 • 202) • 2 • 0.707]
= 5.433 mm

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	5.4329	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

Cone 1

Design Internal Pressure, Small End	0.348	MPa
Design Internal Pressure, Large End	0.348	MPa
Temperature for Internal Pressure	160.0	°C
Design External Pressure	0.103	MPa
Temperature for External Pressure	0.0	°C

CONE Properties:

Cone Material		SA-516 70	
Cone Allowable Stress at Temperature	Ss	138.00	MPa
Cone Allowable Stress At Ambient		138.00	MPa
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	14.0000	mm
Corrosion Allowance for Cone	Cac	1.0000	mm
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	3730.334	mm
Diameter of Large End of Cone	Dl	5506.000	mm
Half Apex Angle for Cone		19.79	degrees
Axial Length of Cone	Lc	2498.000	mm

SMALL-END Properties:

Attached Small End Material Specification		SA-516 70	
Operating Allowable Stress	Ss	138.00	MPa
Ambient Allowable Stress		138.00	MPa
Longitudinal Joint Efficiency	Es,E1	1.00	
Actual Thickness	Ts	14.0000	mm
Corrosion Allowance	Cas	1.0000	mm
Axial Length of Attached Element	Ls	2563.001	mm

The element type attached to the small end of the cone was not a cylinder. Therefore, the material properties printed above reflect those of the cone.

LARGE-END Properties:

Attached Large End Material Specification		SA-516 70	
Operating Allowable Stress	Sl	138.00	MPa
Ambient Allowable Stress		138.00	MPa
Longitudinal Joint Efficiency	El,E1	1.00	
Actual Thickness	Tl	14.0000	mm
Corrosion Allowance	Cal	1.0000	mm
Axial Length of Attached Element	Ll	2563.001	mm

Type of Reinforcement at Large End of Cone: Bar

Large End Reinforcing/Knuckle Material		SA-516 70	
Large Reinforcing/Knuckle Allowable, Operating		138.00	MPa
Large Reinforcing/Knuckle Allowable, Ambient		138.00	MPa
Location of Reinforcement at Large End of Cone:		Shell	
Radial Width of Reinforcing Bar(Large End)	Rwlb	299.9999	mm
Axial Thickness of Reinforcing Bar	Rtlb	20.0000	mm

Type of Reinforcement at Small End of Cone: None

Force Arl & Ars calculation, cone -> flange	No
Force Arl & Ars calculation, cone -> cylinder	No

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 0 °C

Elastic Modulus of Cone Material	0.204E+06 MPa at 0 °C
Elastic Modulus of Small Cylinder Material	0.204E+06 MPa at 0 °C
Elastic Modulus of Large Cylinder Material	0.204E+06 MPa at 0 °C

Elastic Modulus of Large End Reinforcement 0.204E+06 MPa at 0 °C
 Elastic Modulus of Small End Reinforcement 0.204E+06 MPa at 0 °C

Elastic Modulus Data from ASME Section II Part D at 160 °C

Elastic Modulus of Cone Material 0.195E+06 MPa at 160 °C
 Elastic Modulus of Small Cylinder Material 0.195E+06 MPa at 160 °C
 Elastic Modulus of Large Cylinder Material 0.195E+06 MPa at 160 °C
 Elastic Modulus of Large End Reinforcement 0.195E+06 MPa at 160 °C
 Elastic Modulus of Small End Reinforcement 0.195E+06 MPa at 160 °C

Forces and Moments on the ends of the Cone:

Axial Force on Small End of Cone 174250.45 N
 Axial Force on Large End of Cone 267439.41 N
 Moment on Small End of Cone 58663748.00 N-mm
 Moment on Large End of Cone 129450200.00 N-mm

The moments above are computed from the Wind & Seismic loads as well as any user defined moments and include effects due to eccentric masses. The vertical forces include the weight of the elements and details above and below the cone as well as any user defined vertical forces. Vertical forces also include loads cause from vertical accelerations due to earthquake loading.

Note: Neither end of Cone is a Line of Support

Maximum Centroid Reinforcement Distance Large End 47.4151 mm
 Maximum Centroid Reinforcement Distance Small End 39.0746 mm

Note: No ring was found close enough to the small end to be considered.

Reinforcement Calculations for Cone / Large Cylinder:

Required Area of Reinforcement for Large End Under Internal Pressure

Note: Large end is in compression, 1.0 will be used for E1 and E2.

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.00252

Large End Maximum half apex angle w/o reinforcement [δ]:
 $= 326.6 \cdot \sqrt{\frac{P}{(Ss \cdot E1)}} \text{ per 1-5 equation 1}$
 $= 326.6 \cdot \sqrt{\frac{0.348}{(137.997 \cdot 1.0)}}$
 $= 16.404 \text{ degrees}$

Large end actual half apex angle [alpha]: 19.788 degrees

Intermediate Value [k]:
 $= \max(Y / (Sr \cdot Erl), 1)$
 $= \max(26851224 / (138.0 \cdot 194578), 1)$
 $= 1.0000$

where [Y] is:
 $= \text{Large End All. Stress} \cdot \text{Large End Elastic Modulus (Int. temp.)}$
 $= 138.0 \cdot 194578$
 $= 26851224.0 \text{ (MPa)}^2$

Decay Length, Cone Large End:
 $= 2 \cdot \sqrt{\frac{RL(ts - ca)}{}}$
 $= 2 \cdot \sqrt{\frac{2754.0(14.0 - 1.0)}{}}$
 $= 378.428 \text{ mm}$

Required Area of Reinforcement, Large End, Internal [Ar]:
 $= (k \cdot QL \cdot RI / (Ss \cdot E1)) \cdot (1 - \delta/\alpha) \cdot \tan(\alpha)$
 $= (1.0 \cdot 469.3404 \cdot 2754.0 / (138 \cdot 1.0)) \cdot (1 - 16.4/19.79) \cdot 0.3598$
 $= 576.3 \text{ mm}^2$

Force per Length, Cone Large End, tensile side [QL+]:
 $= P(RI/2) + f1$
 $= 0.348(2754.0/2) + -10.02$
 $= 469.340 \text{ N/mm}$



Axial Load per Unit Circumference excluding Pressure [f1]:
 $= \text{Faxial}/(\pi(DI + TI)) + \text{Moment}/(\pi(RI + TI/2)^2)$
 $= -267439.4/(\pi(5508.0+14.0))+.12945E+09/(\pi(2754.0+14.0/2)^2)$
 $= -10.0 \text{ N/mm}$

Force per Length, Cone Large End, compressive side [QL-]:
 $= P(RI/2) + f1$
 $= 0.348(2754.0/2) + -20.82$
 $= 458.534 \text{ N/mm}$

Axial Load per Unit Circumference excluding Pressure [f1]:
 $= \text{Faxial}/(\pi(DI + TI)) - \text{Moment}/(\pi(RI + TI/2)^2)$
 $= -267439.4/(\pi(5508.0+14.0))-0.12945E+09/(\pi(2754.0+14.0/2)^2)$
 $= -20.8 \text{ N/mm}$

Area of Reinforcement Available in Large end Shell [Ael]:
 $= (Ts-t)\sqrt{(RI\cdot Ts)+(Tc-Tr)\sqrt{(RI\cdot Tc/\cos(\alpha))}}$
 $= (13.0 - 6.9585) \cdot \sqrt{(2754.0 \cdot 13.0) + }$
 $(13.0 - 7.3952) \cdot \sqrt{(2754.0 \cdot 13.0/0.941)}$
 $= 2236.4272 \text{ mm}^2$

Summary of Reinforcement area, Large end, Internal Pressure:

Area of reinforcement required per App. 1-5(1)	576.2576	mm ²
Area of reinforcement in shell per App. 1-5(2)	2236.4272	mm ²
Area of reinforcement in stiffening ring	5999.9976	mm ²

Required Area of Reinforcement for Large End Under External Pressure

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.00075

Large End Maximum half apex angle w/o reinforcement [δ]:
 $= 104 \cdot \sqrt{(P / (Ss \cdot E1))}$ per 1-8 equation 1
 $= 104 \cdot \sqrt{(0.103 / (137.997 \cdot 1.0))}$
 $= 2.847 \text{ degrees}$

Large end actual half apex angle [alpha]: 19.788 degrees

Intermediate Value [k]:
 $= \max(Y / (Srl \cdot Erl), 1)$
 $= \max(28165318 / (138.0 \cdot 204100), 1)$
 $= 1.0000$

where [Y] is:
 $= \text{Large End All. Stress} \cdot \text{Large End Elastic Modulus (Ext. temp.)}$
 $= 138.0 \cdot 204100$
 $= 28165318.0 \text{ (MPa)}^2$

Allowable Stress of Large End Material (Ext. Temp)	138.0	MPa
Allowable Stress of Cone Material (Ext. Temp)	138.0	MPa

Required Area of Reinforcement, Large End, External [Arl]:
 $= (k \cdot QL \cdot RI \cdot \tan(\text{angle}) / (Ss \cdot E1)) \cdot (1 - 1/4((P \cdot RI - QL) / QL) \cdot (\delta / \alpha))$
 $= (1.0 \cdot 163.8969 \cdot 2767.0 \cdot 0.36 / (138 \cdot 1.0)) \cdot$
 $(1 - 1/4((0.1 \cdot 2767.0 - 163.897) / 163.897) \cdot (2.847 / 19.788))$
 $= 1151 \text{ mm}^2$

Force per Length, Cone Large End External Pressure, compressive side [QL-]:
 $= \text{Pext}(RI/2) + f1$
 $= 0.103(2767.0/2) + 20.83$
 $= 163.897 \text{ N/mm}$

Axial Load per Unit Circumference excluding Pressure [f1]:
 $= \text{Faxial}/(\pi(DI + TI)) + \text{Moment}/(\pi(RI - TI/2)^2)$
 $= +267439.4/(\pi(5534.0-14.0))+.12945E+09/(\pi(2767.0-14.0/2)^2)$
 $= 20.8 \text{ N/mm}$

Force per Length, Cone Large End External Pressure, tensile side [QL+]:
 $= \text{Pext}(RI/2) + f1$
 $= 0.103(2767.0/2) + 10.01$
 $= 153.083 \text{ N/mm}$



Axial Load per Unit Circumference excluding Pressure [f1]:
 $= F_{axial} / (\pi(DI + TI)) - \text{Moment} / (\pi(RI - TI/2)^2)$
 $= +267439.4 / (\pi(5534.0 - 14.0)) - 0.12945E+09 / (\pi(2767.0 - 14.0/2)^2)$
 $= 10.0 \text{ N/mm}$

Available Area of Reinforcement, Large End, External [Ael]:
 $= 0.55 [\sqrt{(DI \cdot ts)(ts - t)} + \sqrt{(DI \cdot tc / \cos(\alpha))(tc - tr)}]$ per 1-8 eqn. 3
 $= 0.55 [\sqrt{(5534.0 \cdot 13.0)(13.0 - 12.888)} + \sqrt{(5534.0 \cdot 13.0/0.941)(13.0 - 12.885)}]$
 $= 33.9924 \text{ mm}^2$

Summary of Reinforcement Area, Large End, External Pressure:

Area of reinforcement required per App. 1-8(1)	1150.7399	mm ²
Area of reinforcement in shell per App. 1-8(2)	33.9924	mm ²
Area of reinforcement in stiffening ring	5999.9976	mm ²



Because the 1/2 apex angle is less than 30 degrees, the following discontinuity stress evaluation results are for information only.

The following calculations are only required per 1-5(g) and do include external loads due to wind or seismic. These discontinuity stresses are computed at the shell/cone junction and do not include effects of local stiffening from a junction ring.

Results for Discontinuity Stresses per Bednar p. 236 2nd Edition:

Stress Type	Stress	Allowable	Location
Tensile Stress	254.32	460.80	Large Cyl. Long.
Compres. Stress	-181.93	-460.80	Large Cyl. Long.
Membrane Stress	-46.14	207.00	Large End Tang.
Tensile Stress	256.59	460.80	Cone Longitudinal
Compres. Stress	-179.66	-460.80	Cone Longitudinal
Tensile Stress	-41.50	207.00	Cone Tangential

An asterisk (*) denotes that this stress was not applicable for this combination of loads.

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure MPa	Reason for Failure at this Pressure
MAWP	0.615	Thickness due to internal pressure, Cone Large End
MAPnc	0.662	Thickness due to internal pressure, Cone Large End

These pressures were determined by iteration.

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Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

Cone 2

Design Internal Pressure, Small End	0.348	MPa
Design Internal Pressure, Large End	0.348	MPa
Temperature for Internal Pressure	160.0	°C
Design External Pressure	0.103	MPa
Temperature for External Pressure	0.0	°C

CONE Properties:

Cone Material		SA-516 70	
Cone Allowable Stress at Temperature	Ss	138.00	MPa
Cone Allowable Stress At Ambient		138.00	MPa
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	14.0000	mm
Corrosion Allowance for Cone	Cac	1.0000	mm
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	3736.000	mm
Diameter of Large End of Cone	Dl	5504.242	mm
Half Apex Angle for Cone		19.29	degrees
Axial Length of Cone	Lc	2498.000	mm

SMALL-END Properties:

Attached Small End Material Specification		SA-516 70	
Operating Allowable Stress	Ss	138.00	MPa
Ambient Allowable Stress		138.00	MPa
Longitudinal Joint Efficiency	Es,E1	1.00	
Actual Thickness	Ts	12.0000	mm
Corrosion Allowance	Cas	1.0000	mm
Axial Length of Attached Element	Ls	2563.001	mm

LARGE-END Properties:

Attached Large End Material Specification		SA-516 70	
Operating Allowable Stress	Sl	138.00	MPa
Ambient Allowable Stress		138.00	MPa
Longitudinal Joint Efficiency	El,E1	1.00	
Actual Thickness	Tl	14.0000	mm
Corrosion Allowance	Cal	1.0000	mm
Axial Length of Attached Element	Ll	2563.001	mm

The element type attached to the Large end of the cone was not a cylinder. Therefore, the material properties printed above reflect those of the cone.

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: Bar

Small End Reinforcing/Knuckle Material		SA-516 70	
Small Reinforcing/Knuckle Allowable, Operating		138.00	MPa
Small Reinforcing/Knuckle Allowable, Ambient		138.00	MPa
Location of Reinforcement at Small End of Cone:		Shell	
Radial Width of Reinforcing Bar(Small End)	Rwsb	300.0001	mm
Axial Thickness of Reinforcing Bar	Rtsb	14.0000	mm

Force Arl & Ars calculation, cone -> flange No

Force Arl & Ars calculation, cone -> cylinder No

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 0 °C

Elastic Modulus of Cone Material	0.204E+06 MPa at 0 °C
Elastic Modulus of Small Cylinder Material	0.204E+06 MPa at 0 °C
Elastic Modulus of Large Cylinder Material	0.204E+06 MPa at 0 °C

Elastic Modulus of Large End Reinforcement 0.204E+06 MPa at 0 °C
 Elastic Modulus of Small End Reinforcement 0.204E+06 MPa at 0 °C

Elastic Modulus Data from ASME Section II Part D at 160 °C

Elastic Modulus of Cone Material 0.195E+06 MPa at 160 °C
 Elastic Modulus of Small Cylinder Material 0.195E+06 MPa at 160 °C
 Elastic Modulus of Large Cylinder Material 0.195E+06 MPa at 160 °C
 Elastic Modulus of Large End Reinforcement 0.195E+06 MPa at 160 °C
 Elastic Modulus of Small End Reinforcement 0.195E+06 MPa at 160 °C

Forces and Moments on the ends of the Cone:

Axial Force on Small End of Cone 149658.86 N
 Axial Force on Large End of Cone 174250.45 N
 Moment on Small End of Cone 10037767.00 N-mm
 Moment on Large End of Cone 59145468.00 N-mm

The moments above are computed from the Wind & Seismic loads as well as any user defined moments and include effects due to eccentric masses. The vertical forces include the weight of the elements and details above and below the cone as well as any user defined vertical forces. Vertical forces also include loads cause from vertical accelerations due to earthquake loading.

Note: Neither end of Cone is a Line of Support

Maximum Centroid Reinforcement Distance Large End 47.4075 mm
 Maximum Centroid Reinforcement Distance Small End 35.9514 mm

Reinforcement Calculations for Cone / Small Cylinder:

Required Area of Reinforcement for Small End under Internal Pressure

Small end ratio of pressure to allowable stress [P/Ss E1]: 0.00252

Small End Maximum half apex angle w/o reinforcement [δ]:
 $= 89 \cdot \sqrt{P / (Ss \cdot E1)}$ per 1-5 equation 6
 $= 89 \cdot \sqrt{0.348 / (137.997 \cdot 1.0)}$
 $= 4.470$ degrees

Small end actual half apex angle [alpha]: 19.290 degrees

Intermediate Value [k]:
 $= \max(Y / (Sr \cdot Ers), 1)$
 $= \max(26851224 / (138.0 \cdot 194578), 1)$
 $= 1.0000$

where [Y] is:
 $= \text{Small End All. Stress} \cdot \text{Small End Elastic Modulus (Int. temp.)}$
 $= 138.0 \cdot 194578$
 $= 26851224.0 \text{ (MPa)}^2$

Decay Length, Cone Small End:
 $= 1.4 \cdot \sqrt{Rs(ts - ca)}$
 $= 1.4 \cdot \sqrt{1869.0(12.0 - 1.0)}$
 $= 200.738 \text{ mm}$

Required Area of Reinforcement, Small End, Internal [Ars]:
 $= k \cdot QS \cdot Rs / (Ss \cdot E1) \cdot (1 - \delta/\alpha) \cdot \tan(\alpha)$ per 1-5 equation 7
 $= 1.0 \cdot 313.515 \cdot 1869.0 / (138 \cdot 1.0) \cdot (1 - 4.47/19.29) \cdot 0.35$
 $= 1142 \text{ mm}^2$

Force per Length, Cone Small End, tensile side [QS+]:
 $= P(Rs/2) + f2$
 $= 0.348(1869.0/2) + -11.8$
 $= 313.515 \text{ N/mm}$

Axial Load per Unit Circumference excluding Pressure [f2]:
 $= \text{Faxial} / (\pi(Ds + Ts)) + \text{Moment} / (\pi(Rs + Ts/2)^2)$
 $= -149658.9 / (\pi(3738.0 + 12.0)) + 10037767 / (\pi(1869.0 + 12.0/2)^2)$
 $= -11.8 \text{ N/mm}$



Force per Length, Cone Small End, compressive side [QS-]:

$$= P(Rs/2) + f_2$$

$$= 0.348(1869.0/2) + -13.62$$

$$= 311.698 \text{ N/mm}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= Faxial/(\pi(Ds + Ts)) - Moment/(\pi(Rs + Ts/2)^2)$$

$$= -149658.9/(\pi(3738.0+12.0))-10037767/(\pi(1869.0+12.0/2)^2)$$

$$= -13.6 \text{ N/mm}$$

Area of Reinforcement Available in Small End Shell [Aes]:

$$= 0.78[\sqrt{(Rs \cdot ts)}(ts-t) + \sqrt{(Rs \cdot tc/\cos(\alpha))} \cdot (tc - tr)] \text{ per 1-5 eqn. 8}$$

$$= 0.78[\sqrt{(1869.0 \cdot 11.0)}(11.0 - 4.722) + \sqrt{(1869.0 \cdot 13.0/0.944)} \cdot (13.0 - 5.003)]$$

$$= 1702.8489 \text{ mm}^2$$

Summary of Reinforcement Area, Small End, Internal Pressure:

Area of reinforcement required per App. 1-5(7)	1141.8499	mm ²
Area of reinforcement in shell per App. 1-5(8)	1702.8489	mm ²
Area of reinforcement in stiffening ring	4200.0015	mm ²

Required Area of Reinforcement for Small End Under External Pressure

Note: Small end is in compression, 1.0 will be used for E1 and E2.

Allowable Stress of Small End Material (Ext. Temp)	138.0	MPa
Allowable Stress of Cone Material (Ext. Temp)	138.0	MPa

Intermediate Value [k]:

$$= \max(Y / (Srs \cdot Ers), 1)$$

$$= \max(28165318 / (138.0 \cdot 204100), 1)$$

$$= 1.0000$$

where [Y] is:

$$= \text{Small End All. Stress} \cdot \text{Small End Elastic Modulus (Ext. temp.)}$$

$$= 138.0 \cdot 204100$$

$$= 28165318.0 \text{ (MPa)}^2$$

Area of Reinforcement Required in Small End Shell [Ars]:

$$= k \cdot QS \cdot Rs \cdot \tan(\alpha) / (Ss \cdot E1)$$

$$= (1.0 \cdot 110.8252 \cdot 1880.0 \cdot 0.35 / (138 \cdot 1.0))$$

$$= 528.5 \text{ mm}^2$$

Force per Length, Cone Small End External Pressure, compressive side [QS-]:

$$= Pext(Rs/2) + f_2$$

$$= 0.103(1880.0/2) + 13.62$$

$$= 110.825 \text{ N/mm}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= Faxial/(\pi(Ds + Ts)) + Moment/(\pi(Rs - Ts/2)^2)$$

$$= +149658.9/(\pi(3760.0-12.0))+10037767/(\pi(1880.0-12.0/2)^2)$$

$$= 13.6 \text{ N/mm}$$

Force per Length, Cone Small End External Pressure, tensile side [QS+]:

$$= Pext(Rs/2) + f_2$$

$$= 0.103(1880.0/2) + 11.8$$

$$= 109.006 \text{ N/mm}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= Faxial/(\pi(Ds + Ts)) - Moment/(\pi(Rs - Ts/2)^2)$$

$$= +149658.9/(\pi(3760.0-12.0))-10037767/(\pi(1880.0-12.0/2)^2)$$

$$= 11.8 \text{ N/mm}$$

Area of Reinforcement Available in Small End Shell [Aes]:

$$= 0.55[\sqrt{(Ds \cdot ts)}(ts-t) + \sqrt{(Ds \cdot tc/\cos(\alpha))}(tc-tr)]$$

$$= 0.55[\sqrt{(3760.0 \cdot 11.0)}(11.0 - 10.253) + \sqrt{(3760.0 \cdot 13.0/0.944)}(13.0 - 3.646)]$$

$$= 1254.3397 \text{ mm}^2$$



Summary of Reinforcement Area, Small End, External Pressure:

Area of reinforcement required per App. 1-8(6)	528.4822	mm ²
Area of reinforcement in shell per App. 1-8(7)	1254.3397	mm ²
Area of reinforcement in stiffening ring	4200.0015	mm ²



Because the 1/2 apex angle is less than 30 degrees, the following discontinuity stress evaluation results are for information only.

The following calculations are only required per 1-5(g) and do include external loads due to wind or seismic. These discontinuity stresses are computed at the shell/cone junction and do not include effects of local stiffening from a junction ring.

Results for Discontinuity Stresses per Bednar p. 236 2nd Edition:

Stress Type	Stress	Allowable	Location
Tensile Stress	183.06	460.80	Small Cyl. Long.
Compres. Stress	-125.88	-460.80	Small Cyl. Long.
Membrane Stress	132.61	207.00	Small End Tang.
Tensile Stress	136.23	460.80	Cone Longitudinal
Compres. Stress	-84.97	-460.80	Cone Longitudinal
Tensile Stress	126.47	207.00	Cone Tangential

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure MPa	Reason for Failure at this Pressure
MAWP	0.703	Thickness due to internal pressure, Cone Large End
MAPnc	0.760	Thickness due to internal pressure, Cone Large End

These pressures were determined by iteration.

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Input, Nozzle Desc: N1 20"

From: 80

Pressure for Reinforcement Calculations	P	0.4015	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	5506.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm
Head Internal Corrosion Allowance	cas	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		20.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

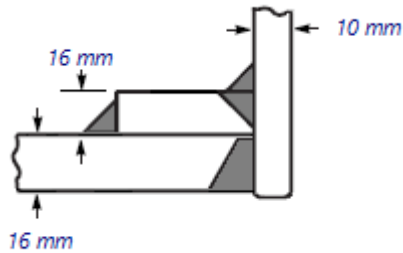
Outside Projection	ho	220.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	750.0000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		111.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		5000.0	mm
Centerline Distance to Tangent		2400.0	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N1 20"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	20.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.4 \cdot 5508.0 \cdot 0.8997 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.4)$
 $= 7.2116 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.4 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 0.4)$
 $= 0.7403 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.1757 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1016.0000 mm
Parallel to Vessel Wall, opening length	d	508.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

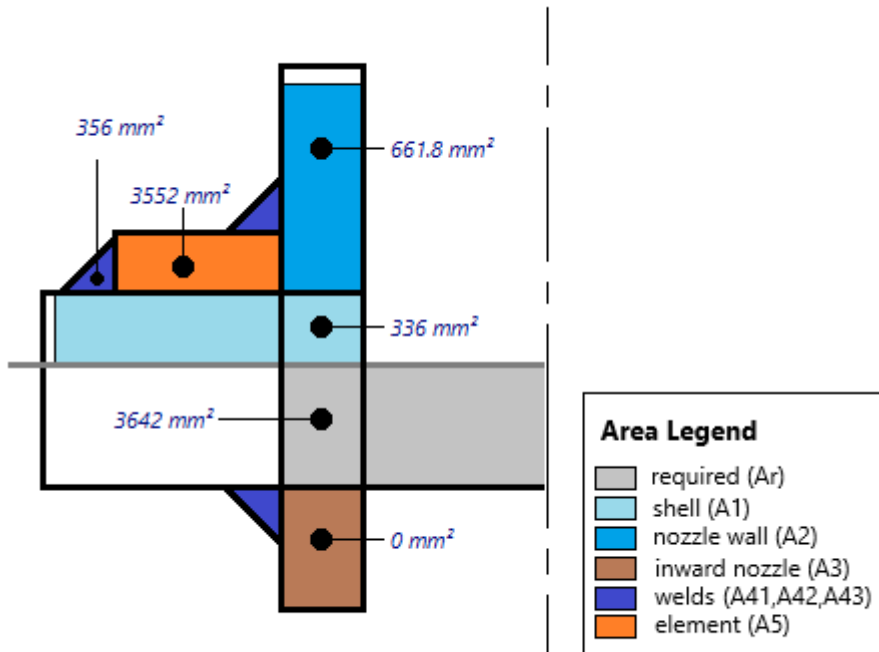


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3663.498	3642.016	NA
Area in Shell	A1	3956.502	335.967	NA
Area in Nozzle Wall	A2	694.474	661.826	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		356.000	356.000	NA
Area in Element	A5	3552.000	3552.000	NA
TOTAL AREA AVAILABLE	Atot	8558.976	4905.793	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 671.0139 16.0000 mm
 Based on given Pad Diameter: 750.0000 10.3073 mm
 Based on Shell or Nozzle Thickness: 756.8223 10.0000 mm

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(508.0 \cdot 14.3386 \cdot 1 + 2 \cdot 10.0 \cdot 14.3386 \cdot 1(1 - 1.0))$
 $= 3642.016 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 508.0(1.0 \cdot 15.0 - 1.0 \cdot 14.339) - 2 \cdot 10.0$
 $(1.0 \cdot 15.0 - 1.0 \cdot 14.3386) \cdot (1 - 1.0)$
 $= 335.967 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (10.0 - 1.18) \cdot 1.0$

= 661.826 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 16.0^2 \cdot 1.0$
 = 356.000 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(750.0 - 528.0) \cdot 16.0 \cdot 1.0$
 = 3552.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.1757 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 9.0118 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 9.0118 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.0631 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0631 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[8.331, max(9.0118, 3.0631)]
 = 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(1.1757, 8.3312)
 = 8.3312 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.6,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	318.4 MPa	Passed
Occasional	:	5.0,	Allowable	:	183.5 MPa	Passed
Shear	:	16.8,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 10.0, tr = 0.74, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.074$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 10.0, tr = 0.74, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.074$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 10.0, tr = 0.74, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.074$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 16.0, tr = 7.212, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.481$, Temp. Reduction = 35 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 10.0, tr = 0.74, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.074$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -48 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.40 / 5.10 = 0.079$

Weld Size Calculations, Description: N1 20"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	16.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm
Pad Weld	8.0000 = $0.5 \cdot T_{minPad}$	11.3120 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (3642.0164 - 335.9672 + 2 \cdot 10.0 \cdot 1.0 \cdot (1.0 \cdot 15.0 - 14.3386)) \cdot 138)$
 $= 458012.91 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (661.8259 + 3552.0 + 356.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 630571.00 \text{ N}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (661.8259 + 0.0 + 100.0 + (300.0)) \cdot 138$
 $= 146516.86 \text{ N}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (661.8259 + 0.0 + 356.0 + 3552.0 + (300.0)) \cdot 138$
 $= 671966.75 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 528.0 \cdot 10.0 \cdot 0.49 \cdot 138$

= 560779. N

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 750.0 \cdot 16.0 \cdot 0.49 \cdot 138$$

$$= 1274476. N$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 259.0) \cdot (10.0 - 0.0) \cdot 0.7 \cdot 138$$

$$= 785941. N$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 528.0 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 1355026. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 528.0 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 1016270. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (1274476 + 785941) = 2060416 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (560779 + 1355026 + 1016270 + 0) = 2932075 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (1274476 + 1016270 + 0) = 2290745 N$$

Summary of Failure Path Calculations:

Path 1-1 = 2060416 N , must exceed W = 458012 N or W1 = 630571 N
 Path 2-2 = 2932074 N , must exceed W = 458012 N or W2 = 146516 N
 Path 3-3 = 2290745 N , must exceed W = 458012 N or W3 = 671966 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 7.0190 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.0191 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.931 %

Input Echo, WRC107/537 Item 1, Description: N1 20" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	9910.800	mm
Vessel Thickness	Tv	16.000	mm
Design Temperature	T1	160.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	ID
Corrosion Allowance for Nozzle	Can	0.0000 mm
Nozzle Diameter	Dn	508.000 mm
Nozzle Thickness	Tn	10.000 mm
Nozzle Material		SA-516 70
Nozzle UNS Number		K02700
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa
Thickness of Reinforcing Pad	Tpad	16.000 mm
Diameter of Reinforcing Pad	Dpad	750.000 mm
Design Internal Pressure	Dp	0.402 MPa
Include Pressure Thrust		No

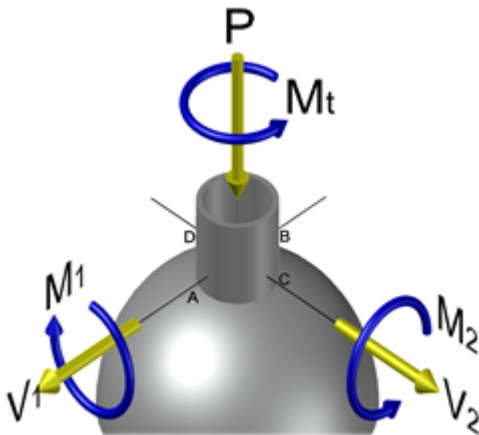
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-37500.0	N
Longitudinal Shear (SUS)	(V1) V1	25000.0	N
Circumferential Shear (SUS)	(Vc) V2	25000.0	N
Circumferential Moment (SUS)	(Mc) M1	35000000.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	35000000.0	N-mm
Torsional Moment (SUS)	Mt	50000000.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$$

$$= 528.0 + 2 \cdot 1.65 \cdot \sqrt{(4963.9 (16.0 - 1.0))}$$

$$= 1428.474 \text{ mm}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	-37500.0	N
Circumferential Shear	(VC) V2	25000.0	N
Longitudinal Shear	(VL) V1	25000.0	N
Circumferential Moment	(MC) M1	35000000.0	N-mm
Longitudinal Moment	(ML) M2	35000000.0	N-mm



Torsional Moment MT 50000000.0 N-mm

Dimensionless Param: U = 0.67 TAU = 25.90 RHO = 4.00 (3.10)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 7	0.02638	(A,B,C,D)
M(x) / P	SP 7	0.01552	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 7	0.05791	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 7	0.03630	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 7	0.05791	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 7	0.03630	(A,B,C,D)
N(y) * T / P	SP 7	0.27856	(A,B,C,D)
M(y) / P	SP 7	0.05103	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 7	0.36086	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 7	0.13916	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 7	0.36086	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 7	0.13916	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Rad. Bend. P		3.6	-3.6	3.6	-3.6	3.6	-3.6	3.6	-3.6
Rad. Memb. MC		0.0	0.0	0.0	0.0	-5.4	-5.4	5.4	5.4
Rad. Bend. MC		0.0	0.0	0.0	0.0	-20.2	20.2	20.2	-20.2
Rad. Memb. ML		-5.4	-5.4	5.4	5.4	0.0	0.0	0.0	0.0
Rad. Bend. ML		-20.2	20.2	20.2	-20.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-20.9	12.2	30.2	-17.4	-20.9	12.2	30.2	-17.4
Tang. Memb. P		10.9	10.9	10.9	10.9	10.9	10.9	10.9	10.9
Tang. Bend. P		11.9	-11.9	11.9	-11.9	11.9	-11.9	11.9	-11.9
Tang. Memb. MC		0.0	0.0	0.0	0.0	-33.5	-33.5	33.5	33.5
Tang. Bend. MC		0.0	0.0	0.0	0.0	-77.4	77.4	77.4	-77.4
Tang. Memb. ML		-33.5	-33.5	33.5	33.5	0.0	0.0	0.0	0.0
Tang. Bend. ML		-77.4	77.4	77.4	-77.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-88.1	42.9	133.7	-45.0	-88.1	42.9	133.7	-45.0
Shear VC		1.0	1.0	-1.0	-1.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear MT		3.7	3.7	3.7	3.7	3.7	3.7	3.7	3.7
Tot. Shear		4.7	4.7	2.7	2.7	2.7	2.7	4.7	4.7
Str. Int.		88.4	43.6	133.8	45.3	88.2	43.1	133.9	45.8

Unitless Prm: U = 1.37 TAU = 0.00 (37.00) RHO = 0.00 (1.50)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.04302	(A,B,C,D)
M(x) / P	SR 2	0.02149	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.04536	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.03524	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.04536	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.03524	(A,B,C,D)
N(y) * T / P	SR 2	0.01293	(A,B,C,D)



M(y) / P SR 2 0.00645 (A,B,C,D)
 N(y) * T * SQRT(Rm * T) / MC SR 3 0.01362 (A,B,C,D)
 M(y) * SQRT(Rm * T) / MC SR 3 0.01059 (A,B,C,D)
 N(y) * T * SQRT(Rm * T) / ML SR 3 0.01362 (A,B,C,D)
 M(y) * SQRT(Rm * T) / ML SR 3 0.01059 (A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		7.2	7.2	7.2	7.2	7.2	7.2	7.2	7.2
Rad. Bend. P		21.5	-21.5	21.5	-21.5	21.5	-21.5	21.5	-21.5
Rad. Memb. MC		0.0	0.0	0.0	0.0	-25.8	-25.8	25.8	25.8
Rad. Bend. MC		0.0	0.0	0.0	0.0	-120.5	120.5	120.5	-120.5
Rad. Memb. ML		-25.8	-25.8	25.8	25.8	0.0	0.0	0.0	0.0
Rad. Bend. ML		-120.5	120.5	120.5	-120.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-117.7	80.3	175.0	-109.0	-117.7	80.3	175.0	-109.0
Tang. Memb. P		2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Tang. Bend. P		6.4	-6.4	6.4	-6.4	6.4	-6.4	6.4	-6.4
Tang. Memb. MC		0.0	0.0	0.0	0.0	-7.8	-7.8	7.8	7.8
Tang. Bend. MC		0.0	0.0	0.0	0.0	-36.2	36.2	36.2	-36.2
Tang. Memb. ML		-7.8	-7.8	7.8	7.8	0.0	0.0	0.0	0.0
Tang. Bend. ML		-36.2	36.2	36.2	-36.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-35.4	24.2	52.6	-32.7	-35.4	24.2	52.6	-32.7
Shear VC		1.4	1.4	-1.4	-1.4	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.4	-1.4	1.4	1.4
Shear MT		3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8
Tot. Shear		5.2	5.2	2.4	2.4	2.4	2.4	5.2	5.2
Str. Int.		118.0	80.8	175.0	109.0	117.7	80.4	175.2	109.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		32.1	32.1	32.1	32.1	32.1	32.1	32.1	32.1
Rad. Pl (SUS)		-4.3	-4.3	6.4	6.4	-4.3	-4.3	6.4	6.4
Rad. Q (SUS)		-16.6	16.6	23.8	-23.8	-16.6	16.6	23.8	-23.8
Long. Pm (SUS)		32.1	32.1	32.1	32.1	32.1	32.1	32.1	32.1
Long. Pl (SUS)		-22.6	-22.6	44.3	44.3	-22.6	-22.6	44.3	44.3
Long. Q (SUS)		-65.5	65.5	89.4	-89.4	-65.5	65.5	89.4	-89.4
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.0	1.0	-1.0	-1.0	-1.0	-1.0	1.0	1.0
Shear Q (SUS)		3.7	3.7	3.7	3.7	3.7	3.7	3.7	3.7
Pm (SUS)		32.1	32.1	32.1	32.1	32.1	32.1	32.1	32.1
Pm+Pl (SUS)		27.8	27.8	76.5	76.5	27.8	27.8	76.5	76.5
Pm+Pl+Q (Total)		67.8	75.7	165.9	28.1	67.4	75.2	166.0	29.1

Vessel Stress Summation Comparison (MPa):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32.10	138.00	Passed
Pm+Pl (SUS)	76.46	207.00	Passed
Pm+Pl+Q (TOTAL)	166.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	66.3	66.3	66.3	66.3	66.3	66.3	66.3	66.3
Rad.	Pl (SUS)	-18.7	-18.7	33.0	33.0	-18.7	-18.7	33.0	33.0
Rad.	Q (SUS)	-99.0	99.0	142.0	-142.0	-99.0	99.0	142.0	-142.0
Long.	Pm (SUS)	66.3	66.3	66.3	66.3	66.3	66.3	66.3	66.3
Long.	Pl (SUS)	-5.6	-5.6	9.9	9.9	-5.6	-5.6	9.9	9.9
Long.	Q (SUS)	-29.8	29.8	42.7	-42.7	-29.8	29.8	42.7	-42.7
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.4	1.4	-1.4	-1.4	-1.4	-1.4	1.4	1.4
Shear	Q (SUS)	3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8
Pm (SUS)		66.3	66.3	66.3	66.3	66.3	66.3	66.3	66.3
Pm+Pl (SUS)		60.9	60.9	99.4	99.4	60.9	60.9	99.4	99.4
Pm+Pl+Q (Total)		82.9	147.1	241.4	76.4	82.4	146.8	241.6	76.9

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	66.34	138.00	Passed
Pm+Pl (SUS)	99.44	207.00	Passed
Pm+Pl+Q (TOTAL)	241.56	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N52"

From: 80

Pressure for Reinforcement Calculations	P	0.4006	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	5506.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm
Head Internal Corrosion Allowance	cas	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1000.0000	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

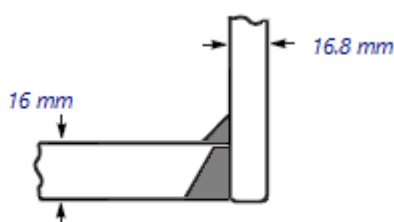
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N52"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.4 \cdot 5508.0 \cdot 0.8997 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.4)$
 $= 7.1951 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.4 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0738 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	115.2948 mm
Parallel to Vessel Wall	Rn+tn+t	57.6474 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

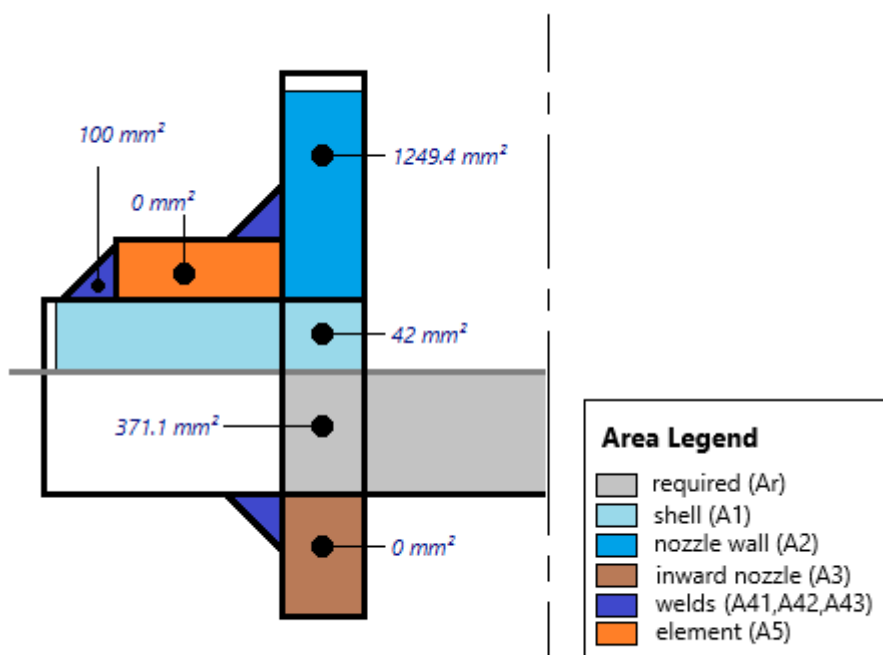


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	372.465	371.133	NA
Area in Shell	A1	495.833	42.014	NA
Area in Nozzle Wall	A2	1273.694	1249.388	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1869.527	1391.402	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 78.91 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(51.7668 \cdot 14.3386 \cdot 1 + 2 \cdot 16.764 \cdot 14.3386 \cdot 1(1 - 1.0))$
 $= 371.133 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 14.339) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 14.3386) \cdot (1 - 1.0)$
 $= 42.014 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 37.5)(16.76 - 0.39)1.0/\sin(79.4)$
 $= 1249.388 \text{ mm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 16.0$, $t_r = 7.195$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E*/(t_g - c) = 0.48$, Temp. Reduction = 35 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.40/5.10 = 0.079$

Weld Size Calculations, Description: N52"

Intermediate Calc. for nozzle/shell Welds Tmin 16.7640 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 \cdot Wo \text{ mm}$



Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (371.133 - 42.0144 + 2 \cdot 16.764 \cdot 1.0 \cdot \\ &\quad (1.0 \cdot 15.0 - 14.3386)) \cdot 138) \\ &= 48473.36 \text{ N} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\ &= (1249.3878 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138 \\ &= 186196.31 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (1249.3878 + 0.0 + 100.0 + (502.92)) \cdot 138 \\ &= 255592.12 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (1249.3878 + 0.0 + 100.0 + 0.0 + (502.92)) \cdot 138 \\ &= 255592.12 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 85.9329 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 91266. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 34.4249) \cdot (16.764 - 0.0) \cdot 0.7 \cdot 138 \\ &= 175122. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 85.9329 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 137833. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (91266 + 175122) = 266388 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (91266 + 0 + 137833 + 0) = 229099 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ &= (91266 + 137833 + 0) = 229099 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 266388 N	, must exceed W = 48473 N	or W1 = 186196 N
Path 2-2 = 229099 N	, must exceed W = 48473 N	or W2 = 255592 N
Path 3-3 = 229099 N	, must exceed W = 48473 N	or W3 = 255592 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N52"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.4 \cdot 5508.0 \cdot 0.8997 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.4)$
 $= 7.1951 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.4 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0738 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 114.3280 mm
 Parallel to Vessel Wall Rn+tn+t 57.1640 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 37.5000 mm

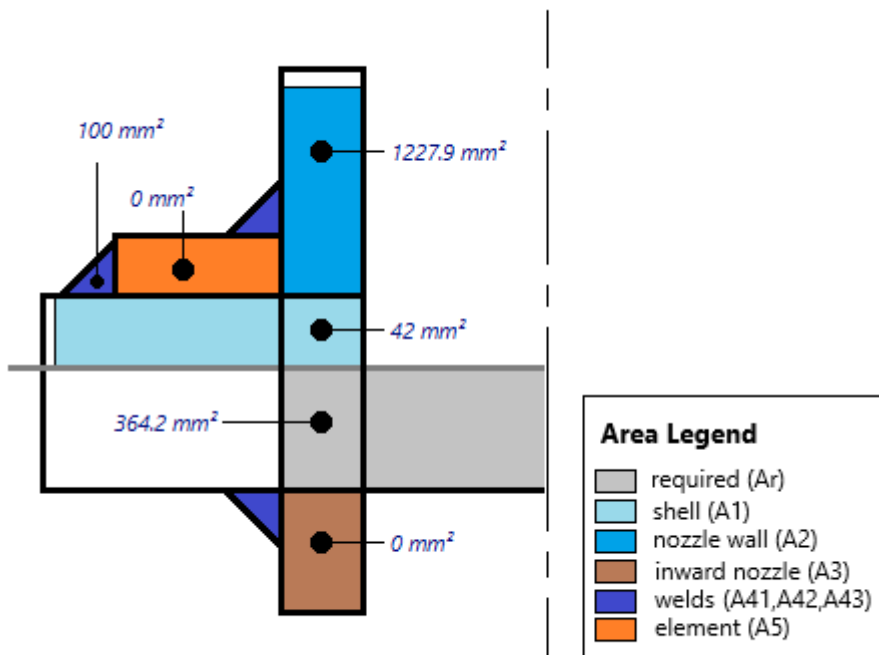


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	365.509	364.202	NA
Area in Shell	A1	495.833	42.014	NA
Area in Nozzle Wall	A2	1251.762	1227.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1847.595	1369.889	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 14.3386 \cdot 1 + 2 \cdot 16.764 \cdot 14.3386 \cdot 1(1-1.0))$
 $= 364.202 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 14.339) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 14.3386) \cdot (1 - 1.0)$
 $= 42.014 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.39) 1.0$
 $= 1227.874 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 0.3923 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.9934 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.9934 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.0631 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0631 mm
Wall Thickness per table UG-45	tb3 = 4.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(8.9934, 3.0631)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Weld Size Calculations, Description: N52"

Intermediate Calc. for nozzle/shell Welds Tmin 16.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) Sv)$
 $= \max(0, (364.2016 - 42.0144 + 2 \cdot 16.764 \cdot 1.0 \cdot$
 $(1.0 \cdot 15.0 - 14.3386)) 138)$
 $= 47516.93 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$

$$= (1227.8741 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$$
$$= 183227.73 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$
$$= (1227.8741 + 0.0 + 100.0 + (502.92)) \cdot 138$$
$$= 252623.53 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$
$$= (1227.8741 + 0.0 + 100.0 + 0.0 + (502.92)) \cdot 138$$
$$= 252623.53 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$
$$= (3.1416/2.0) \cdot 84.328 \cdot 10.0 \cdot 0.49 \cdot 138$$
$$= 89562. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$
$$= (3.1416 \cdot 33.782) \cdot (16.764 - 0.0) \cdot 0.7 \cdot 138$$
$$= 171851. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$
$$= (3.1416/2.0) \cdot 84.328 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 138$$
$$= 135259. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (89562 + 171851) = 261413 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (89562 + 0 + 135259 + 0) = 224820 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$
$$= (89562 + 135259 + 0) = 224820 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 261413 N	, must exceed W = 47516 N	or W1 = 183227 N
Path 2-2 = 224820 N	, must exceed W = 47516 N	or W2 = 252623 N
Path 3-3 = 224820 N	, must exceed W = 47516 N	or W3 = 252623 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 8.8549 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 231.5634 mm

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Input, Nozzle Desc: N9 22"

From: 90

Pressure for Reinforcement Calculations	P	0.3729	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2650.0010	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1550.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		22.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

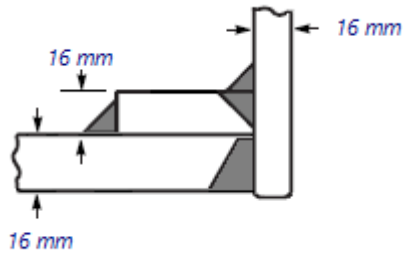
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	220.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1258.8000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		334.0001	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N9 22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	22.000 in.
Actual Thickness Used in Calculation	0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.37 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.37)$
 $= 7.4532 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.37 \cdot 279.4 / (138 \cdot 1.0 - 0.6 \cdot 0.37)$
 $= 0.7561 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.2552 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1117.6000 mm
Parallel to Vessel Wall, opening length	d	558.8000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

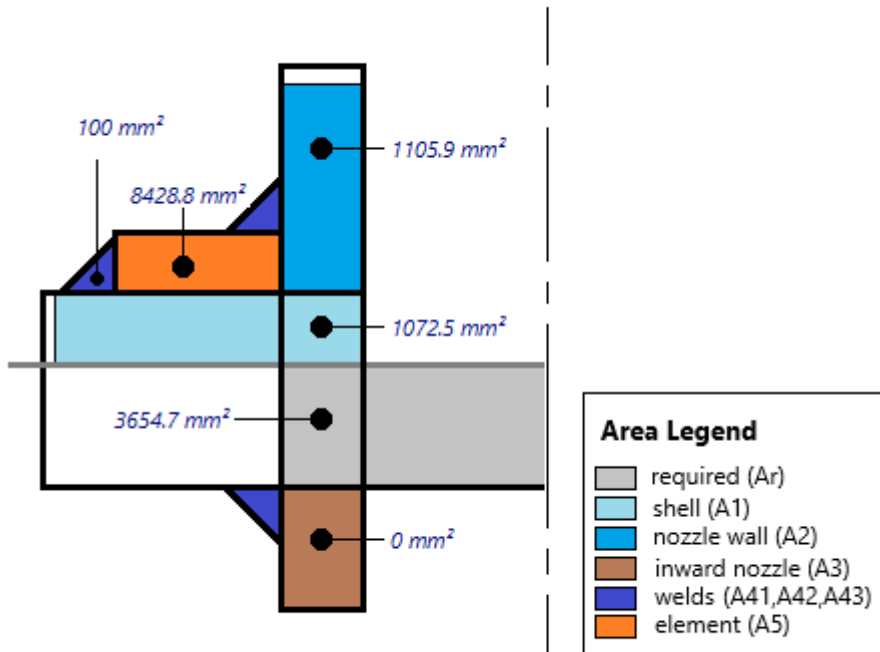


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4164.856	3654.726	NA
Area in Shell	A1	4217.144	1072.547	NA
Area in Nozzle Wall	A2	1143.290	1105.861	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	8428.800	8428.800	NA
TOTAL AREA AVAILABLE	Atot	13889.234	10707.208	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 676.8199 16.0000 mm
 Based on given Pad Diameter: 1258.8000 2.6126 mm
 Based on Shell or Nozzle Thickness: 676.8199 16.0000 mm

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(558.8 \cdot 13.0806 \cdot 1 + 2 \cdot 16.0 \cdot 13.0806 \cdot 1(1 - 1.0))$
 $= 3654.726 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 558.8(1.0 \cdot 15.0 - 1.0 \cdot 13.081) - 2 \cdot 16.0$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.0806) \cdot (1 - 1.0)$
 $= 1072.547 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (16.0 - 1.26) \cdot 1.0$

= 1105.861 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$
= 100.000 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(1117.6 - 590.8) \cdot 16.0 \cdot 1.0$
= 8428.800 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.2552 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.4532 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.4532 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.0648 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0648 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[8.331, max(8.4532, 3.0648)]
= 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.2552, 8.3312)
= 8.3312 mm

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 16.5,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 328.5 MPa	Passed
Occasional	: 3.2,	Allowable	: 183.5 MPa	Passed
Shear	: 10.5,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govern. thk, tg = 16.0, tr = 0.756, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.047$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govern. thk, tg = 16.0, tr = 0.756, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.047$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govern. thk, tg = 16.0, tr = 0.756, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.047$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 16.0$, $t_r = 7.453$, $c = 1.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.497$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 16.0$, $t_r = 7.453$, $c = 1.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.497$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -47 °C
 Gov. MDMT of the Reinforcement Pad : -47 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -47 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.37/1.96 = 0.190$

Weld Size Calculations, Description: N9 22"

Intermediate Calc. for nozzle/shell Welds T_{min} 16.0000 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$7.5000 = 0.5 * T_{minPad}$	$9.8980 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (3654.7263 - 1072.5474 + 2 \cdot 16.0 \cdot 1.0 \cdot (1.0 \cdot 15.0 - 13.0806)) \cdot 138)$
 $= 364779.03$ N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1105.8606 + 8428.7998 + 100.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 1329446.12$ N

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1105.8606 + 0.0 + 100.0 + (480.0)) \cdot 138$
 $= 232624.78$ N

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1105.8606 + 0.0 + 100.0 + 8428.7998 + (480.0)) \cdot 138$
 $= 1395679.25$ N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 590.8 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 627478. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 1258.8 \cdot 14.0 \cdot 0.49 \cdot 138$$

$$= 1871695. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 287.4) \cdot (16.0 - 0.0) \cdot 0.7 \cdot 138$$

$$= 1395394. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 590.8 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 1516192. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 590.8 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 947620. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (1871695 + 1395394) = 3267089 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (627478 + 1516192 + 947620 + 0) = 3091290 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (1871695 + 947620 + 0) = 2819315 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 3267088 N	, must exceed W = 364779 N	or W1 = 1329446 N
Path 2-2 = 3091290 N	, must exceed W = 364779 N	or W2 = 232624 N
Path 3-3 = 2819315 N	, must exceed W = 364779 N	or W3 = 1395679 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 15.8942 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 251.8942 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.784 %

Input Echo, WRC297 Item 1, Description: N9 22" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	5506.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	558.800	mm
Nozzle Thickness	16.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1258.800	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	0.37	MPa
Radial Load	P	-41250.00	N
Circumferential Shear	Vc	27499.98	N
Longitudinal Shear	Vl	27499.98	N
Circumferential Moment	Mc	0.424E+08	N-mm
Longitudinal Moment	Ml	0.424E+08	N-mm
Torsional Moment	Mt	0.605E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 590.8 + 2 \cdot 1.65 \cdot \sqrt{(2769.5(16.0 - 1.0))}$$

$$= 1263.405 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	5539.000	mm
Nozzle Outside Diameter	(d) =	590.800	mm
Vessel Thickness used	(T) =	31.000	mm
Nozzle Thickness used	(t) =	16.000	mm
T / t	=	1.938	
d / t	=	36.925	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.426$$

Nr/P	=	0.078
Mr/P	=	0.112
M0/P	=	0.064
N0/P	=	0.166
MrD/Mc	=	0.207
NrDL/Mc	=	0.110
M0d/Mc	=	0.133
N0DL/Mc	=	0.156
MrD/Ml	=	0.137
NrDL/Ml	=	0.081
M0D/Ml	=	0.078
N0DL/Ml	=	0.278

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-20	-20	20	20
Outplane Bending (ML)	-34	34	34	-34
Normal Pressure Stress	32	33	32	33

Outplane Stress Summary	0	38	112	9
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	28	-28	28	-28
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-61	61	61	-61
Inplane Pressure Stress	16	16	16	16

Inplane Stress Summary	-18	46	116	-64
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	21	48	117	74

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-59	59	59	-59
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16

Outplane Stress Summary	-31	55	111	-40
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	28	-28	28	-28
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-92	92	92	-92
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	32	33	32	33



Inplane Stress Summary -35 92 166 -76

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary 2 2 4 4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress 36 92 166 77

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses in the
 hoop direction)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-20	-20	20	20
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	8	7	8

Hoop Stress Summary -5 -5 35 35

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	70	-70	70	-70
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-162	162	162	-162
Axial Pressure Stress	3	3	3	3

Axial Stress Summary -97 86 249 -217

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress)

	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6

Shear Stress Summary 8 8 5 5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress 97 93 249 253

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses in the

	Cu Left	Cl Left	Du Right	Dl Right
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hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	8	7	8
Hoop Stress Summary	3	3	26	26

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	70	-70	70	-70
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-255	255	255	-255
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-189	179	341	-310

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6
Shear Stress Summary	5	5	8	8

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	193	179	341	337

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 590.8 + 2 \cdot 1.65 \cdot \sqrt{2761.5(16.0 - 1.0)}$
 $= 1262.433 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 5523.000 mm
 Nozzle Outside Diameter (d) = 1258.800 mm
 Vessel Thickness used (T) = 15.000 mm
 Nozzle Thickness used (t) = 350.000 mm
 $T / t = 0.043$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.373$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.051
 Mr/P = 0.078
 M0/P = 0.021
 N0/P = 0.023
 MrD/Mc = 0.246
 NrDL/Mc = 0.109

M0d/Mc = 0.073
 N0DL/Mc = 0.046
 MrD/Ml = 0.053
 NrDL/Ml = 0.060
 M0D/Ml = 0.016
 N0DL/Ml = 0.026

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	68	68	68	68
Outplane Stress Summary	77	60	114	38

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	85	-85	85	-85
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-47	47	47	-47
Inplane Pressure Stress	34	34	34	34
Inplane Stress Summary	72	-3	185	-81

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	78	63	185	119

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-65	65	65	-65
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34
Outplane Stress Summary	-11	74	134	-44

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	85	-85	85	-85
Inplane Membrane (Mc)	-16	-16	16	16
Inplane Bending (Mc)	-220	220	220	-220
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	68	68	68	68
Inplane Stress Summary	-73	196	400	-212

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	73	196	400	212

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	33	32	33	32	33	32	33
Circ. Pl (SUS)	-13	-13	27	27	-5	-5	11	11
Circ. Q (SUS)	-18	18	50	-50	-64	64	120	-120
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-3	-3	9	9	-4	-4	18	18
Long. Q (SUS)	-33	33	89	-89	-43	43	75	-75
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	32.0	33.0	32.0	33.0	32.0	33.0	32.0	33.0
Pm+Pl (SUS)	19.0	20.0	59.0	60.0	27.0	28.0	43.0	44.0
Pm+Pl+Q (Total)	21.8	47.0	115.4	74.2	38.2	92.2	163.2	76.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	60.00	207.00	Passed
Pm+Pl+Q (TOTAL)	163.17	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	68	68	68	68	68	68	68	68
Circ. Pl (SUS)	1	1	7	7	-7	-7	25	25
Circ. Q (SUS)	9	-9	37	-37	-135	135	305	-305
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	1	1	17	17	-2	-2	10	10
Long. Q (SUS)	38	-38	132	-132	-42	42	88	-88
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	68.0	68.0	68.0	68.0	68.0	68.0	68.0	68.0
Pm+Pl (SUS)	69.0	69.0	75.0	75.0	61.0	61.0	93.0	93.0
Pm+Pl+Q (Total)	78.2	63.0	183.0	119.0	74.0	196.0	398.0	212.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	68.00	138.00	Passed
Pm+Pl (SUS)	93.00	207.00	Passed
Pm+Pl+Q (TOTAL)	398.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	8	7	8	7	8	7	8
Circ. Pl (SUS)	-13	-13	27	27	-4	-4	18	18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-9	-9	11	11	-9	-9	11	11
Long. Q (SUS)	-92	92	232	-232	-185	185	325	-325
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	7.0	8.0	7.0	8.0	7.0	8.0	7.0	8.0
Pm+Pl (SUS)	7.0	6.6	34.0	35.0	9.2	10.2	25.1	26.1
Pm+Pl+Q (Total)	98.5	92.1	246.1	253.2	194.3	179.1	339.2	337.3

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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PV Elite 28 U2 Licensee: SL User
FileName : C-1901_R0_Ope
Nozzle Calcs.: N9 22"

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HEXAGON

Pm (SUS)	8.00	138.00	Passed
Pm+Pl (SUS)	35.05	207.00	Passed
Pm+Pl+Q (TOTAL)	339.16	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: M4 24"

From: 90

Pressure for Reinforcement Calculations	P	0.3729	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2650.0010	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1550.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-1.00	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

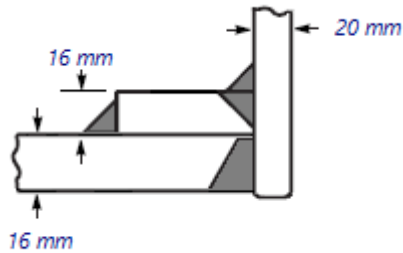
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	220.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1100.0000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		225.2000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M4 24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.37 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.37)$
 $= 7.4532 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.37 \cdot 304.8 / (138 \cdot 1.0 - 0.6 \cdot 0.37)$
 $= 0.8249 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3281 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1219.2000 mm
Parallel to Vessel Wall, opening length	d	609.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

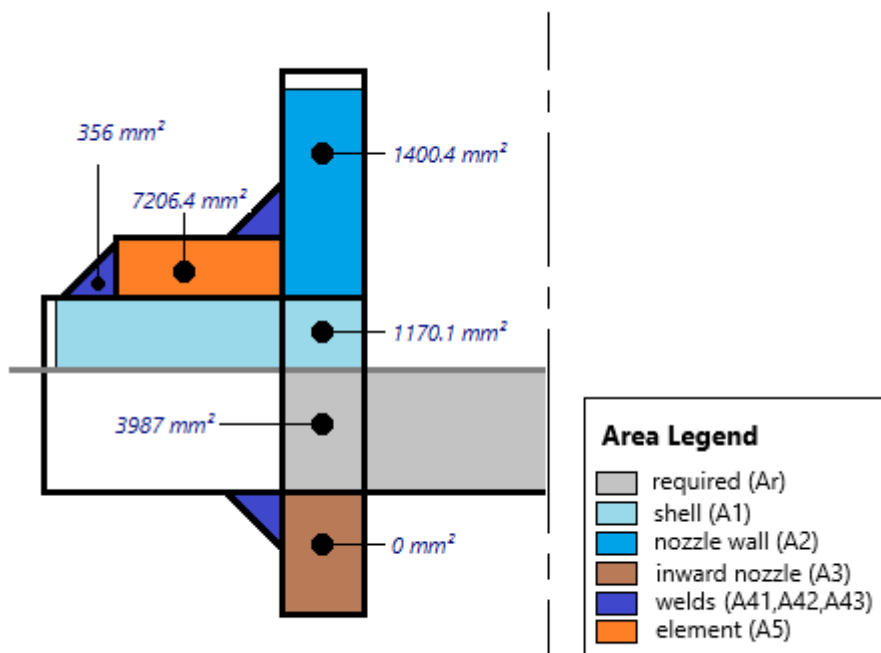


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4543.480	3986.974	NA
Area in Shell	A1	4600.520	1170.052	NA
Area in Nozzle Wall	A2	1438.135	1400.393	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		356.000	356.000	NA
Area in Element	A5	7206.400	7206.400	NA
TOTAL AREA AVAILABLE	Atot	13601.054	10132.846	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 715.8831 16.0000 mm
 Based on given Pad Diameter: 1100.0000 2.3546 mm
 Based on Shell or Nozzle Thickness: 715.8831 16.0000 mm

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(609.6 \cdot 13.0806 \cdot 1 + 2 \cdot 20.0 \cdot 13.0806 \cdot 1(1 - 1.0))$
 $= 3986.974 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 609.6(1.0 \cdot 15.0 - 1.0 \cdot 13.081) - 2 \cdot 20.0$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.0806) \cdot (1 - 1.0)$
 $= 1170.052 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (20.0 - 1.33) \cdot 1.0$

= 1400.393 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - c_{an}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 16.0^2 \cdot 1.0$
= 356.000 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(1100.0 - 649.6) \cdot 16.0 \cdot 1.0$
= 7206.400 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 1.3281$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
= 1.3281 mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 20.0$, $t_r = 0.825$, $c = 0.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.041$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 12 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 16.0$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.497$, Temp. Reduction = 33 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 16.0$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.497$, Temp. Reduction = 33 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 16.0$, $t_r = 7.453$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.497$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 16.0$, $t_r = 7.453$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.497$, Temp. Reduction = 33 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
Min Metal Temp. at Required thickness (UCS 66.1) -47 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -47 °C

Gov. MDMT of the Reinforcement Pad : -47 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -47 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.37/1.96 = 0.190

Weld Size Calculations, Description: M4 24"

Intermediate Calc. for nozzle/shell Welds Tmin 16.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	7.5000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (3986.9741 - 1170.0518 + 2 \cdot 20.0 \cdot 1.0 \cdot (1.0 \cdot 15.0 - 13.0806)) \cdot 138)$
 $= 399289.06 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - Can/.707)^2 \cdot f_r2) \cdot S_v$
 $= (1400.3933 + 7206.3999 + 356.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 1236737.88 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (1400.3933 + 0.0 + 100.0 + (600.0)) \cdot 138$
 $= 289824.38 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (1400.3933 + 0.0 + 356.0 + 7206.3999 + (600.0)) \cdot 138$
 $= 1319529.50 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 649.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 689928. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1100.0 \cdot 16.0 \cdot 0.49 \cdot 138$
 $= 1869231. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 314.8) \cdot (20.0 - 0.0) \cdot 0.7 \cdot 138$
 $= 1910534. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 649.6 \cdot 16.0 \cdot 0.74 \cdot 138$
 $= 1667093. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$



$$= (3.1416/2.0) \cdot 649.6 \cdot (14.0 - 0.0) \cdot 0.74 \cdot 138 \\ = 1458706. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1869231 + 1910534) = 3779765 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (689929 + 1667093 + 1458706 + 0) = 3815727 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (1869231 + 1458706 + 0) = 3327937 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 3779764 N	, must exceed W = 399289 N	or W1 = 1236737 N
Path 2-2 = 3815727 N	, must exceed W = 399289 N	or W2 = 289824 N
Path 3-3 = 3327936 N	, must exceed W = 399289 N	or W3 = 1319529 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 19.2271 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 255.2271 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 3.177 %

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Input, Nozzle Desc: N6A 2"

From: 90

Pressure for Reinforcement Calculations	P	0.3807	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2650.0010	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		750.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

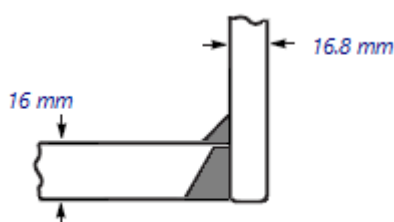
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N6A 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.38 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.38)$
 $= 7.6102 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.38 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0702 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3280 mm
Parallel to Vessel Wall	Rn+tn+t	57.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

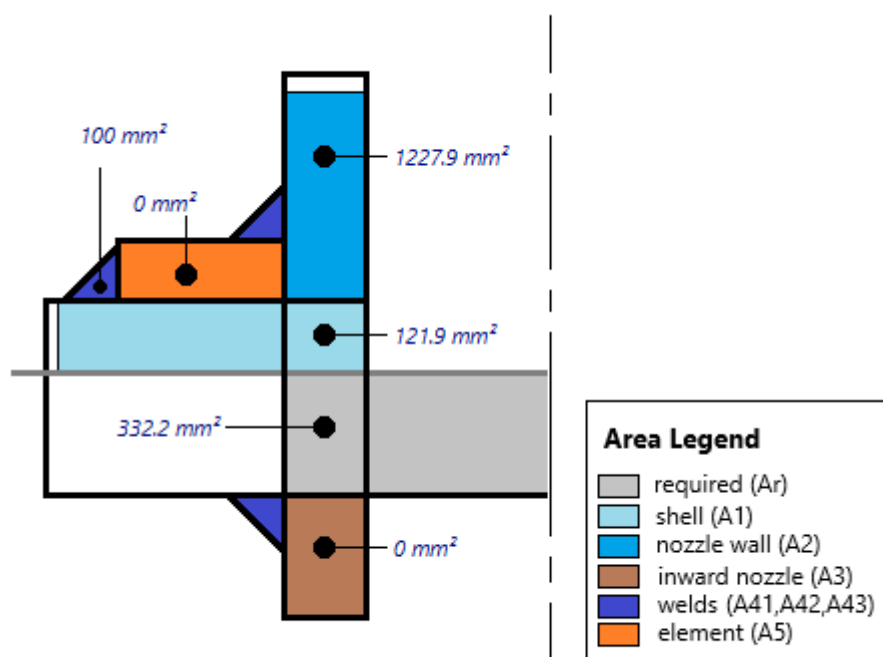


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	386.600	332.248	NA
Area in Shell	A1	469.457	121.934	NA
Area in Nozzle Wall	A2	1252.037	1227.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1821.494	1449.808	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 13.0806 \cdot 1 + 2 \cdot 16.764 \cdot 13.0806 \cdot 1(1 - 1.0))$
 $= 332.248 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 13.081) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.0806) \cdot (1 - 1.0)$
 $= 121.934 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.39) 1.0$
 $= 1227.874 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 8.6102 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 8.6102 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(8.6102, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.0$, $tr = 7.61$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.507$, Temp. Reduction = 31 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -26 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -26 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.38/5.10 = 0.075

Weld Size Calculations, Description: N6A 2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (332.2478 - 121.9341 + 2 • 16.764 • 1.0 •
(1.0 • 15.0 - 13.0806))138)
= 37900.08 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1227.8741 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 183227.73 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1227.8741 + 0.0 + 100.0 + (502.92)) • 138
= 252623.53 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1227.8741 + 0.0 + 100.0 + 0.0 + (502.92)) • 138
= 252623.53 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (14.0 - 0.0) • 0.74 • 138
= 189362. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 189362 + 0) = 278924 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 189362 + 0) = 278924 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 37900 N or W1 = 183227 N

Path 2-2 = 278923 N , must exceed W = 37900 N or W2 = 252623 N

Path 3-3 = 278923 N , must exceed W = 37900 N or W3 = 252623 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3228 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.6978 mm

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Input, Nozzle Desc: K4B 2"

From: 100

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	3000.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		5750.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

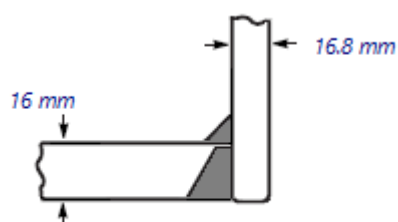
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K4B 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3280	mm
Parallel to Vessel Wall	Rn+tn+t	57.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

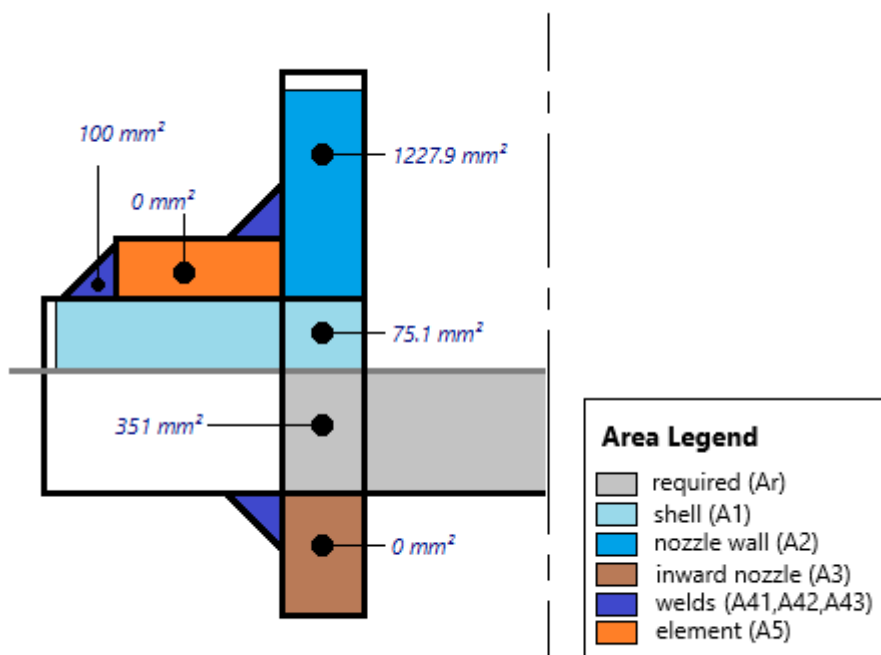


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	350.968	NA
Area in Shell	A1	510.861	75.114	NA
Area in Nozzle Wall	A2	1252.488	1227.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.349	1402.988	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 13.8176 \cdot 1 + 2 \cdot 16.764 \cdot 13.8176 \cdot 1(1-1.0))$
 $= 350.968 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 13.818) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.8176) \cdot (1 - 1.0)$
 $= 75.114 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.39) 1.0$
 $= 1227.874 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.464$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: K4B 2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (350.9678 - 75.1136 + 2 • 16.764 • 1.0 •
(1.0 • 15.0 - 13.8176))138)
= 43534.06 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1227.8741 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 183227.73 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1227.8741 + 0.0 + 100.0 + (502.92)) • 138
= 252623.53 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1227.8741 + 0.0 + 100.0 + 0.0 + (502.92)) • 138
= 252623.53 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (10.0 - 0.0) • 0.74 • 138
= 135259. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 135259 + 0) = 224820 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 135259 + 0) = 224820 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 43534 N or W1 = 183227 N

Path 2-2 = 224820 N , must exceed W = 43534 N or W2 = 252623 N

Path 3-3 = 224820 N , must exceed W = 43534 N or W3 = 252623 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3228 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.6978 mm

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Input, Nozzle Desc: K3B 2"

From: 100

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	3000.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		5550.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

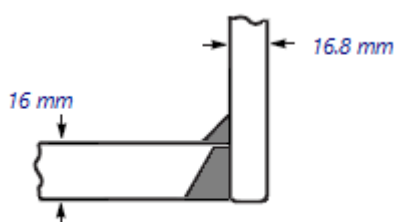
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K3B 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3280 mm
Parallel to Vessel Wall	Rn+tn+t	57.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

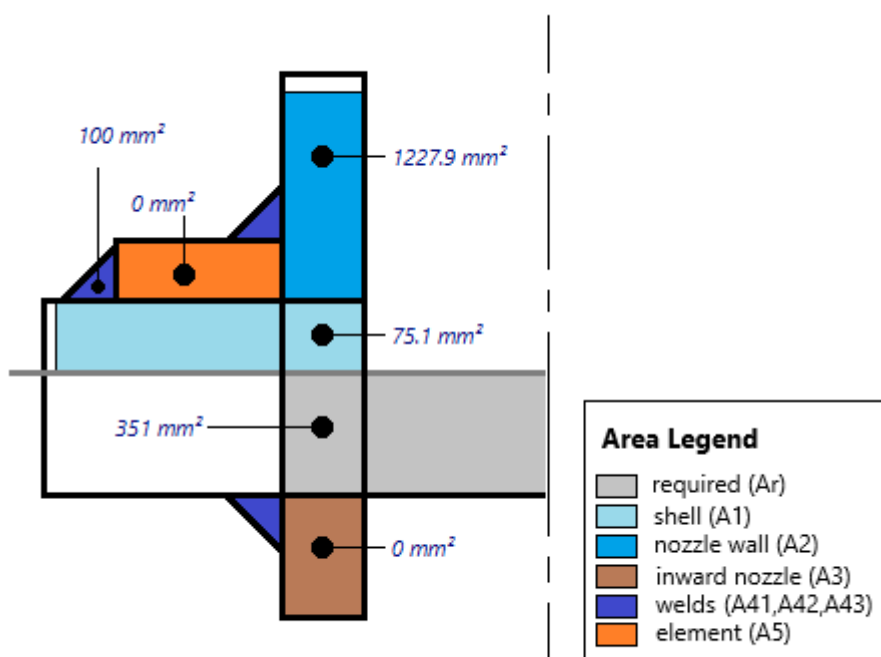


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	350.968	NA
Area in Shell	A1	510.861	75.114	NA
Area in Nozzle Wall	A2	1252.488	1227.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.349	1402.988	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 13.8176 \cdot 1 + 2 \cdot 16.764 \cdot 13.8176 \cdot 1(1-1.0))$
 $= 350.968 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 13.818) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.8176) \cdot (1 - 1.0)$
 $= 75.114 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.39) 1.0$
 $= 1227.874 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.464$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: K3B 2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (350.9678 - 75.1136 + 2 • 16.764 • 1.0 •
(1.0 • 15.0 - 13.8176))138)
= 43534.06 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1227.8741 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 183227.73 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1227.8741 + 0.0 + 100.0 + (502.92)) • 138
= 252623.53 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1227.8741 + 0.0 + 100.0 + 0.0 + (502.92)) • 138
= 252623.53 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (10.0 - 0.0) • 0.74 • 138
= 135259. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 135259 + 0) = 224820 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 135259 + 0) = 224820 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 43534 N or W1 = 183227 N

Path 2-2 = 224820 N , must exceed W = 43534 N or W2 = 252623 N

Path 3-3 = 224820 N , must exceed W = 43534 N or W3 = 252623 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3228 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.6978 mm

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Input, Nozzle Desc: K2B 2"

From: 100

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	3000.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		5350.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

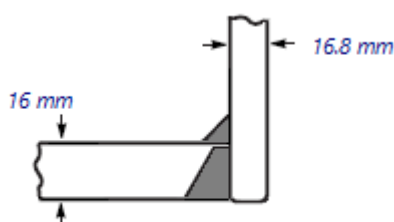
Flange Material		SA-106 B	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	220.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2B 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4035 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3280	mm
Parallel to Vessel Wall	Rn+tn+t	57.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

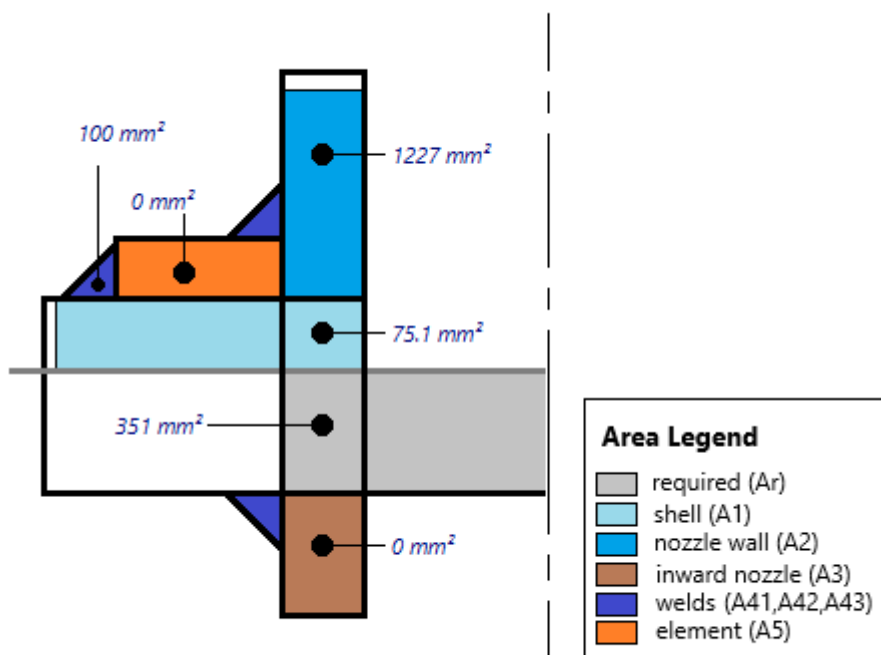


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	350.968	NA
Area in Shell	A1	510.861	75.114	NA
Area in Nozzle Wall	A2	1252.488	1227.037	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.349	1402.151	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 13.8176 \cdot 1 + 2 \cdot 16.764 \cdot 13.8176 \cdot 1(1-1.0))$
 $= 350.968 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 13.818) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.8176) \cdot (1 - 1.0)$
 $= 75.114 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.4) 1.0$
 $= 1227.037 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.4035 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.4035, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E*/(tg - c) = 0.464$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: K2B 2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (350.9678 - 75.1136 + 2 • 16.764 • 1.0 •
(1.0 • 15.0 - 13.8176))138)
= 43534.06 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1227.0372 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 183112.25 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1227.0372 + 0.0 + 100.0 + (502.92)) • 138
= 252508.08 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1227.0372 + 0.0 + 100.0 + 0.0 + (502.92)) • 138
= 252508.08 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (10.0 - 0.0) • 0.74 • 138
= 135259. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 135259 + 0) = 224820 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 135259 + 0) = 224820 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 43534 N or W1 = 183112 N

Path 2-2 = 224820 N , must exceed W = 43534 N or W2 = 252508 N

Path 3-3 = 224820 N , must exceed W = 43534 N or W3 = 252508 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3228 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 236.3228 mm

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Input, Nozzle Desc: K1 B2"

From: 100

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	3000.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		5050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

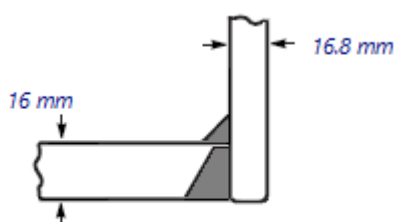
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K1 B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3280	mm
Parallel to Vessel Wall	Rn+tn+t	57.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

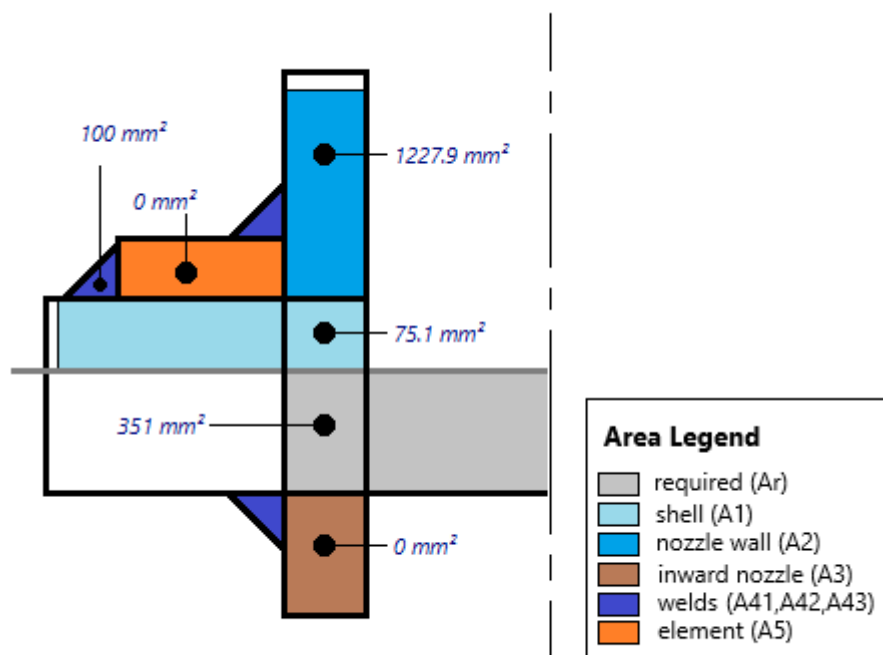


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	350.968	NA
Area in Shell	A1	510.861	75.114	NA
Area in Nozzle Wall	A2	1252.488	1227.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.349	1402.988	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 13.8176 \cdot 1 + 2 \cdot 16.764 \cdot 13.8176 \cdot 1(1-1.0))$
 $= 350.968 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 13.818) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.8176) \cdot (1 - 1.0)$
 $= 75.114 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.39) 1.0$
 $= 1227.874 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.464$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: K1 B2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (350.9678 - 75.1136 + 2 • 16.764 • 1.0 •
(1.0 • 15.0 - 13.8176))138)
= 43534.06 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1227.8741 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 183227.73 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1227.8741 + 0.0 + 100.0 + (502.92)) • 138
= 252623.53 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1227.8741 + 0.0 + 100.0 + 0.0 + (502.92)) • 138
= 252623.53 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (10.0 - 0.0) • 0.74 • 138
= 135259. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 135259 + 0) = 224820 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 135259 + 0) = 224820 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 43534 N or W1 = 183227 N

Path 2-2 = 224820 N , must exceed W = 43534 N or W2 = 252623 N

Path 3-3 = 224820 N , must exceed W = 43534 N or W3 = 252623 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3228 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.6978 mm

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Input, Nozzle Desc: K1A 2"

From: 100

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	3000.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		5050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

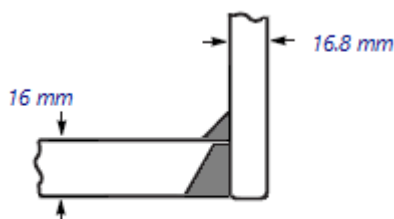
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K1A 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3280	mm
Parallel to Vessel Wall	Rn+tn+t	57.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

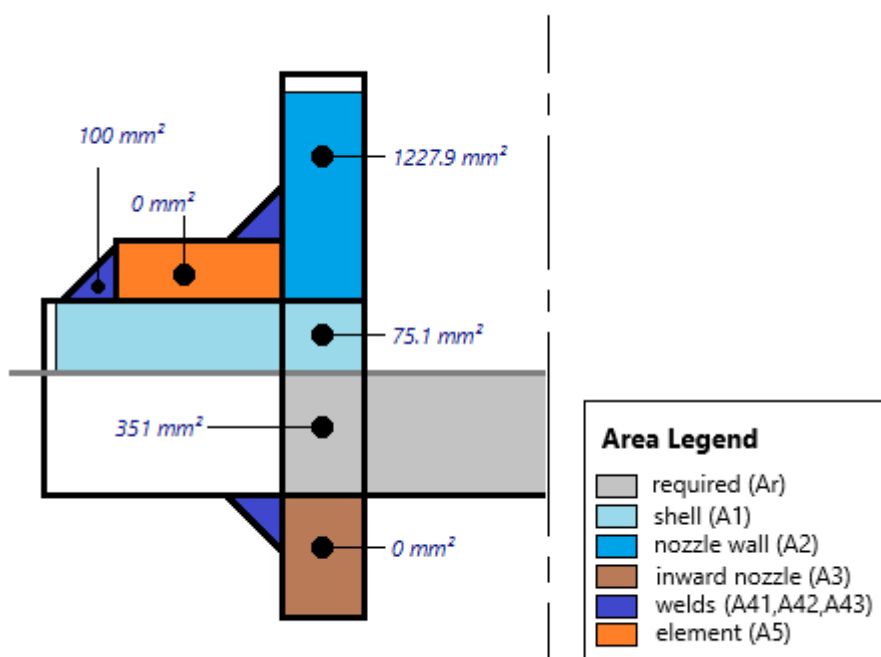


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	350.968	NA
Area in Shell	A1	510.861	75.114	NA
Area in Nozzle Wall	A2	1252.488	1227.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.349	1402.988	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 13.8176 \cdot 1 + 2 \cdot 16.764 \cdot 13.8176 \cdot 1(1-1.0))$
 $= 350.968 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 13.818) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.8176) \cdot (1 - 1.0)$
 $= 75.114 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.39) 1.0$
 $= 1227.874 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.464$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: K1A 2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (350.9678 - 75.1136 + 2 • 16.764 • 1.0 •
(1.0 • 15.0 - 13.8176))138)
= 43534.06 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1227.8741 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 183227.73 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1227.8741 + 0.0 + 100.0 + (502.92)) • 138
= 252623.53 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1227.8741 + 0.0 + 100.0 + 0.0 + (502.92)) • 138
= 252623.53 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (10.0 - 0.0) • 0.74 • 138
= 135259. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 135259 + 0) = 224820 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 135259 + 0) = 224820 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 43534 N or W1 = 183227 N

Path 2-2 = 224820 N , must exceed W = 43534 N or W2 = 252623 N

Path 3-3 = 224820 N , must exceed W = 43534 N or W3 = 252623 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3228 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.6978 mm

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Input, Nozzle Desc: K2A 2"

From: 100

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	3000.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4850.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

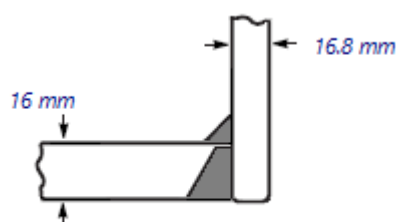
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K2A 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3280 mm
Parallel to Vessel Wall	Rn+tn+t	57.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

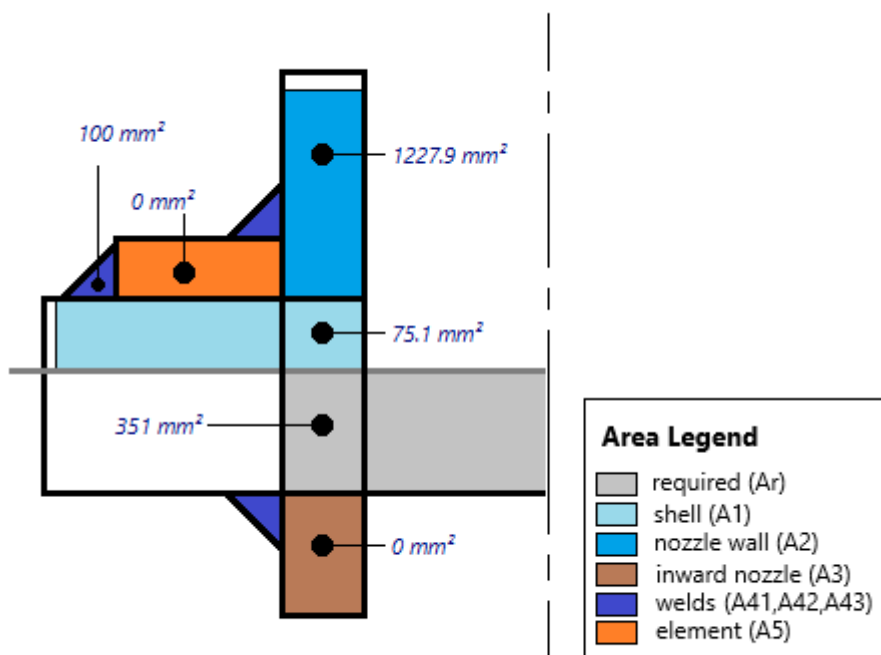


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	350.968	NA
Area in Shell	A1	510.861	75.114	NA
Area in Nozzle Wall	A2	1252.488	1227.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.349	1402.988	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 13.8176 \cdot 1 + 2 \cdot 16.764 \cdot 13.8176 \cdot 1(1-1.0))$
 $= 350.968 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 13.818) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.8176) \cdot (1 - 1.0)$
 $= 75.114 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.39) 1.0$
 $= 1227.874 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.464$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: K2A 2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (350.9678 - 75.1136 + 2 • 16.764 • 1.0 •
(1.0 • 15.0 - 13.8176))138)
= 43534.06 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1227.8741 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 183227.73 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1227.8741 + 0.0 + 100.0 + (502.92)) • 138
= 252623.53 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1227.8741 + 0.0 + 100.0 + 0.0 + (502.92)) • 138
= 252623.53 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (10.0 - 0.0) • 0.74 • 138
= 135259. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 135259 + 0) = 224820 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 135259 + 0) = 224820 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 43534 N or W1 = 183227 N

Path 2-2 = 224820 N , must exceed W = 43534 N or W2 = 252623 N

Path 3-3 = 224820 N , must exceed W = 43534 N or W3 = 252623 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3228 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.6978 mm

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Input, Nozzle Desc: K3A 2"

From: 100

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	3000.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4650.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

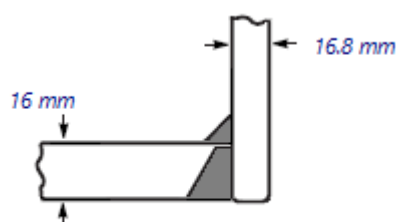
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K3A 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3280 mm
Parallel to Vessel Wall	Rn+tn+t	57.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

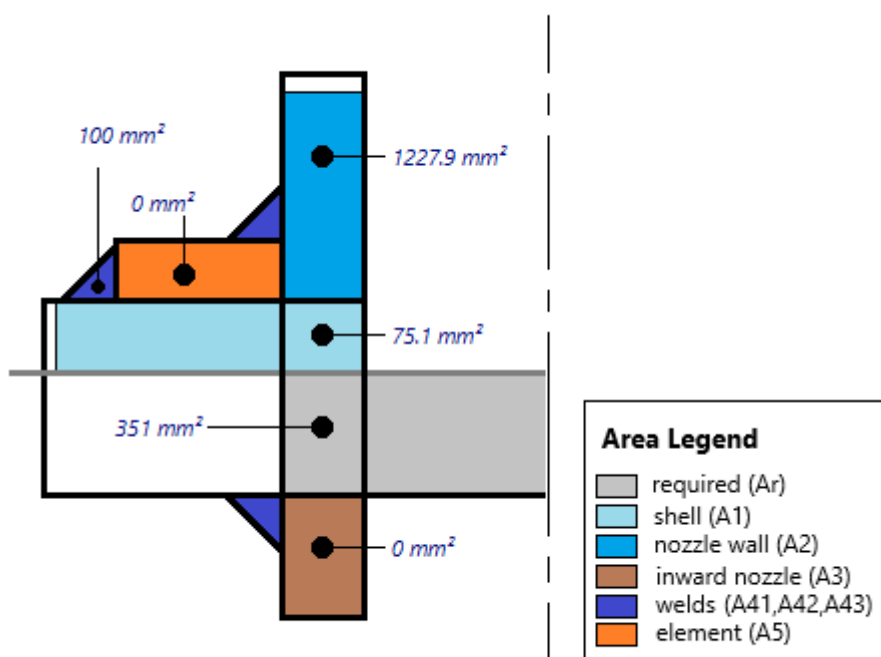


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	350.968	NA
Area in Shell	A1	510.861	75.114	NA
Area in Nozzle Wall	A2	1252.488	1227.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.349	1402.988	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 13.8176 \cdot 1 + 2 \cdot 16.764 \cdot 13.8176 \cdot 1(1-1.0))$
 $= 350.968 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 13.818) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.8176) \cdot (1 - 1.0)$
 $= 75.114 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.39) 1.0$
 $= 1227.874 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.464$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: K3A 2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (350.9678 - 75.1136 + 2 • 16.764 • 1.0 •
(1.0 • 15.0 - 13.8176))138)
= 43534.06 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1227.8741 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 183227.73 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1227.8741 + 0.0 + 100.0 + (502.92)) • 138
= 252623.53 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1227.8741 + 0.0 + 100.0 + 0.0 + (502.92)) • 138
= 252623.53 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (10.0 - 0.0) • 0.74 • 138
= 135259. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 135259 + 0) = 224820 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 135259 + 0) = 224820 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 43534 N or W1 = 183227 N

Path 2-2 = 224820 N , must exceed W = 43534 N or W2 = 252623 N

Path 3-3 = 224820 N , must exceed W = 43534 N or W3 = 252623 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3228 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.6978 mm

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Input, Nozzle Desc: K4A 2"

From: 100

Pressure for Reinforcement Calculations	P	0.3484	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	3000.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

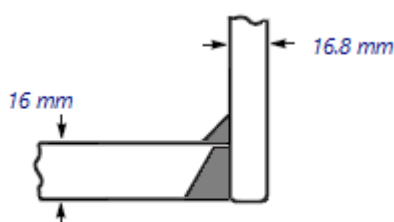
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K4A 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9626 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3280	mm
Parallel to Vessel Wall	Rn+tn+t	57.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

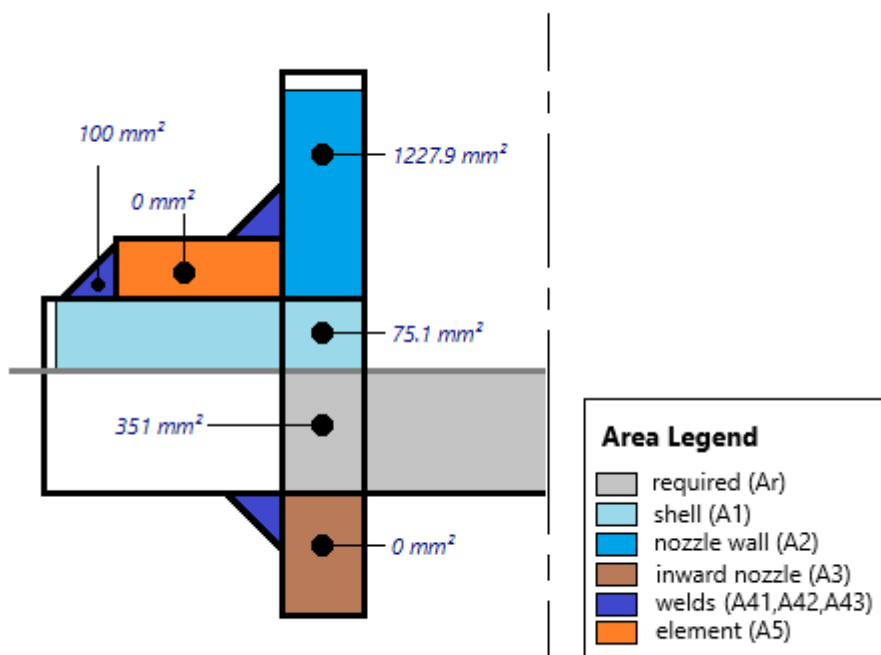


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.700	350.968	NA
Area in Shell	A1	510.600	75.114	NA
Area in Nozzle Wall	A2	1252.485	1227.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.085	1402.988	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 13.8176 \cdot 1 + 2 \cdot 16.764 \cdot 13.8176 \cdot 1(1-1.0))$
 $= 350.968 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 13.818) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.8176) \cdot (1 - 1.0)$
 $= 75.114 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.39) 1.0$
 $= 1227.874 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9626 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9626 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9626, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.0$, $tr = 6.963$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.464$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: K4A 2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (350.9678 - 75.1136 + 2 • 16.764 • 1.0 •
(1.0 • 15.0 - 13.8176))138)
= 43534.06 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1227.8741 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 183227.73 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1227.8741 + 0.0 + 100.0 + (502.92)) • 138
= 252623.53 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1227.8741 + 0.0 + 100.0 + 0.0 + (502.92)) • 138
= 252623.53 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (10.0 - 0.0) • 0.74 • 138
= 135259. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 135259 + 0) = 224820 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 135259 + 0) = 224820 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 43534 N or W1 = 183227 N

Path 2-2 = 224820 N , must exceed W = 43534 N or W2 = 252623 N

Path 3-3 = 224820 N , must exceed W = 43534 N or W3 = 252623 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3228 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.6978 mm

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Input, Nozzle Desc: N2A 104"

From: 100

Pressure for Reinforcement Calculations	P	0.3582	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	3000.0000	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	3899.9998	mm
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Distance from Bottom/Left Tangent		3050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

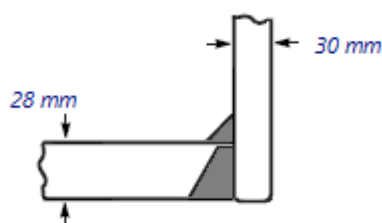
Diameter Basis (for tr calc only)		Outside	
Layout Angle		320.00	deg
Diameter		104.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	30.0000	mm

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2A 104"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 104.000 in.
 Actual Thickness Used in Calculation 1.181 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.36 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.36)$
 $= 7.1588 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.36 \cdot 1320.7999 / (138 \cdot 1.0 + 0.4 \cdot 0.36)$
 $= 3.4244 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 4.2096 mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 3899.9998 mm
 Parallel to Vessel Wall, opening length d 1949.9999 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 67.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

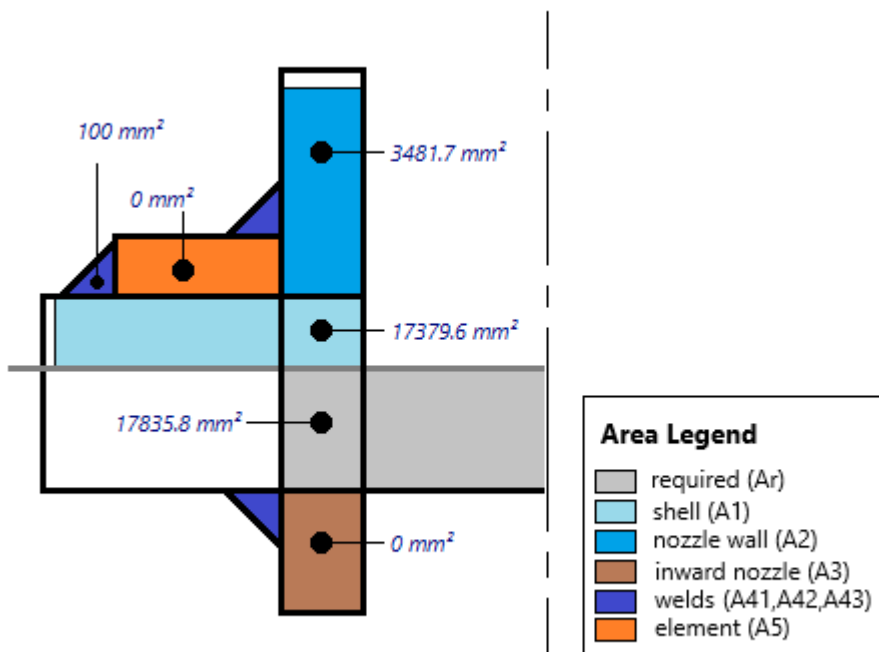


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	18481.250	17835.797	NA
Area in Shell	A1	26158.594	17379.637	NA
Area in Nozzle Wall	A2	3587.710	3481.701	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	29846.303	20961.338	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(2581.5999 \cdot 13.8176 \cdot 1 + 2 \cdot 30.0 \cdot 13.8176 \cdot 1(1-1.0))$
 $= 17835.797 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 1318.4(1.0 \cdot 27.0 - 1.0 \cdot 13.818) - 2 \cdot 30.0$
 $(1.0 \cdot 27.0 - 1.0 \cdot 13.8176) \cdot (1 - 1.0)$
 $= 17379.637 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 67.5)(30.0 - 4.21) 1.0$
 $= 3481.701 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.2096 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 8.1588 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 8.1588 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 8.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.331, \max(8.1588, 3.0648)]$
 $= 8.1588 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.2096, 8.1588)$
 $= 8.1588 \text{ mm}$

Available Nozzle Neck Thickness = 30.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	12.1,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	332.9 MPa	Passed
Occasional	:	7.6,	Allowable	:	183.5 MPa	Passed
Shear	:	3.8,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 28.0$, $t_r = 7.159$, $c = 1.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.265$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 3872.3999 mm
 Parallel to Vessel Wall, opening length d 1936.2000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 67.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [External Pressure]	AR	11891.127	11891.127
Area Available in Shell	A1	17015.805	17015.805
Area Available in Nozzle Wall	A2	3481.701	3481.701
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	100.000	100.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	20597.506	20597.506

The area available without a pad is Sufficient.

The following Calculation is performed using the internal design pressure (P) in the corroded condition.

Warning: The flange could not be considered in the moment of inertia calculations.

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

	Area	Distance	Inertia	d	I+Ad ²
Nozzle	15210.0	253.5	0.3258E+09	104.166	0.4908E+09
Shell	11664.0	13.5	0.7086E+06	135.834	0.2159E+09
Sum	26874.0				0.7068E+09

Membrane stress per Appendix 1-7 (1 or 2) S_m 118.66 MPa
 Allowable Membrane Stress S 138.00 MPa
 Bending stress at the inside of the Nozzle S_b 63.67 MPa
 Combined Membrane plus Bending Stress $S_m + S_b$ 182.32 MPa
 Allowable Membrane + Bending Stress $1.5 \cdot S$ 207.00 MPa
 M.A.W.P for the Appendix 1-7 Stress calc 0.41 MPa

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle = 6747.727 mm²
 Area from Shell = 7380.563 mm²
 Total Area, A_s = 14128.289 mm²

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [S_m]:

$$= P [R (R_n + t_n + \sqrt{(r_m \cdot t)}) + R_n (t + \sqrt{(R_n m \cdot t_n)})] / A_s$$

$$= 0.358 [2754.0 (1290.8 + 30.0 + 273.354) + 1290.8 (224.924)] / 14128.289$$

$$= 118.655 \text{ MPa}$$

The area A_s was computed per figure 1-7-1 for the S_m calculation.

Moment [M]:

$$= (R_n^3 / 6 + R \cdot R_n \cdot e) P$$

$$= (1290.8^3 / 6 + 2754.0 \cdot 1290.8 \cdot 135.834) 0.358$$

$$= 0.30142E+09 \text{ N-mm}$$



Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= (M(e + t/2)) / I$$

$$= (0.30142E+09(135.834 + 27.0/2)) / .70677E+09$$

$$= 63.667 \text{ MPa}$$

Weld Size Calculations, Description: N2A 104"

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (17835.7969 - 17379.6367 + 2 \cdot 30.0 \cdot 1.0 \cdot (1.0 \cdot 27.0 - 13.8176)) \cdot 138)$$

$$= 172082.45 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (3481.7007 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$$

$$= 494223.72 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (3481.7007 + 0.0 + 100.0 + (1620.0)) \cdot 138$$

$$= 717760.69 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (3481.7007 + 0.0 + 100.0 + 0.0 + (1620.0)) \cdot 138$$

$$= 717760.69 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 2641.5999 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 2805546. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 1305.7999) \cdot (30.0 - 0.0) \cdot 0.7 \cdot 138$$

$$= 11887430. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 2641.5999 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 4237022. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (2805546 + 11887430) = 14692976 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (2805546 + 0 + 4237023 + 0) = 7042568 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (2805546 + 4237023 + 0) = 7042568 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 14692976N , must exceed W = 172082 N or W1 = 494223 N
 Path 2-2 = 7042568 N , must exceed W = 172082 N or W2 = 717760 N
 Path 3-3 = 7042568 N , must exceed W = 172082 N or W3 = 717760 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 337.5298 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 865.5298 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.149 %

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N2A 104" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	5506.000	mm	
Shell Thickness	28.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	0.0000	mm	
Nozzle Diameter	2641.600	mm	
Nozzle Thickness	30.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-516 70		
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	0.36	MPa
Radial Load	P	-49030.75	N
Circumferential Shear	Vc	104121.89	N
Longitudinal Shear	Vl	237250.98	N
Circumferential Moment	Mc	0.475E+09	N-mm
Longitudinal Moment	Ml	0.567E+09	N-mm
Torsional Moment	Mt	0.529E+09	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)} \\
 &= 2641.6 + 2 \cdot 1.65 \cdot \sqrt{2767.5(28.0 - 1.0)} \\
 &= 3543.669 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 5535.000 mm
 Nozzle Outside Diameter (d) = 2641.600 mm
 Vessel Thickness used (T) = 27.000 mm
 Nozzle Thickness used (t) = 30.000 mm
 T / t = 0.900
 d / t = 88.053

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 6.833$$

Nr/P = 0.023
 Mr/P = 0.072
 M0/P = 0.022
 N0/P = 0.142
 MrD/Mc = 0.189
 NrDL/Mc = 0.040
 M0d/Mc = 0.064
 N0DL/Mc = 0.132
 MrD/M1 = 0.044
 NrDL/M1 = 0.022
 M0D/M1 = 0.014
 N0DL/M1 = 0.087

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-25	-25	25	25
Outplane Bending (ML)	-24	24	24	-24
Normal Pressure Stress	36	36	36	36
Outplane Stress Summary	4	36	104	38

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-77	77	77	-77
Inplane Pressure Stress	18	18	18	18
Inplane Stress Summary	-35	61	132	-80

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 40 62 132 119

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-32	-32	32	32
Outplane Bending (Mc)	-94	94	94	-94
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	18	18	18	18
Outplane Stress Summary	-90	80	163	-42

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-279	279	279	-279
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	36	36	36	36
Inplane Stress Summary	-222	278	356	-260

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	3	3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	222	278	356	260

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-25	-25	25	25
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18	18	18	18
Hoop Stress Summary	2	2	53	53

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-47	47	47	-47
Axial Pressure Stress	9	9	9	9

Axial Stress Summary	-21	33	80	-54
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	2	2	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	24	33	80	108

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-32	-32	32	32
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18	18	18	18

Hoop Stress Summary	-4	-4	60	60
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-202	202	202	-202
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9	9	9	9

Axial Stress Summary	-176	188	234	-210
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	1	1	1	1

Shear Stress Summary	0	0	3	3
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress 176 192 234 270

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	36	36	36	36	36	36	36	36
Circ. Pl (SUS)	-16	-16	34	34	-8	-8	10	10
Circ. Q (SUS)	-16	16	32	-32	-250	250	308	-308
Long. Pm (SUS)	18	18	18	18	18	18	18	18
Long. Pl (SUS)	-5	-5	7	7	-23	-23	41	41
Long. Q (SUS)	-48	48	106	-106	-86	86	102	-102
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-2	-2	2	2
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0
Pm+Pl (SUS)	20.0	20.0	70.0	70.0	33.2	33.2	59.3	59.3
Pm+Pl+Q (Total)	39.1	61.0	131.0	119.0	222.0	278.0	354.0	262.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	138.00	Passed
Pm+Pl (SUS)	70.00	207.00	Passed
Pm+Pl+Q (TOTAL)	354.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	18	18	18	18	18	18	18	18
Circ. Pl (SUS)	-16	-16	34	34	-23	-23	41	41
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-3	-3	3	3	-2	-2	2	2
Long. Q (SUS)	-28	28	66	-66	-183	183	221	-221
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	18.0	18.0	18.0	18.0	18.0	18.0	18.0	18.0
Pm+Pl (SUS)	6.0	6.0	52.0	52.0	12.2	12.2	59.0	59.0
Pm+Pl+Q (Total)	24.1	34.0	78.0	106.0	176.0	195.0	232.0	269.0

Nozzle Stress Summation Comparison (MPa):

Type of	Max. S.I.	S.I. Allowable	Result
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PV Elite 28 U2 Licensee: SL User
FileName : C-1901_R0_Ope
Nozzle Calcs.: N2A 104"

Nozl: 72 2:08pm

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HEXAGON

Stress Int.			
Pm (SUS)	18.00	138.00	Passed
Pm+Pl (SUS)	59.02	207.00	Passed
Pm+Pl+Q (TOTAL)	269.03	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N2B 104"

From: 100

Pressure for Reinforcement Calculations	P	0.3582	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	3000.0000	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	3899.9998	mm
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Distance from Bottom/Left Tangent		3050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

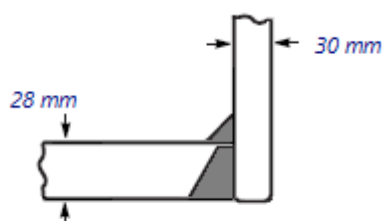
Diameter Basis (for tr calc only)		Outside	
Layout Angle		225.00	deg
Diameter		104.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	30.0000	mm

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2B 104"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 104.000 in.
 Actual Thickness Used in Calculation 1.181 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.36 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.36)$
 $= 7.1588 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.36 \cdot 1320.7999 / (138 \cdot 1.0 + 0.4 \cdot 0.36)$
 $= 3.4244 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 4.2096 mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 3899.9998 mm
 Parallel to Vessel Wall, opening length d 1949.9999 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 67.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

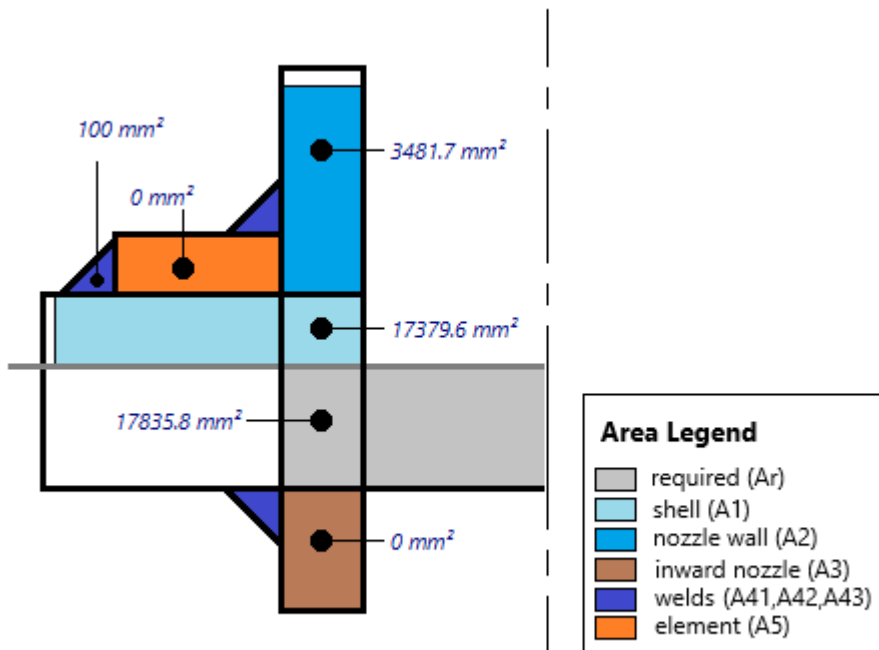


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	18481.250	17835.797	NA
Area in Shell	A1	26158.594	17379.637	NA
Area in Nozzle Wall	A2	3587.710	3481.701	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	29846.303	20961.338	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(2581.5999 \cdot 13.8176 \cdot 1 + 2 \cdot 30.0 \cdot 13.8176 \cdot 1(1-1.0))$
 $= 17835.797 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 1318.4(1.0 \cdot 27.0 - 1.0 \cdot 13.818) - 2 \cdot 30.0$
 $(1.0 \cdot 27.0 - 1.0 \cdot 13.8176) \cdot (1 - 1.0)$
 $= 17379.637 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 67.5)(30.0 - 4.21) 1.0$
 $= 3481.701 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.2096 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 8.1588 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 8.1588 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 8.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.331, \max(8.1588, 3.0648)]$
 $= 8.1588 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.2096, 8.1588)$
 $= 8.1588 \text{ mm}$

Available Nozzle Neck Thickness = 30.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 12.1,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 332.9 MPa	Passed
Occasional	: 7.6,	Allowable	: 183.5 MPa	Passed
Shear	: 3.8,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 28.0$, $t_r = 7.159$, $c = 1.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.265$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 3872.3999 mm
 Parallel to Vessel Wall, opening length d 1936.2000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 67.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [External Pressure]	AR	11891.127	11891.127
Area Available in Shell	A1	17015.805	17015.805
Area Available in Nozzle Wall	A2	3481.701	3481.701
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	100.000	100.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	20597.506	20597.506

The area available without a pad is Sufficient.

The following Calculation is performed using the internal design pressure (P) in the corroded condition.

Warning: The flange could not be considered in the moment of inertia calculations.

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

	Area	Distance	Inertia	d	I+Ad ²
Nozzle	15210.0	253.5	0.3258E+09	104.166	0.4908E+09
Shell	11664.0	13.5	0.7086E+06	135.834	0.2159E+09
Sum	26874.0				0.7068E+09

Membrane stress per Appendix 1-7 (1 or 2) S_m 118.66 MPa
 Allowable Membrane Stress S 138.00 MPa
 Bending stress at the inside of the Nozzle S_b 63.67 MPa
 Combined Membrane plus Bending Stress $S_m + S_b$ 182.32 MPa
 Allowable Membrane + Bending Stress $1.5 \cdot S$ 207.00 MPa
 M.A.W.P for the Appendix 1-7 Stress calc 0.41 MPa

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle = 6747.727 mm²
 Area from Shell = 7380.563 mm²
 Total Area, A_s = 14128.289 mm²

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [S_m]:

$$= P [R (R_n + t_n + \sqrt{(r_m \cdot t)}) + R_n (t + \sqrt{(R_n m \cdot t_n)})] / A_s$$

$$= 0.358 [2754.0 (1290.8 + 30.0 + 273.354) + 1290.8 (224.924)] / 14128.289$$

$$= 118.655 \text{ MPa}$$

The area A_s was computed per figure 1-7-1 for the S_m calculation.

Moment [M]:

$$= (R_n^3 / 6 + R \cdot R_n \cdot e) P$$

$$= (1290.8^3 / 6 + 2754.0 \cdot 1290.8 \cdot 135.834) 0.358$$

$$= 0.30142E+09 \text{ N-mm}$$



Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$\begin{aligned} &= (M(e + t/2)) / I \\ &= (0.30142E+09(135.834 + 27.0/2)) / .70677E+09 \\ &= 63.667 \text{ MPa} \end{aligned}$$

Weld Size Calculations, Description: N2B 104"

Intermediate Calc. for nozzle/shell welds T_{min} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (17835.7969 - 17379.6367 + 2 \cdot 30.0 \cdot 1.0 \cdot (1.0 \cdot 27.0 - 13.8176)) \cdot 138) \\ &= 172082.45 \text{ N} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A_2 + A_5 + A_4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v \\ &= (3481.7007 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138 \\ &= 494223.72 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (3481.7007 + 0.0 + 100.0 + (1620.0)) \cdot 138 \\ &= 717760.69 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (3481.7007 + 0.0 + 100.0 + 0.0 + (1620.0)) \cdot 138 \\ &= 717760.69 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 2641.5999 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 2805546. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [S_{nw}]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 1305.7999) \cdot (30.0 - 0.0) \cdot 0.7 \cdot 138 \\ &= 11887430. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [T_{ngw}]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 2641.5999 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 4237022. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{onw} + S_{nw}) = (2805546 + 11887430) = 14692976 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw}) \\ &= (2805546 + 0 + 4237023 + 0) = 7042568 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (S_{onw} + T_{ngw} + S_{inw}) \\ &= (2805546 + 4237023 + 0) = 7042568 \text{ N} \end{aligned}$$



Summary of Failure Path Calculations:

Path 1-1 = 14692976N , must exceed W = 172082 N or W1 = 494223 N
 Path 2-2 = 7042568 N , must exceed W = 172082 N or W2 = 717760 N
 Path 3-3 = 7042568 N , must exceed W = 172082 N or W3 = 717760 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 337.5298 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 865.5298 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.149 %

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N2B 104" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	5506.000	mm	
Shell Thickness	28.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	0.0000	mm	
Nozzle Diameter	2641.600	mm	
Nozzle Thickness	30.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-516 70		
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	0.36	MPa
Radial Load	P	-49030.75	N
Circumferential Shear	Vc	104121.89	N
Longitudinal Shear	Vl	237250.98	N
Circumferential Moment	Mc	0.475E+09	N-mm
Longitudinal Moment	Ml	0.567E+09	N-mm
Torsional Moment	Mt	0.529E+09	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))} \\
 &= 2641.6 + 2 \cdot 1.65 \cdot \sqrt{(2767.5(28.0 - 1.0))} \\
 &= 3543.669 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 5535.000 mm
 Nozzle Outside Diameter (d) = 2641.600 mm
 Vessel Thickness used (T) = 27.000 mm
 Nozzle Thickness used (t) = 30.000 mm
 T / t = 0.900
 d / t = 88.053

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 6.833$$

Nr/P = 0.023
 Mr/P = 0.072
 M0/P = 0.022
 N0/P = 0.142
 MrD/Mc = 0.189
 NrDL/Mc = 0.040
 M0d/Mc = 0.064
 N0DL/Mc = 0.132
 MrD/M1 = 0.044
 NrDL/M1 = 0.022
 M0D/M1 = 0.014
 N0DL/M1 = 0.087

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-25	-25	25	25
Outplane Bending (ML)	-24	24	24	-24
Normal Pressure Stress	36	36	36	36
Outplane Stress Summary	4	36	104	38

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-77	77	77	-77
Inplane Pressure Stress	18	18	18	18
Inplane Stress Summary	-35	61	132	-80

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 40 62 132 119

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-32	-32	32	32
Outplane Bending (Mc)	-94	94	94	-94
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	18	18	18	18
Outplane Stress Summary	-90	80	163	-42

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-279	279	279	-279
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	36	36	36	36
Inplane Stress Summary	-222	278	356	-260

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	3	3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	222	278	356	260

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-25	-25	25	25
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18	18	18	18
Hoop Stress Summary	2	2	53	53

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside



Axial Membrane (P)	0	0	0	0
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-47	47	47	-47
Axial Pressure Stress	9	9	9	9

Axial Stress Summary	-21	33	80	-54
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	2	2	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	24	33	80	108

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-32	-32	32	32
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	18	18	18	18

Hoop Stress Summary	-4	-4	60	60
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-202	202	202	-202
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	9	9	9	9

Axial Stress Summary	-176	188	234	-210
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	1	1	1	1

Shear Stress Summary	0	0	3	3
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress 176 192 234 270

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	36	36	36	36	36	36	36	36
Circ. Pl (SUS)	-16	-16	34	34	-8	-8	10	10
Circ. Q (SUS)	-16	16	32	-32	-250	250	308	-308
Long. Pm (SUS)	18	18	18	18	18	18	18	18
Long. Pl (SUS)	-5	-5	7	7	-23	-23	41	41
Long. Q (SUS)	-48	48	106	-106	-86	86	102	-102
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-2	-2	2	2
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0
Pm+Pl (SUS)	20.0	20.0	70.0	70.0	33.2	33.2	59.3	59.3
Pm+Pl+Q (Total)	39.1	61.0	131.0	119.0	222.0	278.0	354.0	262.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	138.00	Passed
Pm+Pl (SUS)	70.00	207.00	Passed
Pm+Pl+Q (TOTAL)	354.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	18	18	18	18	18	18	18	18
Circ. Pl (SUS)	-16	-16	34	34	-23	-23	41	41
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	9	9	9	9	9	9	9	9
Long. Pl (SUS)	-3	-3	3	3	-2	-2	2	2
Long. Q (SUS)	-28	28	66	-66	-183	183	221	-221
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	18.0	18.0	18.0	18.0	18.0	18.0	18.0	18.0
Pm+Pl (SUS)	6.0	6.0	52.0	52.0	12.2	12.2	59.0	59.0
Pm+Pl+Q (Total)	24.1	34.0	78.0	106.0	176.0	195.0	232.0	269.0

Nozzle Stress Summation Comparison (MPa):

Type of	Max. S.I.	S.I. Allowable	Result
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PV Elite 28 U2 Licensee: SL User
FileName : C-1901_R0_Ope
Nozzle Calcs.: N2B 104"

Nozl: 73 2:08pm

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HEXAGON

Stress Int.			
Pm (SUS)	18.00	138.00	Passed
Pm+Pl (SUS)	59.02	207.00	Passed
Pm+Pl+Q (TOTAL)	269.03	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: M3 24"

From: 110

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2649.9980	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		6950.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-1.00	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

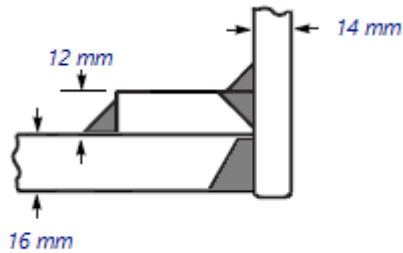
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	220.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1149.6000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		256.0000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M3 24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.551 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 304.8 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 0.7701 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3133 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1219.2000 mm
Parallel to Vessel Wall, opening length	d	609.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

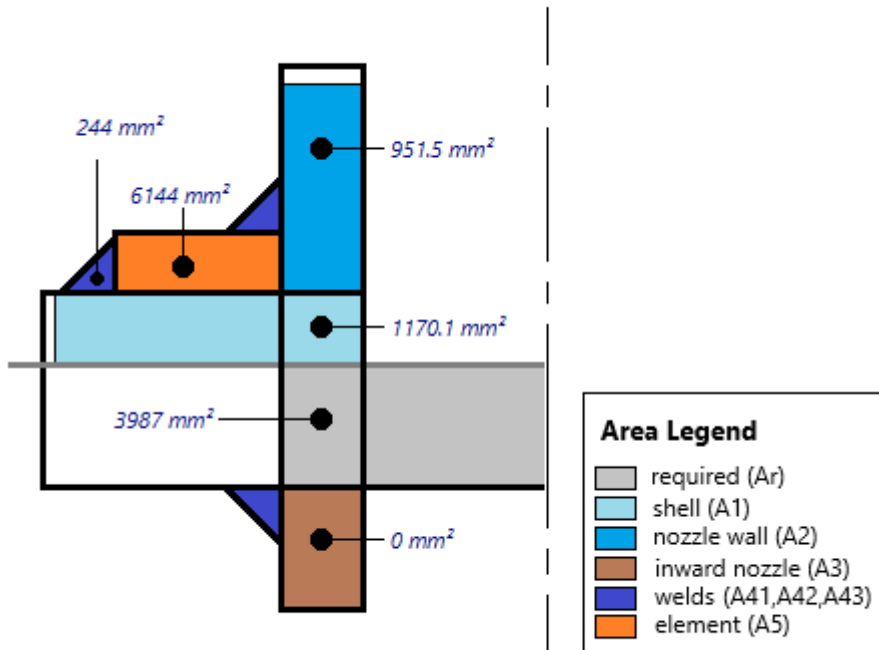


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4241.891	3986.972	NA
Area in Shell	A1	4902.108	1170.056	NA
Area in Nozzle Wall	A2	992.241	951.505	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		244.000	244.000	NA
Area in Element	A5	6144.000	6144.000	NA
TOTAL AREA AVAILABLE	Atot	12282.351	8509.562	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 772.7175 12.0000 mm
 Based on given Pad Diameter: 1149.6000 3.1668 mm
 Based on Shell or Nozzle Thickness: 753.4150 14.0000 mm

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(609.6 \cdot 13.0806 \cdot 1 + 2 \cdot 14.0 \cdot 13.0806 \cdot 1(1 - 1.0))$
 $= 3986.972 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 609.6(1.0 \cdot 15.0 - 1.0 \cdot 13.081) - 2 \cdot 14.0$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.0806) \cdot (1 - 1.0)$
 $= 1170.056 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (14.0 - 1.31) \cdot 1.0$

= 951.505 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 12.0^2 \cdot 1.0$
= 244.000 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(1149.6 - 637.6) \cdot 12.0 \cdot 1.0$
= 6144.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 1.3133$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
= 1.3133 mm

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 14.0$, $t_r = 0.77$, $c = 0.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.055$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 12.0$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.464$, Temp. Reduction = 38 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 12.0$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.464$, Temp. Reduction = 38 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 12.0$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.464$, Temp. Reduction = 38 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 14.0$, $t_r = 0.77$, $c = 0.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.055$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle : -48 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.35/1.96 = 0.178

Weld Size Calculations, Description: M3 24"

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 12.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	6.0000 = 0.5*TminPad	8.4840 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (3986.9719 - 1170.0563 + 2 \cdot 14.0 \cdot 1.0 \cdot (1.0 \cdot 15.0 - 13.0806)) \cdot 138)$
 $= 396110.03 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (951.5051 + 6144.0 + 244.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 1012747.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (951.5051 + 0.0 + 100.0 + (420.0)) \cdot 138$
 $= 203046.77 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (951.5051 + 0.0 + 244.0 + 6144.0 + (420.0)) \cdot 138$
 $= 1070701.25 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 637.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 677184. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1149.6 \cdot 12.0 \cdot 0.49 \cdot 138$
 $= 1465137. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 311.8) \cdot (14.0 - 0.0) \cdot 0.7 \cdot 138$
 $= 1324629. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 637.6 \cdot 12.0 \cdot 0.74 \cdot 138$
 $= 1227222. \text{ N}$

Tension, Shell Groove Weld [Tngw]:



$$\begin{aligned} &= (\pi/2) \cdot D_o \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 637.6 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 1227222. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (1465137 + 1324629) = 2789766 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (677184 + 1227223 + 1227223 + 0) = 3131629 \text{ N}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (1465137 + 1227223 + 0) = 2692360 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 2789766 N	, must exceed W = 396110 N	or W1 = 1012747 N
Path 2-2 = 3131628 N	, must exceed W = 396110 N	or W2 = 203046 N
Path 3-3 = 2692359 N	, must exceed W = 396110 N	or W3 = 1070701 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 18.5210 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 254.5210 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.245 %

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Input, Nozzle Desc: N6B 2"

From: 110

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2649.9980	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		6050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

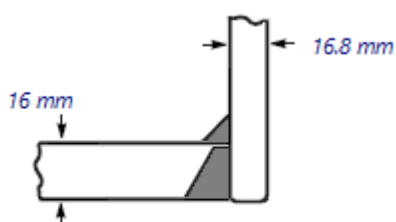
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N6B 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3280	mm
Parallel to Vessel Wall	Rn+tn+t	57.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

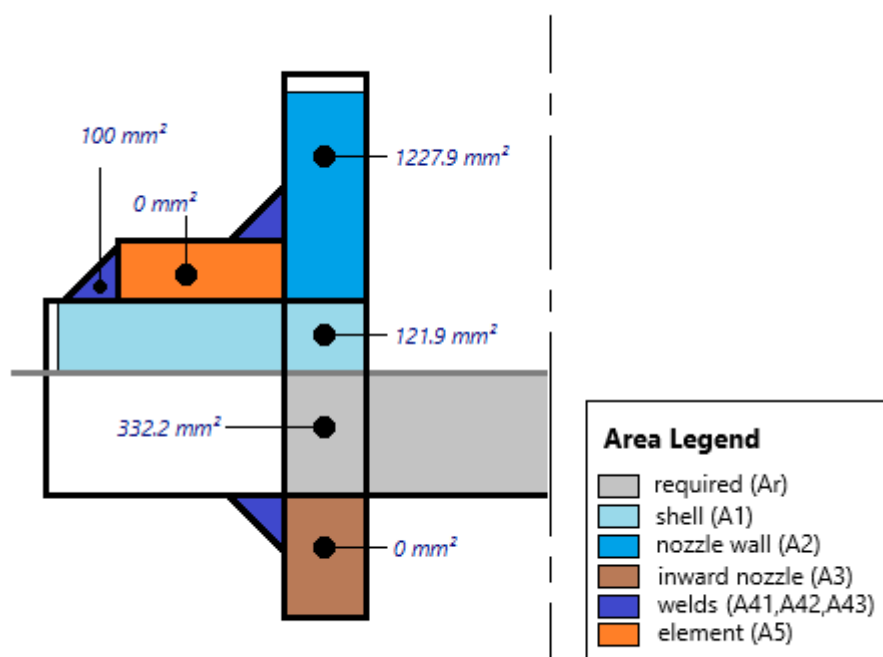


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	332.248	NA
Area in Shell	A1	510.861	121.935	NA
Area in Nozzle Wall	A2	1252.488	1227.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.349	1449.809	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 13.0806 \cdot 1 + 2 \cdot 16.764 \cdot 13.0806 \cdot 1(1-1.0))$
 $= 332.248 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 13.081) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.0806) \cdot (1 - 1.0)$
 $= 121.935 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.39) 1.0$
 $= 1227.874 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.464$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: N6B 2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (332.2477 - 121.9346 + 2 • 16.764 • 1.0 •
(1.0 • 15.0 - 13.0806))138)
= 37900.02 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1227.8741 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 183227.73 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1227.8741 + 0.0 + 100.0 + (502.92)) • 138
= 252623.53 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1227.8741 + 0.0 + 100.0 + 0.0 + (502.92)) • 138
= 252623.53 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (12.0 - 0.0) • 0.74 • 138
= 162311. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 162311 + 0) = 251872 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 162311 + 0) = 251872 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 37900 N or W1 = 183227 N

Path 2-2 = 251872 N , must exceed W = 37900 N or W2 = 252623 N

Path 3-3 = 251872 N , must exceed W = 37900 N or W3 = 252623 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3228 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.6978 mm

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Input, Nozzle Desc: K5A2"

From: 110

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2649.9980	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		8550.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

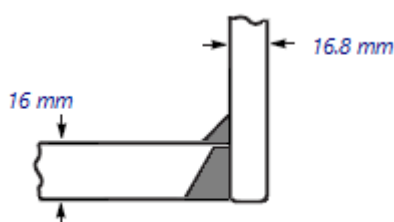
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K5A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3280 mm
Parallel to Vessel Wall	Rn+tn+t	57.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

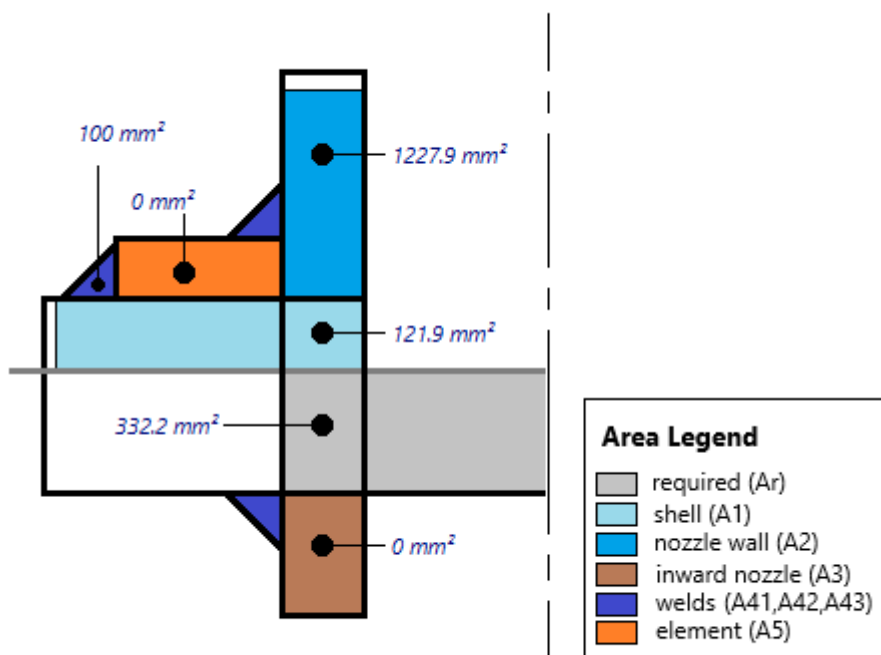


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	332.248	NA
Area in Shell	A1	510.861	121.935	NA
Area in Nozzle Wall	A2	1252.488	1227.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.349	1449.809	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 13.0806 \cdot 1 + 2 \cdot 16.764 \cdot 13.0806 \cdot 1(1-1.0))$
 $= 332.248 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 13.081) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.0806) \cdot (1 - 1.0)$
 $= 121.935 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.39) 1.0$
 $= 1227.874 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.464$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: K5A2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (332.2477 - 121.9346 + 2 • 16.764 • 1.0 •
(1.0 • 15.0 - 13.0806))138)
= 37900.02 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1227.8741 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 183227.73 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1227.8741 + 0.0 + 100.0 + (502.92)) • 138
= 252623.53 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1227.8741 + 0.0 + 100.0 + 0.0 + (502.92)) • 138
= 252623.53 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (12.0 - 0.0) • 0.74 • 138
= 162311. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 162311 + 0) = 251872 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 162311 + 0) = 251872 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 37900 N or W1 = 183227 N

Path 2-2 = 251872 N , must exceed W = 37900 N or W2 = 252623 N

Path 3-3 = 251872 N , must exceed W = 37900 N or W3 = 252623 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3228 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.6978 mm

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Input, Nozzle Desc: K6A2"

From: 110

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2649.9980	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		8550.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

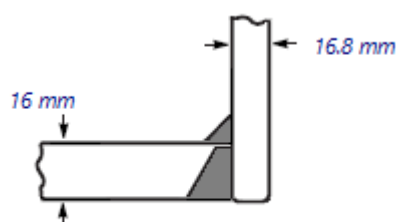
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K6A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3280 mm
Parallel to Vessel Wall	Rn+tn+t	57.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

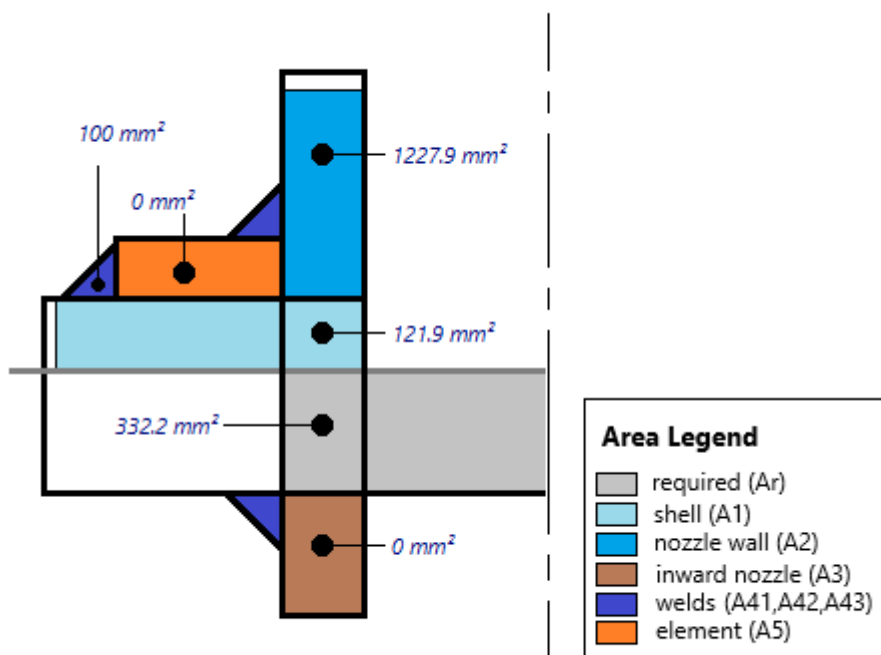


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	332.248	NA
Area in Shell	A1	510.861	121.935	NA
Area in Nozzle Wall	A2	1252.488	1227.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.349	1449.809	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 13.0806 \cdot 1 + 2 \cdot 16.764 \cdot 13.0806 \cdot 1(1-1.0))$
 $= 332.248 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 13.081) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.0806) \cdot (1 - 1.0)$
 $= 121.935 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.39) 1.0$
 $= 1227.874 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.464$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: K6A2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (332.2477 - 121.9346 + 2 • 16.764 • 1.0 •
(1.0 • 15.0 - 13.0806))138)
= 37900.02 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1227.8741 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 183227.73 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1227.8741 + 0.0 + 100.0 + (502.92)) • 138
= 252623.53 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1227.8741 + 0.0 + 100.0 + 0.0 + (502.92)) • 138
= 252623.53 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (12.0 - 0.0) • 0.74 • 138
= 162311. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 162311 + 0) = 251872 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 162311 + 0) = 251872 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 37900 N or W1 = 183227 N

Path 2-2 = 251872 N , must exceed W = 37900 N or W2 = 252623 N

Path 3-3 = 251872 N , must exceed W = 37900 N or W3 = 252623 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3228 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.6978 mm

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Input, Nozzle Desc: N10A2"

From: 110

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2649.9980	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		6550.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

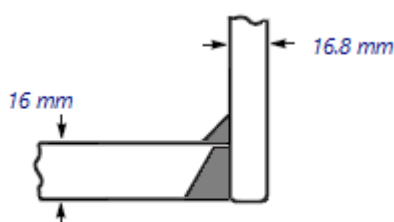
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N10A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3280	mm
Parallel to Vessel Wall	Rn+tn+t	57.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

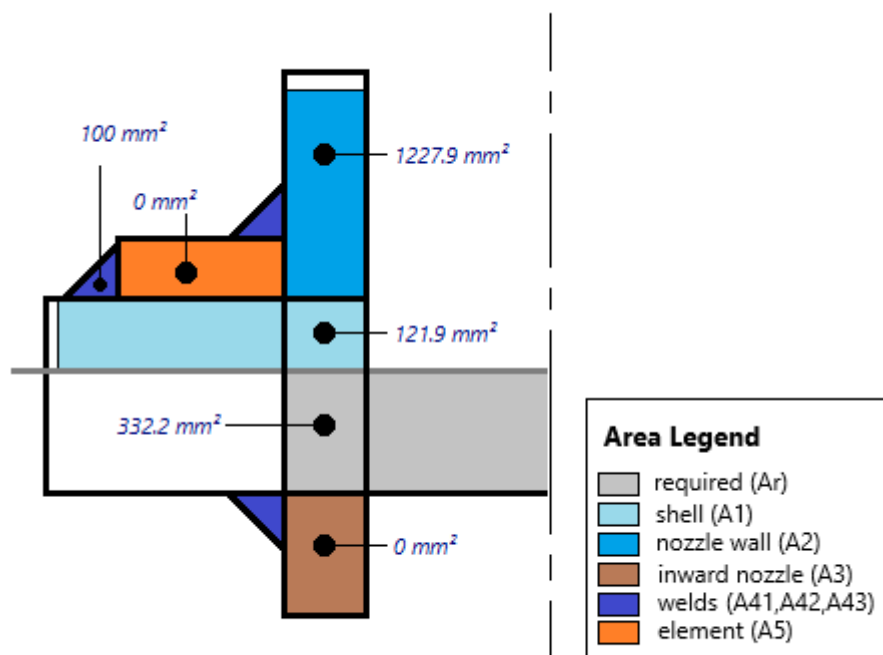


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	332.248	NA
Area in Shell	A1	510.861	121.935	NA
Area in Nozzle Wall	A2	1252.488	1227.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.349	1449.809	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 13.0806 \cdot 1 + 2 \cdot 16.764 \cdot 13.0806 \cdot 1(1-1.0))$
 $= 332.248 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 13.081) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.0806) \cdot (1 - 1.0)$
 $= 121.935 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.39) 1.0$
 $= 1227.874 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.464$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: N10A2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (332.2477 - 121.9346 + 2 • 16.764 • 1.0 •
(1.0 • 15.0 - 13.0806))138)
= 37900.02 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1227.8741 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 183227.73 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1227.8741 + 0.0 + 100.0 + (502.92)) • 138
= 252623.53 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1227.8741 + 0.0 + 100.0 + 0.0 + (502.92)) • 138
= 252623.53 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (12.0 - 0.0) • 0.74 • 138
= 162311. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 162311 + 0) = 251872 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 162311 + 0) = 251872 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 37900 N or W1 = 183227 N

Path 2-2 = 251872 N , must exceed W = 37900 N or W2 = 252623 N

Path 3-3 = 251872 N , must exceed W = 37900 N or W3 = 252623 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3228 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.6978 mm

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Input, Nozzle Desc: N10B2"

From: 110

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2649.9980	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

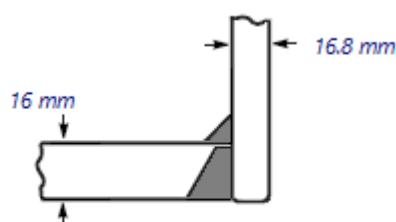
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N10B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3280	mm
Parallel to Vessel Wall	Rn+tn+t	57.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	37.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

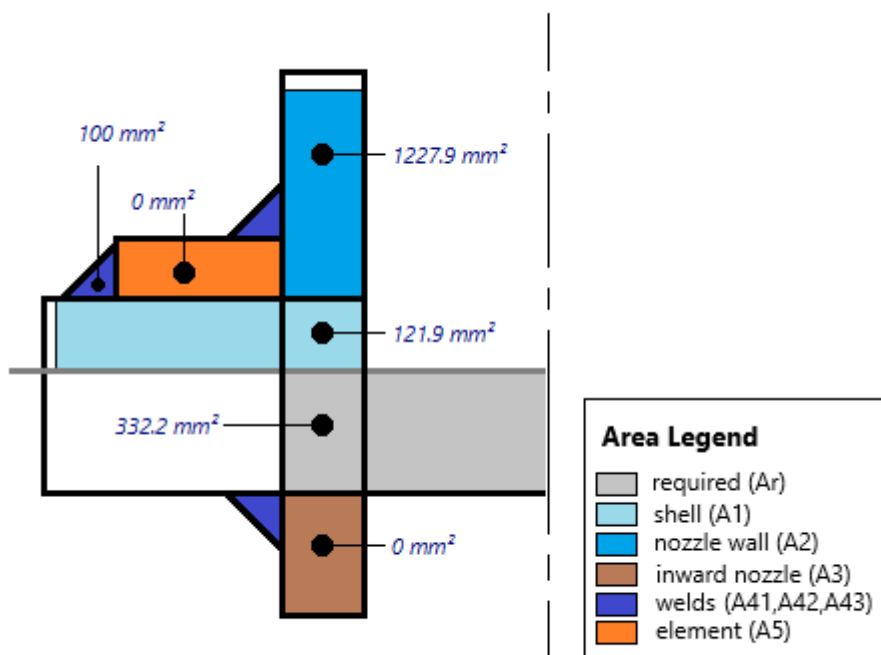


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	332.248	NA
Area in Shell	A1	510.861	121.935	NA
Area in Nozzle Wall	A2	1252.488	1227.874	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1863.349	1449.809	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 13.0806 \cdot 1 + 2 \cdot 16.764 \cdot 13.0806 \cdot 1(1-1.0))$
 $= 332.248 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 63.528(1.0 \cdot 15.0 - 1.0 \cdot 13.081) - 2 \cdot 16.764$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.0806) \cdot (1 - 1.0)$
 $= 121.935 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 37.5)(16.76 - 0.39) 1.0$
 $= 1227.874 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.464$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: N10B2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (332.2477 - 121.9346 + 2 • 16.764 • 1.0 •
(1.0 • 15.0 - 13.0806))138)
= 37900.02 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1227.8741 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 183227.73 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1227.8741 + 0.0 + 100.0 + (502.92)) • 138
= 252623.53 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1227.8741 + 0.0 + 100.0 + 0.0 + (502.92)) • 138
= 252623.53 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (12.0 - 0.0) • 0.74 • 138
= 162311. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 162311 + 0) = 251872 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 162311 + 0) = 251872 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 37900 N or W1 = 183227 N

Path 2-2 = 251872 N , must exceed W = 37900 N or W2 = 252623 N

Path 3-3 = 251872 N , must exceed W = 37900 N or W3 = 252623 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3228 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.6978 mm

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Input, Nozzle Desc: M2 24"

From: 130

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2204.9993	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		9050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

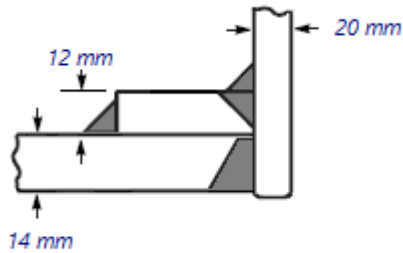
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	220.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1100.0000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		225.2000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M2 24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 304.8 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 0.7701 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3281 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1219.2000 mm
Parallel to Vessel Wall, opening length	d	609.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

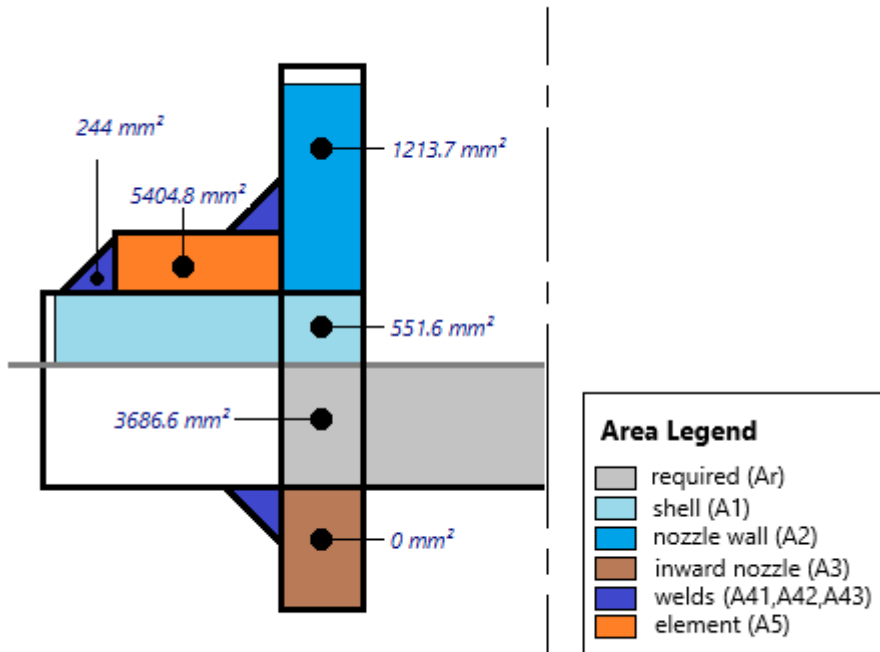


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4241.891	3686.611	NA
Area in Shell	A1	3682.909	551.578	NA
Area in Nozzle Wall	A2	1249.942	1213.674	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		244.000	244.000	NA
Area in Element	A5	5404.800	5404.800	NA
TOTAL AREA AVAILABLE	Atot	10581.652	7414.053	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 789.3799 12.0000 mm
 Based on given Pad Diameter: 1100.0000 3.7242 mm
 Based on Shell or Nozzle Thickness: 769.4113 14.0000 mm

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(609.6 \cdot 12.0952 \cdot 1 + 2 \cdot 20.0 \cdot 12.0952 \cdot 1(1-1.0))$
 $= 3686.611 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 609.6(1.0 \cdot 13.0 - 1.0 \cdot 12.095) - 2 \cdot 20.0$
 $(1.0 \cdot 13.0 - 1.0 \cdot 12.0952) \cdot (1 - 1.0)$
 $= 551.578 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.5) \cdot (20.0 - 1.33) \cdot 1.0$

= 1213.674 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 12.0^2 \cdot 1.0$
= 244.000 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(1100.0 - 649.6) \cdot 12.0 \cdot 1.0$
= 5404.800 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 1.3281$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
= 1.3281 mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 20.0$, $t_r = 0.77$, $c = 0.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.039$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 12.0$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.535$, Temp. Reduction = 28 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 12.0$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.535$, Temp. Reduction = 28 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 12.0$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.535$, Temp. Reduction = 28 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 14.0$, $t_r = 6.958$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.535$, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-46 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle : -46 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.35/1.96 = 0.178

Weld Size Calculations, Description: M2 24"

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 12.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	6.0000 = 0.5*TminPad	8.4840 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (3686.6113 - 551.5778 + 2 \cdot 20.0 \cdot 1.0 \cdot (1.0 \cdot 13.0 - 12.0952)) \cdot 138)$
 $= 437584.12 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1213.6743 + 5404.8003 + 244.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 946923.81 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1213.6743 + 0.0 + 100.0 + (520.0)) \cdot 138$
 $= 253020.97 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1213.6743 + 0.0 + 244.0 + 5404.8003 + (520.0)) \cdot 138$
 $= 1018676.44 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 649.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 689928. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1100.0 \cdot 12.0 \cdot 0.49 \cdot 138$
 $= 1401923. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 314.8) \cdot (20.0 - 0.0) \cdot 0.7 \cdot 138$
 $= 1910534. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 649.6 \cdot 12.0 \cdot 0.74 \cdot 138$
 $= 1250320. \text{ N}$

Tension, Shell Groove Weld [Tngw]:



$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 649.6 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 1250320. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (1401923 + 1910534) = 3312457 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (689929 + 1250320 + 1250320 + 0) = 3190567 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (1401923 + 1250320 + 0) = 2652243 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 3312457 N	, must exceed W = 437584 N	or W1 = 946923 N
Path 2-2 = 3190567 N	, must exceed W = 437584 N	or W2 = 253020 N
Path 3-3 = 2652242 N	, must exceed W = 437584 N	or W3 = 1018676 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 19.2271 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 253.2270 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.177 %

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Input, Nozzle Desc: M1 24"

From: 130

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2204.9993	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		10700.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

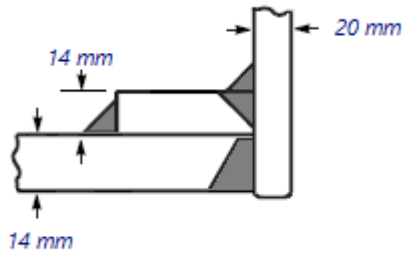
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	220.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1100.0000	mm
Thickness of Pad	te	14.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm
Reinforcing Pad Width		225.2000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M1 24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 304.8 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 0.7701 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3281 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1219.2000 mm
Parallel to Vessel Wall, opening length	d	609.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

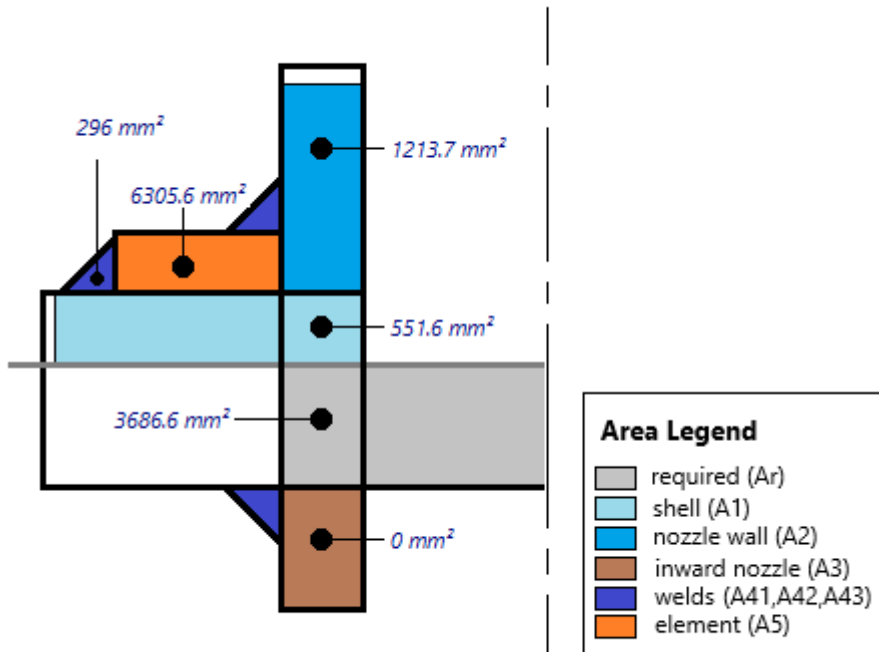


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4241.891	3686.611	NA
Area in Shell	A1	3682.909	551.578	NA
Area in Nozzle Wall	A2	1249.942	1213.674	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		296.000	296.000	NA
Area in Element	A5	6305.601	6305.601	NA
TOTAL AREA AVAILABLE	Atot	11534.452	8366.853	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 765.6970 14.0000 mm
 Based on given Pad Diameter: 1100.0000 3.6087 mm
 Based on Shell or Nozzle Thickness: 765.6970 14.0000 mm

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(609.6 \cdot 12.0952 \cdot 1 + 2 \cdot 20.0 \cdot 12.0952 \cdot 1(1-1.0))$
 $= 3686.611 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 609.6(1.0 \cdot 13.0 - 1.0 \cdot 12.095) - 2 \cdot 20.0$
 $(1.0 \cdot 13.0 - 1.0 \cdot 12.0952) \cdot (1 - 1.0)$
 $= 551.578 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 32.5) \cdot (20.0 - 1.33) \cdot 1.0$

= 1213.674 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - c_{an}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0$
= 296.000 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(1100.0 - 649.6) \cdot 14.0 \cdot 1.0$
= 6305.601 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 1.3281$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
= 1.3281 mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 20.0$, $t_r = 0.77$, $c = 0.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.039$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 12 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 14.0$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.535$, Temp. Reduction = 28 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
Min Metal Temp. at Required thickness (UCS 66.1) -46 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 14.0$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.535$, Temp. Reduction = 28 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
Min Metal Temp. at Required thickness (UCS 66.1) -46 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 14.0$, $t_r = 6.958$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.535$, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
Min Metal Temp. at Required thickness (UCS 66.1) -46 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 14.0$, $t_r = 6.958$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.535$, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
Min Metal Temp. at Required thickness (UCS 66.1) -46 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -46 °C



Gov. MDMT of the Reinforcement Pad : -46 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/1.96 = 0.178

Weld Size Calculations, Description: M1 24"

Intermediate Calc. for nozzle/shell Welds Tmin 14.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	6.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
= max(0, (3686.6113 - 551.5778 + 2 * 20.0 * 1.0 *
(1.0 * 13.0 - 12.0952))138)
= 437584.12 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²*fr2)*Sv
= (1213.6743 + 6305.6006 + 296.0 - 0.0 * 1.0) * 138
= 1078396.75 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 * tn * t * fr1)) * Sv
= (1213.6743 + 0.0 + 100.0 + (520.0)) * 138
= 253020.97 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2*tn*t*fr1))*S
= (1213.6743 + 0.0 + 296.0 + 6305.6006 + (520.0)) * 138
= 1150149.38 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) * Dlo * Wo * 0.49 * Snw
= (3.1416/2.0) * 649.6 * 10.0 * 0.49 * 138
= 689928. N

Shear, Pad Element Weld [Spew]:
= (π/2) * DP * WP * 0.49 * SEW
= (3.1416/2.0) * 1100.0 * 14.0 * 0.49 * 138
= 1635577. N

Shear, Nozzle Wall [Snw]:
= (π * (Dlr + Dlo)/4) * (Thk - Can) * 0.7 * Sn
= (3.1416 * 314.8) * (20.0 - 0.0) * 0.7 * 138
= 1910534. N

Tension, Pad Groove Weld [Tpgw]:
= (π/2) * Dlo * Wgpn * 0.74 * Seg
= (3.1416/2) * 649.6 * 14.0 * 0.74 * 138
= 1458706. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) * Dlo * Wgnvi * 0.74 * Sng



$$= (3.1416/2.0) \cdot 649.6 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138 \\ = 1250320. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1635577 + 1910534) = 3546111 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (689929 + 1458706 + 1250320 + 0) = 3398954 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (1635577 + 1250320 + 0) = 2885896 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 3546110 N	, must exceed W = 437584 N	or W1 = 1078396 N
Path 2-2 = 3398954 N	, must exceed W = 437584 N	or W2 = 253020 N
Path 3-3 = 2885896 N	, must exceed W = 437584 N	or W3 = 1150149 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 19.2271 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 253.2270 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.177 %

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Input, Nozzle Desc: N3 20"

From: 130

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2204.9993	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		9950.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		20.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm

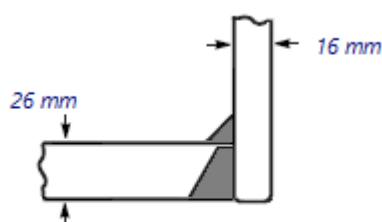
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	220.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N3 20"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 20.000 in.
 Actual Thickness Used in Calculation 0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.35 \cdot 254.0 / (138 \cdot 1.0 + 0.4 \cdot 0.35)$
 $= 0.6401 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.1496 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	952.0000	mm
Parallel to Vessel Wall, opening length	d	476.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	40.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

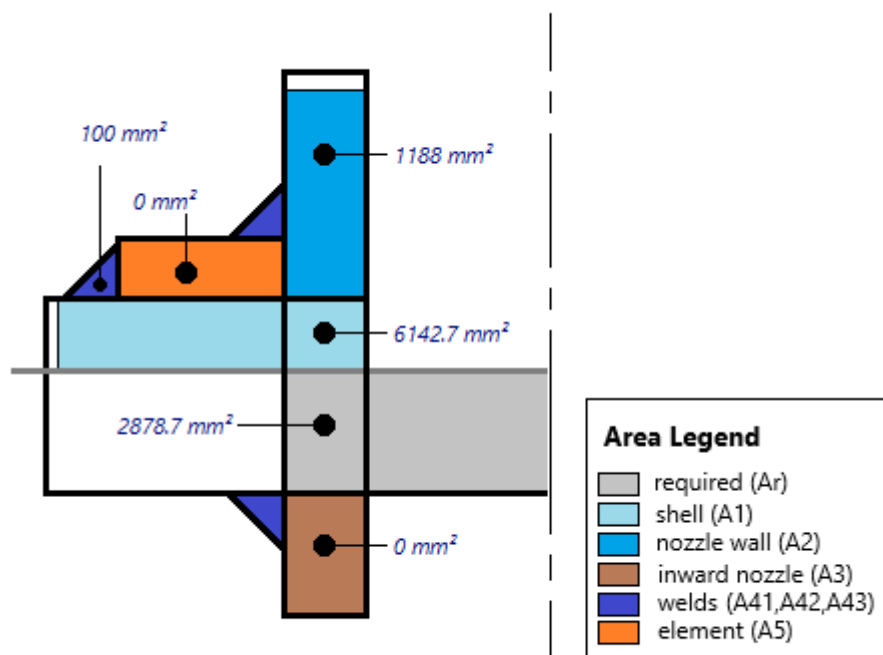


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	3312.238	2878.653	NA
Area in Shell	A1	8587.762	6142.693	NA
Area in Nozzle Wall	A2	1228.788	1188.031	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	9916.550	7430.724	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(476.0 \cdot 12.0952 \cdot 1 + 2 \cdot 16.0 \cdot 12.0952 \cdot 1(1 - 1.0))$
 $= 2878.653 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 476.0(1.0 \cdot 25.0 - 1.0 \cdot 12.095) - 2 \cdot 16.0$
 $(1.0 \cdot 25.0 - 1.0 \cdot 12.0952) \cdot (1 - 1.0)$
 $= 6142.693 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 40.0)(16.0 - 1.15) 1.0$
 $= 1188.031 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.1496 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 8.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.331, \max(7.9585, 3.0648)]$
 $= 7.9585 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.1496, 7.9585)$
 $= 7.9585 \text{ mm}$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 17.8,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 327.2 MPa	Passed
Occasional	: 2.5,	Allowable	: 183.5 MPa	Passed
Shear	: 11.7,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 16.0, tr = 0.64, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.04$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.0, tr = 0.64, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.04$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.35/5.10 = 0.068$

Weld Size Calculations, Description: N3 20"

Intermediate Calc. for nozzle/shell Welds Tmin 16.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (2878.6533 - 6142.6929 + 2 \cdot 16.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 12.0952)) \cdot 138)$
 $= \max(0, -393409.22) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1188.0305 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 177729.88 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1188.0305 + 0.0 + 100.0 + (800.0)) \cdot 138$
 $= 288118.50 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1188.0305 + 0.0 + 100.0 + 0.0 + (800.0)) \cdot 138$
 $= 288118.50 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 508.0 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 539528. \text{ N}$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 246.0) \cdot (16.0 - 0.0) \cdot 0.7 \cdot 138$$

$$= 1194387. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 508.0 \cdot (18.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 1466662. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (539528 + 1194387) = 1733915 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (539528 + 0 + 1466662 + 0) = 2006190 N$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (539528 + 1466662 + 0) = 2006190 N$$

Summary of Failure Path Calculations:

Path 1-1 = 1733915 N	, must exceed W = 0 N	or W1 = 177729 N
Path 2-2 = 2006189 N	, must exceed W = 0 N	or W2 = 288118 N
Path 3-3 = 2006189 N	, must exceed W = 0 N	or W3 = 288118 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 11.7425 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 257.7425 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.252 %

Input Echo, WRC297 Item 1, Description: N3 20" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	5506.000	mm
Shell Thickness	26.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	508.000	mm
Nozzle Thickness	16.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	0.35	MPa
Radial Load	P	-37500.02	N
Circumferential Shear	Vc	24999.98	N
Longitudinal Shear	Vl	24999.98	N
Circumferential Moment	Mc	0.350E+08	N-mm
Longitudinal Moment	Ml	0.350E+08	N-mm
Torsional Moment	Mt	0.500E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(2766.5(26.0 - 1.0))}$$

$$= 1375.859 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	5533.000 mm
Nozzle Outside Diameter	(d) =	508.000 mm
Vessel Thickness used	(T) =	25.000 mm
Nozzle Thickness used	(t) =	16.000 mm
T / t	=	1.562
d / t	=	31.750

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.366$$

Nr/P	=	0.077
Mr/P	=	0.136
M0/P	=	0.062
N0/P	=	0.157
MrD/Mc	=	0.234
NrDL/Mc	=	0.105
M0d/Mc	=	0.122
N0DL/Mc	=	0.126
MrD/Ml	=	0.151
NrDL/Ml	=	0.079
M0D/Ml	=	0.071
N0DL/Ml	=	0.240

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-26	-26	26	26
Outplane Bending (ML)	-47	47	47	-47
Normal Pressure Stress	38	38	38	38
Outplane Stress Summary	-3	46	143	4

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	48	-48	48	-48



Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-100	100	100	-100
Inplane Pressure Stress	19	19	19	19

Inplane Stress Summary	-36	66	181	-116
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	6	6	3	3
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	37	67	181	121
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	22	-22	22	-22
Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-80	80	80	-80
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	19	19	19	19

Outplane Stress Summary	-43	72	145	-60
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	48	-48	48	-48
Inplane Membrane (Mc)	-11	-11	11	11
Inplane Bending (Mc)	-154	154	154	-154
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	38	38	38	38

Inplane Stress Summary	-74	137	257	-148
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	6	6
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right



	Outside	Inside	Outside	Inside
Two * Max Shear Stress	74	137	258	149

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-26	-26	26	26
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	6	6	6
Hoop Stress Summary	-11	-10	41	42

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	85	-85	85	-85
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-11	-11	11	11
Axial Bending (ML)	-180	180	180	-180
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-101	86	282	-249

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	7	7	7	7
Shear Stress Summary	9	9	5	5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	102	99	282	292

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	6	6	6
Hoop Stress Summary	1	1	29	29

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------



Axial Membrane (P)	1	1	1	1
Axial Bending (P)	85	-85	85	-85
Axial Membrane (Mc)	-11	-11	11	11
Axial Bending (Mc)	-293	293	293	-293
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-214	200	395	-362
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	7	7	7	7
Shear Stress Summary	5	5	9	9

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	216	200	395	392

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	38	38	38	38	38	38	38	38
Circ. Pl (SUS)	-17	-17	35	35	-7	-7	15	15
Circ. Q (SUS)	-25	25	69	-69	-106	106	202	-202
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	-4	-4	12	12	-4	-4	22	22
Long. Q (SUS)	-52	52	148	-148	-58	58	102	-102
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	38.0	38.0	38.0	38.0	38.0	38.0	38.0	38.0
Pm+Pl (SUS)	21.2	21.2	73.0	73.0	31.1	31.1	53.1	53.1
Pm+Pl+Q (Total)	37.7	68.1	179.2	121.1	75.3	137.1	255.2	149.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38.00	138.00	Passed
Pm+Pl (SUS)	73.02	207.00	Passed
Pm+Pl+Q (TOTAL)	255.22	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int.	Stress Values at
------------------------	------------------



Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	6	6	6	6	6	6	6
Circ. Pl (SUS)	-17	-17	35	35	-4	-4	22	22
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-10	-10	12	12	-10	-10	12	12
Long. Q (SUS)	-95	95	265	-265	-208	208	378	-378
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Pm+Pl (SUS)	11.3	11.3	41.0	41.0	10.2	10.2	28.1	28.1
Pm+Pl+Q (Total)	103.7	99.3	279.2	292.2	218.3	200.2	392.2	392.3

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	138.00	Passed
Pm+Pl (SUS)	41.04	207.00	Passed
Pm+Pl+Q (TOTAL)	392.33	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N6 C 2"

From: 130

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2204.9993	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		9250.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		220.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

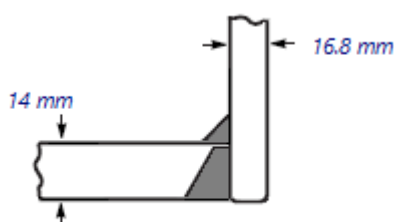
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N6 C 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	110.3280 mm
Parallel to Vessel Wall	Rn+tn+t	55.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

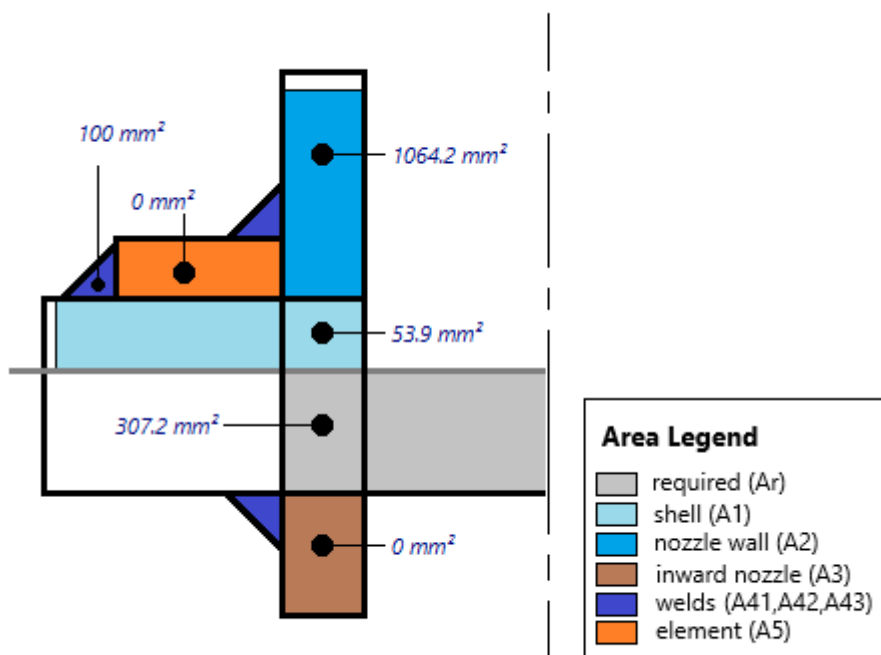


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	307.218	NA
Area in Shell	A1	359.639	53.862	NA
Area in Nozzle Wall	A2	1085.490	1064.158	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1545.129	1218.020	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 12.0952 \cdot 1 + 2 \cdot 16.764 \cdot 12.0952 \cdot 1(1 - 1.0))$
 $= 307.218 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 59.528(1.0 \cdot 13.0 - 1.0 \cdot 12.095) - 2 \cdot 16.764$
 $(1.0 \cdot 13.0 - 1.0 \cdot 12.0952) \cdot (1 - 1.0)$
 $= 53.862 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 32.5)(16.76 - 0.39) 1.0$
 $= 1064.158 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 14.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.535$, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -26 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -26 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: N6 C 2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (307.2176 - 53.8621 + 2 • 16.764 • 1.0 •
(1.0 • 13.0 - 12.0952))138)
= 39145.50 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1064.1576 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 160637.19 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1064.1576 + 0.0 + 100.0 + (435.864)) • 138
= 220780.23 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1064.1576 + 0.0 + 100.0 + 0.0 + (435.864)) • 138
= 220780.23 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (12.0 - 0.0) • 0.74 • 138
= 162311. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 162311 + 0) = 251872 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 162311 + 0) = 251872 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 39145 N or W1 = 160637 N

Path 2-2 = 251872 N , must exceed W = 39145 N or W2 = 220780 N

Path 3-3 = 251872 N , must exceed W = 39145 N or W3 = 220780 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3230 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.6980 mm

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Input, Nozzle Desc: K6B 2"

From: 130

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2204.9993	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		9050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

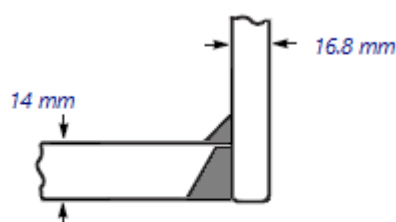
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K6B 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	110.3280 mm
Parallel to Vessel Wall	Rn+tn+t	55.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

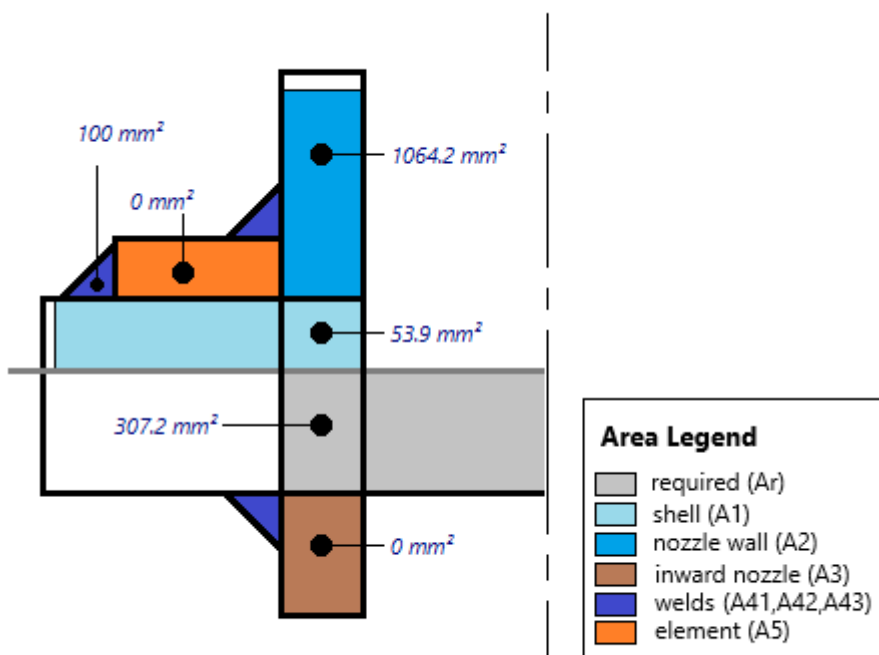


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	307.218	NA
Area in Shell	A1	359.639	53.862	NA
Area in Nozzle Wall	A2	1085.490	1064.158	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1545.129	1218.020	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 12.0952 \cdot 1 + 2 \cdot 16.764 \cdot 12.0952 \cdot 1(1-1.0))$
 $= 307.218 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 59.528(1.0 \cdot 13.0 - 1.0 \cdot 12.095) - 2 \cdot 16.764$
 $(1.0 \cdot 13.0 - 1.0 \cdot 12.0952) \cdot (1 - 1.0)$
 $= 53.862 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 32.5)(16.76 - 0.39) 1.0$
 $= 1064.158 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 14.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.535$, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -26 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -26 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: K6B 2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (307.2176 - 53.8621 + 2 • 16.764 • 1.0 •
(1.0 • 13.0 - 12.0952))138)
= 39145.50 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1064.1576 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 160637.19 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1064.1576 + 0.0 + 100.0 + (435.864)) • 138
= 220780.23 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1064.1576 + 0.0 + 100.0 + 0.0 + (435.864)) • 138
= 220780.23 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (12.0 - 0.0) • 0.74 • 138
= 162311. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 162311 + 0) = 251872 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 162311 + 0) = 251872 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 39145 N or W1 = 160637 N

Path 2-2 = 251872 N , must exceed W = 39145 N or W2 = 220780 N

Path 3-3 = 251872 N , must exceed W = 39145 N or W3 = 220780 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3230 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.6980 mm

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Input, Nozzle Desc: K5B 2"

From: 130

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2204.9993	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		9050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		10.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

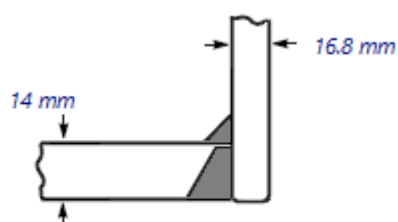
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K5B 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	110.3280 mm
Parallel to Vessel Wall	Rn+tn+t	55.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

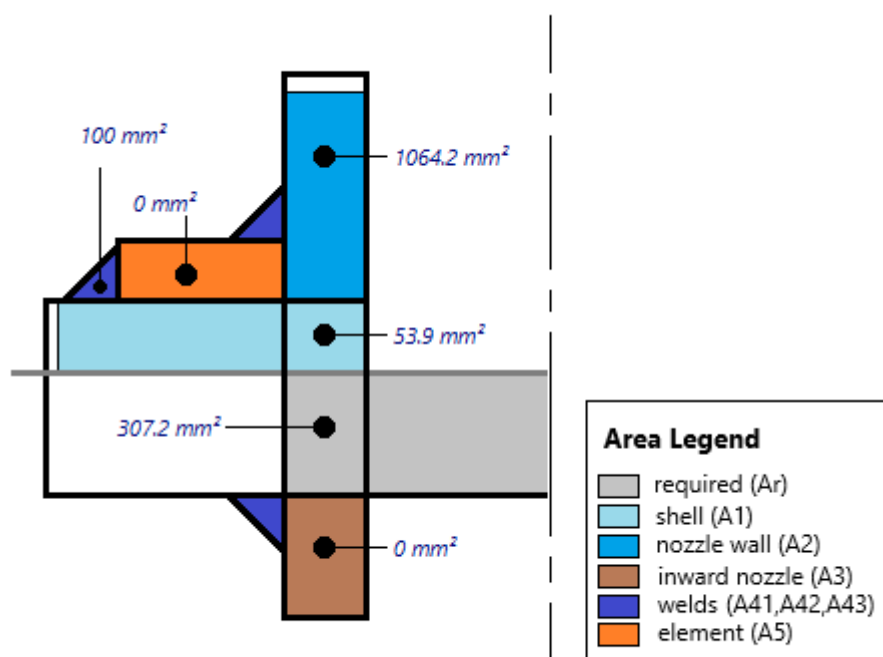


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	307.218	NA
Area in Shell	A1	359.639	53.862	NA
Area in Nozzle Wall	A2	1085.490	1064.158	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1545.129	1218.020	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 12.0952 \cdot 1 + 2 \cdot 16.764 \cdot 12.0952 \cdot 1(1-1.0))$
 $= 307.218 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 59.528(1.0 \cdot 13.0 - 1.0 \cdot 12.095) - 2 \cdot 16.764$
 $(1.0 \cdot 13.0 - 1.0 \cdot 12.0952) \cdot (1 - 1.0)$
 $= 53.862 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 32.5)(16.76 - 0.39) 1.0$
 $= 1064.158 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 14.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.535$, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -26 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -26 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: K5B 2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (307.2176 - 53.8621 + 2 • 16.764 • 1.0 •
(1.0 • 13.0 - 12.0952))138)
= 39145.50 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1064.1576 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 160637.19 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1064.1576 + 0.0 + 100.0 + (435.864)) • 138
= 220780.23 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1064.1576 + 0.0 + 100.0 + 0.0 + (435.864)) • 138
= 220780.23 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (12.0 - 0.0) • 0.74 • 138
= 162311. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 162311 + 0) = 251872 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 162311 + 0) = 251872 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 39145 N or W1 = 160637 N

Path 2-2 = 251872 N , must exceed W = 39145 N or W2 = 220780 N

Path 3-3 = 251872 N , must exceed W = 39145 N or W3 = 220780 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3230 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.6980 mm

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Input, Nozzle Desc: N72"

From: 130

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	5506.00	mm
Design Length of Section	L	2204.9993	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		10050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

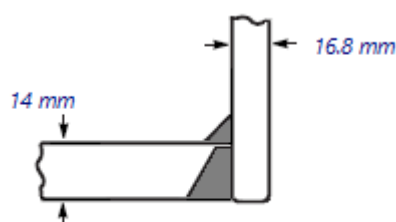
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N72"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.35 \cdot 2754.0 / (138 \cdot 1.0 - 0.6 \cdot 0.35)$
 $= 6.9585 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
 $= 0.0642 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	110.3280 mm
Parallel to Vessel Wall	Rn+tn+t	55.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

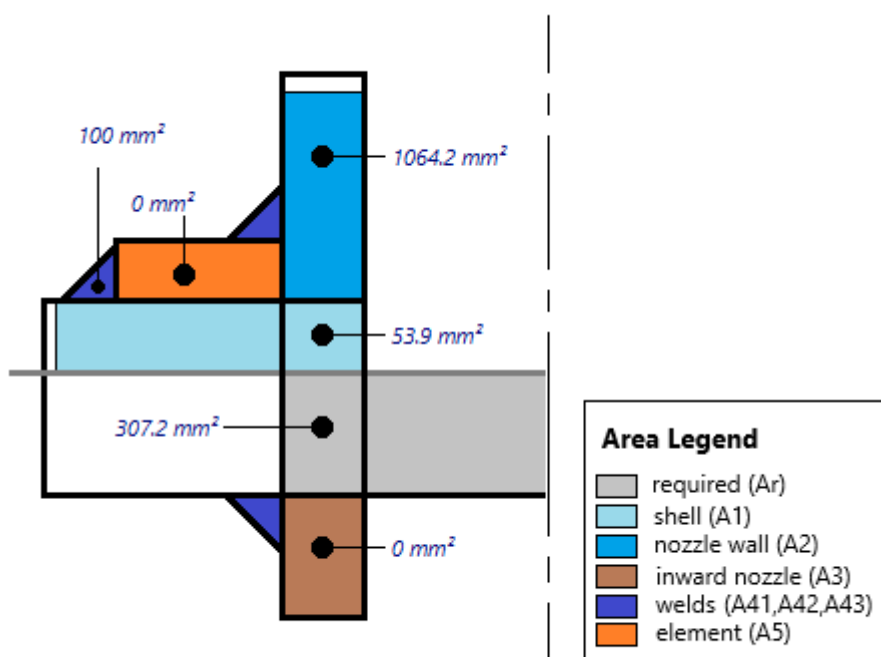


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5	Design	External	Mapnc
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Area Required	Ar	353.491	307.218	NA
Area in Shell	A1	359.639	53.862	NA
Area in Nozzle Wall	A2	1085.490	1064.158	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1545.129	1218.020	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(50.8 \cdot 12.0952 \cdot 1 + 2 \cdot 16.764 \cdot 12.0952 \cdot 1(1-1.0))$
 $= 307.218 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 59.528(1.0 \cdot 13.0 - 1.0 \cdot 12.095) - 2 \cdot 16.764$
 $(1.0 \cdot 13.0 - 1.0 \cdot 12.0952) \cdot (1 - 1.0)$
 $= 53.862 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 32.5)(16.76 - 0.39) 1.0$
 $= 1064.158 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.3923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 7.9585 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 7.9585 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 3.0648 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0648 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 4.8000 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.9585, 3.0648)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.3923, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 14.0$, $tr = 6.958$, $c = 1.0 \text{ mm}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E* / (tg - c) = 0.535$, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -26 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -26 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: N72"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (307.2176 - 53.8621 + 2 • 16.764 • 1.0 •
(1.0 • 13.0 - 12.0952))138)
= 39145.50 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1064.1576 + 0.0 + 100.0 - 0.0 • 1.0) • 138
= 160637.19 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1064.1576 + 0.0 + 100.0 + (435.864)) • 138
= 220780.23 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1064.1576 + 0.0 + 100.0 + 0.0 + (435.864)) • 138
= 220780.23 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 84.328 • 10.0 • 0.49 • 138
= 89562. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 33.782) • (16.764 - 0.0) • 0.7 • 138
= 171851. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 84.328 • (12.0 - 0.0) • 0.74 • 138
= 162311. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (89562 + 171851) = 261413 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (89562 + 0 + 162311 + 0) = 251872 N



Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (89562 + 162311 + 0) = 251872 N

Summary of Failure Path Calculations:

Path 1-1 = 261413 N , must exceed W = 39145 N or W1 = 160637 N

Path 2-2 = 251872 N , must exceed W = 39145 N or W2 = 220780 N

Path 3-3 = 251872 N , must exceed W = 39145 N or W3 = 220780 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3230 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.6980 mm

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Input, Nozzle Desc: N6 D 2"

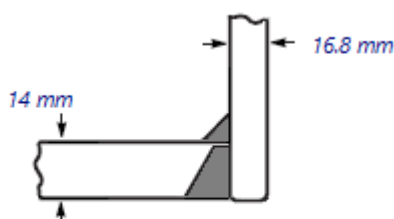
From: 150

Pressure for Reinforcement Calculations	P	0.3481	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	0	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cone at Nozzle Location	D	5000.54	mm
Equivalent Length of Conical Section	Le	2563.0010	mm
Cone Half Apex Angle	Alpha	19.79	Degrees
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		11500.00	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Conical Transition, tr [Int. Press]
= $P \cdot D / (2 \cdot \cos(a) \cdot (S_v \cdot E - 0.6 \cdot P))$ Appendix 1-4(e)
= $0.348 \cdot 5002.668 / (2 \cdot \cos(19.8) \cdot (137.997 \cdot 1.0 - 0.6 \cdot 0.348))$
= 6.7167 mm

Reqd Cone thickness at Nozzle Location under External Pres. : 13.8854 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $25.4 \cdot (\exp([0.35 / (138.0 \cdot 1.0)] - 1))$
= 0.0642 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.5159	mm
Parallel to Vessel Wall	Rn+tn+t	56.7579	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138.0/138.0)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000

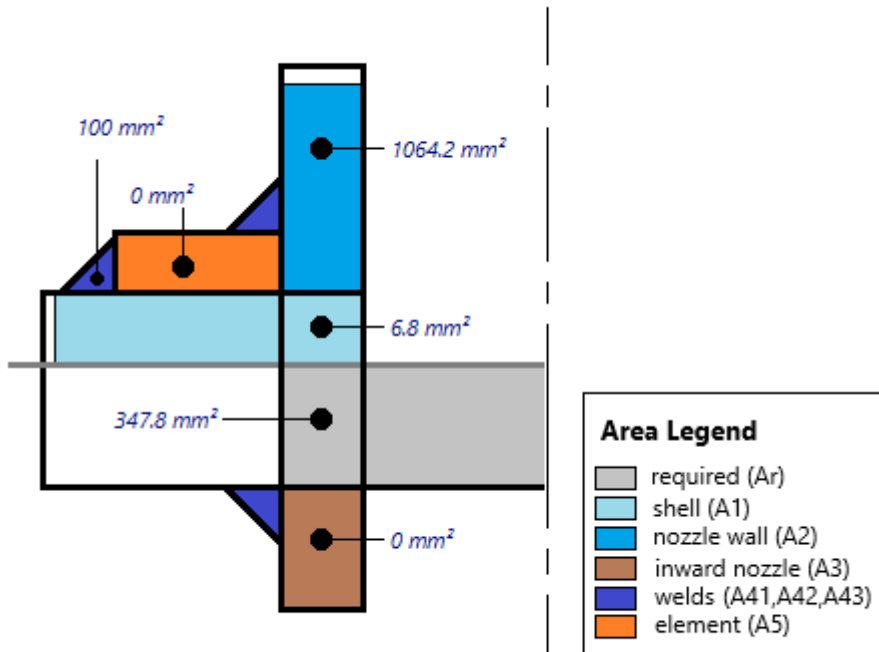


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	362.619	347.828	NA
Area in Shell	A1	374.033	6.822	NA
Area in Nozzle Wall	A2	1085.545	1064.212	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1559.578	1171.034	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 70.21 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F (1 - fr1))$ per UG-37(d)
 $= 0.5 (53.9879 \cdot 12.8854 \cdot 1 + 2 \cdot 16.764 \cdot 12.8854 \cdot 1 (1 - 1.0))$
 $= 347.828 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 59.528 (1.0 \cdot 13.0 - 1.0 \cdot 12.885) - 2 \cdot 16.764$
 $(1.0 \cdot 13.0 - 1.0 \cdot 12.885) \cdot (1 - 1.0)$
 $= 6.822 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2 / \sin(\alpha3)$
 $= (2 \cdot 32.5) (16.76 - 0.39) 1.0 / \sin(90.6)$
 $= 1064.212 \text{ mm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$
$$= 100.000 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 0.3923 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.7167 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.7167 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.9931 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.9931 mm
Wall Thickness per table UG-45	tb3 = 4.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$
$$= \min[4.8, \max(7.7167, 2.9931)]$$
$$= 4.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$
$$= \max(0.3923, 4.8)$$
$$= 4.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 14.0, tr = 6.717, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.517, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-28 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -28 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.35/5.10 = 0.068

Weld Size Calculations, Description: N6 D 2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$
$$= \max(0, (347.8277 - 6.8219 + 2 \cdot 16.764 \cdot 1.0 \cdot (1.0 \cdot 13.0 - 12.8854)) \cdot 138)$$
$$= 47584.12 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$
$$= (1064.2119 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$$
$$= 160644.67 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$
$$= (1064.2119 + 0.0 + 100.0 + (435.864)) \cdot 138$$



= 220787.72 N

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S \\ &= (1064.2119 + 0.0 + 100.0 + 0.0 + (435.864)) \cdot 138 \\ &= 220787.72 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 89.6199 \cdot 10.0 \cdot 0.49 \cdot 138 \\ &= 95182. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 35.9019) \cdot (16.764 - 0.0) \cdot 0.7 \cdot 138 \\ &= 182636. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 89.6199 \cdot (12.0 - 0.0) \cdot 0.74 \cdot 138 \\ &= 172496. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (95182 + 182636) = 277818 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (95182 + 0 + 172496 + 0) = 267678 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (95182 + 172496 + 0) = 267678 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 277817 N	, must exceed W = 47584 N	or W1 = 160644 N
Path 2-2 = 267678 N	, must exceed W = 47584 N	or W2 = 220787 N
Path 3-3 = 267678 N	, must exceed W = 47584 N	or W3 = 220787 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.3556 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.7305 mm

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: